SURVIVAL MODELS

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1 Introduction to Survival Models

1.1 Module information

The Survival Models course is a one semester course, F79SU. The module contributes to exemption from the Institute and Faculty of Actuaries Subject CS2: Risk Modelling and Survival Analysis.

Assessment: 2 hour exam (60%), project work (40%).
Information about the project work, including dates, will be available through the course page on Canvas.

Course web site: The course web site can be found on Canvas, providing information on:

- **The statistical language, R:** R is central to computation in this course and the web site holds much information specifically on using R in Survival Models. Numerous example programs are also available.

- **Projects, Tutorial solutions, Past papers**

1.2 Notation and basic formulae

The future lifetime of a new born person is a random variable, denoted T, which is continuously distributed on an interval $[0, w]$, $0 < w < \infty$. The value of w is called the **limiting age** and in practice is usually taken as some value in the range 100-120 (although Jeanne Calment, the longest-lived documented person, lived to 122).

The **distribution function** of T is written

$$F(t) = \Pr(T \leq t)$$

and the **survival function** is denoted by

$$S(t) = \Pr(T > t) = 1 - F(t).$$

Suppose we know that a policyholder with lifetime T has survived to age x. We define T_x to be the **future** lifetime after age x, of a life who has survived to age x, $0 \leq x \leq w$. Note that $T_0 = T$. We write

$$F_x(t) = \Pr(T_x \leq t)$$

for the distribution function of T_x, and

$$S_x(t) = \Pr(T_x > t) = 1 - F_x(t)$$

for the **survival function** of T_x.

We have the following relation between $F_x(t)$ and $F(t)$.

$$\begin{aligned} F_x(t) &= \Pr(T_x \leq t) \\ &= \Pr(T \leq t + x \mid T > x) \\ &= \frac{F(x+t) - F(x)}{S(x)}. \end{aligned}$$

Actuaries have their own notation for $F_x(t)$ so we define

$$\begin{aligned} {}_tq_x &= F_x(t) \\ {}_tp_x &= 1 - {}_tq_x = S_x(t) \end{aligned}$$

and in particular

$$\begin{aligned} q_x &= {}_1q_x \\ p_x &= {}_1p_x \end{aligned}$$

which are called **rates of mortality**.

A fundamental quantity in a survival model is the **force of mortality** (or **hazard rate** or **instantaneous death rate**). For age x, $0 \le x \le w$, we write

$$\mu_x = \lim_{h \to 0+} \frac{1}{h} \times \Pr(T \le x + h \mid T > x).$$

Now we know that

$$\Pr(T \le x + h \mid T > x) = {}_hq_x$$

so for **small** h we have the important formula

$$ {}_hq_x \approx h \cdot \mu_x. \tag{1}$$

Further relationships

(a) $S_x(t) = \dfrac{S(x+t)}{S(x)}$.

(b) The chance a life aged x survives for a further $s+t$ is first, the chance that a life aged x survives for t, followed by the chance that a life now aged $x+t$ survives for a further time s, i.e.

$$ {}_{s+t}p_x = {}_tp_x \times {}_sp_{x+t}.$$

(c) The chance a life aged x dies in $x+t$ to $x+t+dt$ is approximately $f_x(t)\,dt$. We can also think of this as the chance the life aged x survives for a further t, i.e. ${}_tp_x$, and then dies in $x+t$ to $x+t+dt$, i.e. $\mu_{x+t}dt$. Thus,

$$f_x(t) = {}_tp_x \cdot \mu_{x+t}.$$

(d) We also have the following integral forms for ${}_tq_x$ and ${}_tp_x$.

$$\begin{aligned} {}_tq_x &= \int_0^t {}_sp_x \cdot \mu_{x+s}\, ds \\ {}_tp_x &= \exp\left\{ -\int_0^t \mu_{x+s}\, ds \right\}. \end{aligned}$$

When μ_{x+s} is constant for $0 \le s < t$ and equal to μ_x, say, this last formula gives rise to the fundamental expression

$$\begin{aligned} {}_tp_x &= \exp\left\{ -\int_0^t \mu_{x+s}\, ds \right\} \\ &= \exp\left\{ -\int_0^t \mu_x\, ds \right\} \\ &= \exp(-\mu_x t) \\ \Rightarrow \quad {}_tq_x &= 1 - \exp(-\mu_x t). \end{aligned} \tag{2}$$

Example: (HW, 2008) Suppose that a particular female life has a piecewise constant force of mortality as follows: $\mu_t = 0.010$ for $60 \leq t < 65$, $\mu_t = 0.016$ for $65 \leq t < 70$ and $\mu_t = 0.026$ for $70 \leq t < 75$.

(a) Evaluate ${}_5p_{60}$, ${}_5p_{65}$ and ${}_5p_{70}$.

(b) Suppose we have three independent female lives all aged 60 and all subject to the above mortality. Find the probability that all of the following three events occur: one dies before age 65, one dies between 70 and 75 and one survives beyond 75.

(a) We have ${}_5p_{60} = \exp(-5\mu) = \exp(-0.05) = 0.9512$, ${}_5p_{65} = \exp(-5\mu) = \exp(-0.08) = 0.9231$ and ${}_5p_{70} = \exp(-5\mu) = \exp(-0.13) = 0.8781$.

(b) Pr(Die before age 65) $= 1 - {}_5p_{60} = 0.0488$.
Pr(Die between age 70 & 75) $= {}_5p_{60} * {}_5p_{65} * (1 - {}_5p_{70}) = 0.9512 * 0.9231 * 0.1219 = 0.1070$.
Pr(Survive beyond age 75) $= {}_5p_{60} * {}_5p_{65} * {}_5p_{70} = 0.9512 * 0.9231 * 0.8781 = 0.7710$.
Required probability is $6 * 0.0488 * 0.1070 * 0.7710 = 0.02416$. ■

Example: Suppose that $T \sim \mathcal{E}(\lambda)$. Find μ_x.

We know that $T \sim \mathcal{E}(\lambda) \Rightarrow F(t) = 1 - \exp(-\lambda t)$. Using the definition of the force of mortality we have

$$\begin{aligned}
& \Pr(T \leq x + h \mid T > x) \\
= \; & \Pr(x < T \leq x + h) / \Pr(T > x) \\
= \; & \frac{F(x+h) - F(x)}{S(x)} \\
= \; & \frac{1 - \exp(-\lambda(x+h)) - 1 + \exp(-\lambda x)}{\exp(-\lambda x)} \\
= \; & \frac{\exp(-\lambda x)(1 - \exp(-\lambda h))}{\exp(-\lambda x)} \\
= \; & (1 - \exp(-\lambda h)) \\
= \; & 1 - (1 - \lambda h + \tfrac{1}{2}\lambda^2 h^2 - \ldots) \\
= \; & \lambda h + \text{higher powers of } h.
\end{aligned}$$

So dividing by h and letting $h \to 0$ we get

$$\mu_x = \lambda.$$

Thus, if T is exponential, the life is subject to a **constant** force of mortality.

The converse is also true and follows immediately from equation (2). Thus, if a life is subject to a constant force of mortality λ then its lifetime follows an exponential distribution with parameter λ and mean lifetime $1/\lambda$.

Tutorial 1: Revision

1. Prove that
$$S_x(t) = \frac{S(x+t)}{S(x)}.$$

2. Prove that
$${}_tq_x = \int_0^t {}_sp_x\, \mu_{x+s}\, ds$$
and give a simple interpretation of this result.

3. Prove that
$${}_tp_x = \exp\left\{-\int_0^t \mu_{x+s}\, ds\right\}.$$

4. (a) (i) (HW, 2007) Suppose that the mortality of a life has a piecewise constant force of mortality as follows: μ_0 for $70 \le t < 71$ and μ_1 for $71 \le t < 72$. Evaluate ${}_2p_{70}$.
   (ii) Suppose that the mortality of a second life has a constant force of mortality $(\mu_0 + \mu_1)/2$ for $70 \le t < 72$. Evaluate ${}_2p_{70}$.
   (iii) What general rule do (i) and (ii) illustrate?

   (b) Consider two independent lives, A and B. Life A is age 70 exact and is subject to the mortality in ELT15(Males), and life B is age 65 exact and is subject to the mortality in ELT15(Females).
   (i) Calculate ${}_5p_{70}$ for life A.
   (ii) Calculate ${}_5p_{65}$ for life B.
   (iii) Calculate the probability both lives survive for 5 years.
   (iv) Calculate the probability exactly one life survives for 5 years.

5. (HW, 2006) The mortality of a set of male lives is approximately constant over single years as follows:
$$\begin{aligned} \mu_{70+t} &= 0.04, \quad 0 \le t < 1 \\ \mu_{71+t} &= 0.045, \quad 0 \le t < 1 \\ \mu_{72+t} &= 0.05, \quad 0 \le t < 1 \end{aligned}$$
Consider a set of lives aged exactly 70. Evaluate the probability that
   (a) a single life dies before age 73
   (b) of three such lives one dies before age 73 and the other two survive.

6. Computing: see the web site: section *R-Scripts*, document *Introduction to R*.

2 Estimating the Lifetime Distribution

2.1 Cohort Studies

We want to estimate the distribution of the lifetime variables T, T_0 and T_x. We consider first data that arises from a **cohort study**. Later we will look at the **cross-sectional study**.

We observe a **large number of new-born lives**. The proportion alive at age t is an estimate of the survivor function $S(t)$. This will be a step function, and the larger the number of lives the smoother the estimate of $S(t)$ will be. Equivalently, we can take the proportion who have died up to and including time t. This estimate of $F(t)$ is the **empirical distribution function**, $\hat{F}(t)$. A typical graph of $\hat{F}(t)$ (for a small sample) would look like this.

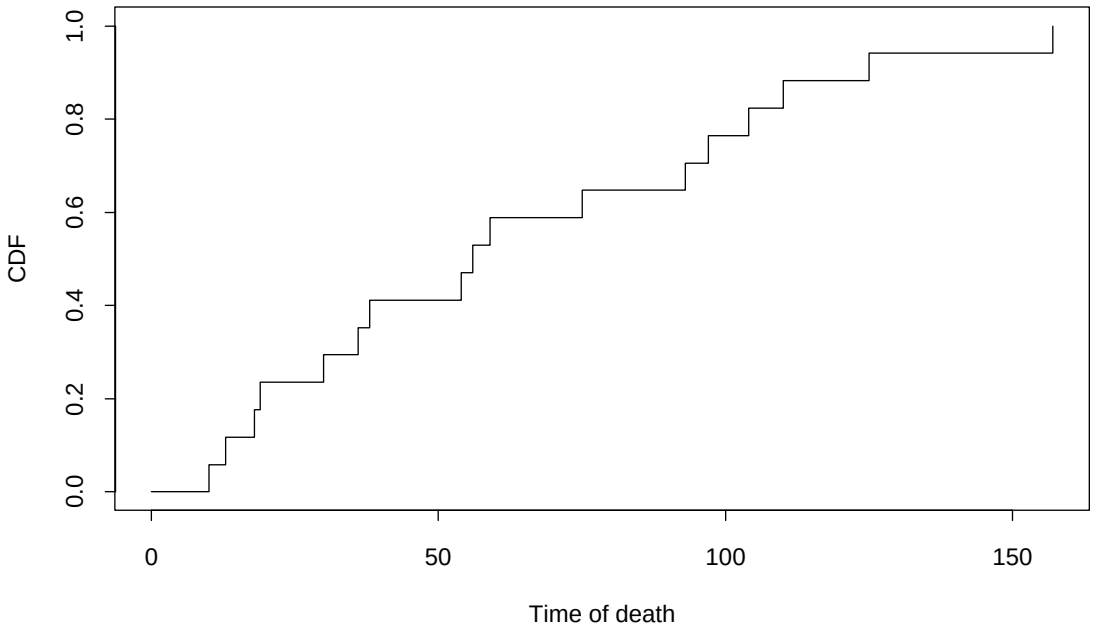


Figure 1: Empirical distribution function, $\hat{F}(t)$

This graph could be smoothed or **graduated**, if required.

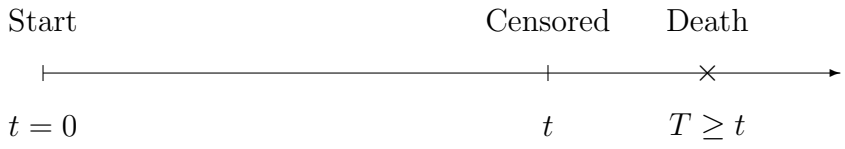
Problems: the cohort study suffers from a number of problems which make it unsuitable in insurance.

- The study will take about 100 years to complete.
- In practice, we will lose track of a large number of individuals in the cohort. This may produce a biased sample, since the mortality of the lost lives may be different from those lives that remain in the study. We call this problem **censoring**. For censored lives, all we know is that these lives had lifetimes that exceed a certain age.

2.2 Censoring

Censoring is a key feature of survival data and has a profound impact on the estimation procedure. Censoring comes in a number of different forms.

1. **Right-censoring** occurs if the censoring mechanism cuts short observations in progress.



   Example: Right-censoring will occur if a mortality investigation ends on a fixed date. All we can say about the lifetime is that it **exceeds the censoring time** t, i.e. $T \geq t$.

2. **Left-censoring** occurs if we do not know the time of entry into the state of interest.



   Example: Left-censoring often occurs in medical studies when only an examination reveals the illness. This time we can say that the survival time from t_1, the **unknown time of onset**, exceeds $t_3 - t_2$, i.e. $T \geq t_3 - t_2$.

3. **Interval-censoring** occurs when the event of interest occurs somewhere between two times t_1 and t_2. In the diagram we know the life is alive at time t_1, dead at time t_2, but we do not know the exact time of death t. All we can say is that $t_1 \leq T \leq t_2$.



   Example: Interval-censoring will occur if a mortality investigation records only the the calendar year of death.

Further terminology: There are a number of ways in which any of the above forms of censoring can arise.

1. **Random censoring** occurs if the censoring time C_i for life i is a random variable. The observation is then censored if $C_i < T_i$, where T_i is the random lifetime of life i.

   Example: In medical studies a patient moves away from the study area at a random time, and the study loses track of the patient.

2. **Non-informative censoring** occurs if the censoring gives no information about the lifetimes $\{T_i\}$. Thus if each member of the pair $\{T_i, C_i\}$ is independent then censoring is non-informative.

Example: In medical studies a patient moves away from the study area at a time unconnected with the progress of their illness. This is non-informative. However, if the patient moves away from the study area in order to receive a different treatment this is informative. Informative censoring may introduce bias into the observed survival times and must be treated with extra care.

3. **Type I censoring** occurs when the censoring times are known in advance, i.e. censoring occurs at fixed times.

   Example: A mortality investigation may end on a particular date.

4. **Type II censoring** occurs if observation continues until a fixed known number of deaths occurs. In this case the number of events is non-random.

   Example: Type II censoring occurs in reliability testing when components may be tested until a certain number fail. This kind of censoring is uncommon in mortality studies.

2.3 Cross-sectional Studies

Instead of following a **cohort** of lives for a long time we can use a **cross-sectional** study. We divide the investigation into single years of age, say, $x \to x+1$, and then follow each "mini-cohort" for three or four years. There are a number of points to note:

- We are now mixing mortality since the mortality of a 60 year old now is not the same as the mortality of a 57 year old in three years time.
- The method introduces type I censoring in the most obvious way, since **most lives will not die** (we suppose).

2.4 The Kaplan-Meier estimate of the survivor function

Suppose we have n lives with survival times $t_1, \ldots, t_n$ where some of the observations are right-censored (non-informative). Suppose there are r distinct death times, $r \leq n$, which we arrange in ascending order $t_{(1)}, \ldots, t_{(r)}$. We form the set of intervals

$$I_j = [t_{(j)},\, t_{(j+1)}), \quad j = 0, 1, \ldots, r$$

where $t_{(0)} = 0$ and $t_{(r+1)} = \infty$. Let

$$n_j = \text{number of patients alive at } t_{(j)}-$$

i.e., **immediately before** time $t_{(j)}$ (the jth death-time), and let

$$d_j = \text{number of deaths at } t_{(j)}.$$

Consider the interval $(t_{(j)}-\delta,\, t_{(j)}]$. There are n_j patients alive at the beginning of the interval and d_j deaths at $t_{(j)}$ so the probability that a patient dies in $(t_{(j)} - \delta,\, t_{(j)}]$ is estimated by d_j/n_j. So the maximum likelihood estimate of the probability of surviving from $t_{(j)} - \delta$ to $t_{(j)}$ is $1 - d_j/n_j = (n_j - d_j)/n_j$. Further, since there are no deaths in $(t_{(j)},\, t_{(j+1)} - \delta]$, the probability of surviving from $t_{(j)} - \delta$ to $t_{(j+1)} - \delta$ is also $(n_j - d_j)/n_j$. If we let $\delta \to 0$ then we can estimate the probability of surviving the interval $t_{(j)}$ to $t_{(j+1)}$ as $(n_j - d_j)/n_j$.

We now assume that all deaths are independent of each other. Let t lie in I_k. Then we estimate the probability of survival to time t as the probability of surviving through $t_{(k)}$ to $t_{(k+1)}$, and all preceding intervals. This gives the maximum likelihood estimate of the survivor function, usually known as the *Kaplan-Meier* estimate,

$$\hat{S}(t) = \prod_{j=1}^{k} \left(\frac{n_j - d_j}{n_j} \right), \; t \in [t_{(k)}, t_{(k+1)}) = I_k$$

with $\hat{S}(0) = 1$. Note the following

- If the largest observation corresponds to a death then $n_r = d_r$ so $\hat{S}(t) = 0$ for $t \geq t_{(r)}$.
- If the largest observation, say t^*, is censored then $\hat{S}(t)$ is undefined for $t > t^*$.

Example: The data refer to an experiment to study the use of a medical device, which may prove to be effective or ineffective. The time origin corresponds to the first day on which the medical advice is used. The observation is the time to discontinuation of use of the medical device because it is proving ineffective. Notice that some observations are *censored*, denoted *. In this example, censoring could occur for a number of reasons: (a) no further need of the device, (b) lost to follow-up, (c) the study ended.

Table 1: Time in weeks to discontinuation of the use of a medical device

10	13*	18*	19	23*	30	36	38*	54*
56*	59	75	93	97	104*	107	107*	107*

We set out the calculation of the Kaplan-Meier estimate as follows.

Table 2: Kaplan-Meier estimate of the survivor function

Interval	n_j	d_j	$(n_j - d_j)/n_j$	$\hat{S}(t)$
0−	18	0	1.0000	1.0000
10−	18	1	0.9444	0.9444
19−	15	1	0.9333	0.8815
30−	13	1	0.9231	0.8137
36−	12	1	0.9167	0.7459
59−	8	1	0.8750	0.6526
75−	7	1	0.8571	0.5594
93−	6	1	0.8333	0.4662
97−	5	1	0.8000	0.3729
107−	3	1	0.6667	0.2486

The table is constructed in the following steps:

1. First order the data. If there are any censored observations with the same times as a death time **place them after the deaths**.
2. Column one consists of the **distinct** event times, $t_{(j)}$, $j = 1, \ldots, r$.

3. Column two consists of the number of observations greater than or equal to the event time $t_{(j)}$ (including the events at time $t_{(j)}$).

4. Column three consists of the number of events at time $t_{(j)}$; if there are no multiple events the entries will all be 1 (except the first which is 0).

5. Columns four and five compute $\hat{S}(t)$.

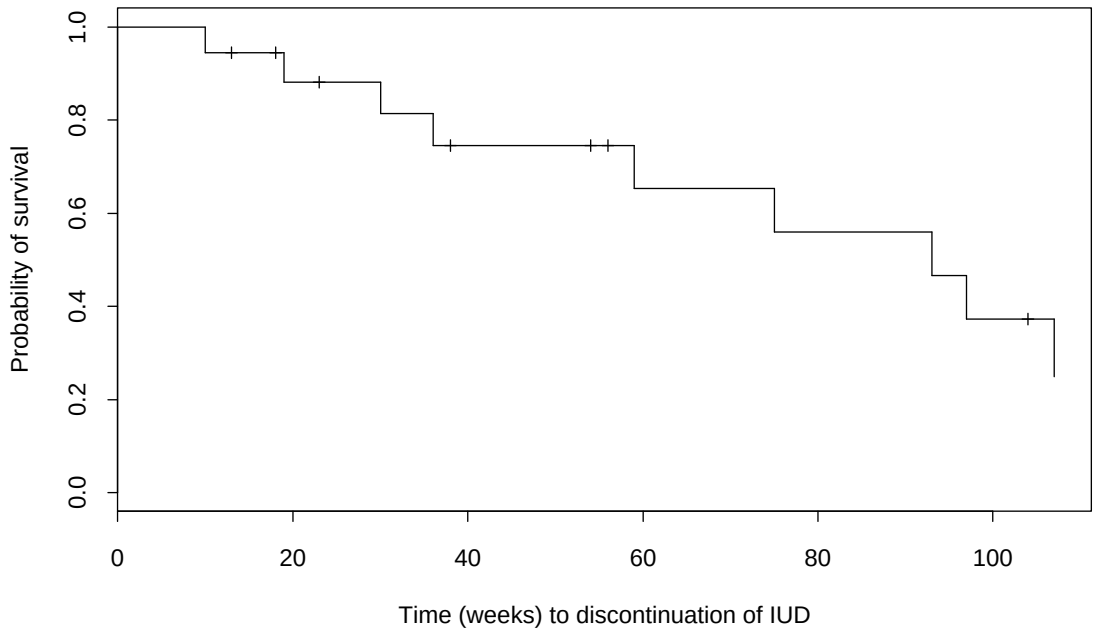


Figure 2: Kaplan-Meier estimate of survivor function, $\hat{S}(t)$, for medical device data

The standard error of the Kaplan-Meier estimate

How good is the Kaplan-Meier estimate? We can answer this by computing its standard error. We could use this to compare two lifetime distributions, for example. The variance is given by Greenwood's formula (proof not required)

$$\text{Var}\left(\hat{S}(t)\right) \approx \left(\hat{S}(t)\right)^2 \sum_{j=1}^{k} \frac{d_j}{n_j(n_j - d_j)},\ t \in [t_{(k)},\, t_{(k+1)}).$$

Using `survfit( )` in R to fit the Kaplan-Meier model

See web site: section: *R-Scripts*, code `Kaplan_Meier.r`.

```
library(survival)
Time <- c(10,13,18,19,23,30,36,38,54,56,59,75,93,97,104,rep(107,3))
Censor <- c(1,0,0,1,0,1,1,0,0,0,1,1,1,1,0,1,0,0)
Surv(Time, Censor)
KM.surv <- survfit(Surv(Time, Censor) ~ 1, conf.type = "plain")
names(KM.surv)
summary(KM.surv)
plot(KM.surv, main = "Time to discontinuation of IUD",
xlab = "Time (in weeks)", ylab = "Survival function with 95% CI")
```

2.5 The Cox Regression Model

The non-parametric approach of the Kaplan-Meier estimate is not well suited to answering questions concerning the **effects of covariates** on the survival function. If a covariate partitions the lives into a small number of homogeneous groups, eg: Sex, then we may find a K-M estimate for each group. But, if the covariate assumes many values, eg: Age, then this approach is not possible. We want a regression type model. In this section we consider a general class of models where the explanatory variables act **multiplicatively** on the force of mortality or hazard function. This model is known as the **proportional hazards model** or the **Cox model** (after D. R. Cox who invented it (1972)).

Definition: A survival time T follows a **Cox proportional hazards** model if the hazard function (force of mortality) for the ith life with covariate $z_i = (z_{i1}, z_{i2}, \ldots, z_{ip})'$ can be written as

$$\lambda(t; z_i) = \lambda_0(t) \exp(\beta' z_i) = \lambda_0(t) \exp(\beta_1 z_{i1} + \beta_2 z_{i2} + \ldots + \beta_p z_{ip}).$$

Comments

(a) The function $\lambda_0(t)$ is known as the **baseline hazard**.

(b) The covariates act **multiplicatively** on the baseline hazard; equivalently they act **additively** on the log scale.

(c) We set $\beta_0 = 0$, since any fixed effect can simply be absorbed into the baseline hazard.

(d) Cox showed that it is possible to estimate the effects of the covariates without specifying the baseline hazard.

Under the Cox model, the hazards of different lives with covariate vectors z_1 and z_2 are in the same proportion at all times t:

$$\frac{\lambda(t; z_1)}{\lambda(t; z_2)} = \frac{\exp(\beta' z_1)}{\exp(\beta' z_2)}.$$

This means that the general shape of the hazard is determined by the baseline hazard, while the exponential term accounts for differences between lives. Thus, if we know the β's we can compare the lives, since the β's tell us which lives have high/low hazards, and this without knowing the hazard function! The baseline hazard, $\lambda_0(t)$, is a nuisance parameter, but it has been eliminated by the proportional hazards model. So, how do we estimate β?

The partial likelihood function, $L(\beta)$

First, we set up some notation. We suppose that data are available on n lives, amongst whom there are r distinct death times and $n - r$ right-censored survival times. We will assume there are *no ties in the death times*; ties lead to complications so we will avoid this point. We denote the r ordered death times by $t_{(1)} < t_{(2)} < \ldots < t_{(r)}$. The set of lives who are at risk at time $t_{(j)}$ is denoted $R(t_{(j)})$, so that $R(t_{(j)})$ is the set of lives who are alive and uncensored *just prior to* $t_{(j)}$. We call $R(t_{(j)})$ the *risk set* at time $t_{(j)}$. Suppose the life with explanatory variable $z_{(j)}$ dies at time $t_{(j)}$. We prove

Result: The (partial) likelihood function for the Cox proportional hazards model is

$$L(\beta) = \prod_{j=1}^{r} \frac{\exp(\beta' z_{(j)})}{\sum_{\ell \epsilon R(t_{(j)})} \exp(\beta' z_\ell)}$$

Proof: The baseline hazard has an arbitrary form so it is possible, for example, that $\lambda_0(t)$ is zero in those time intervals in which there are no deaths. Thus, there can be no information on the β's from the intervals with no deaths. The information on the β's comes entirely from the death times and we can restrict our attention to the death times. Just before time $t_{(j)}$ there are $R(t_{(j)})$ lives at risk. We have

$$\begin{aligned}
&\Pr(\text{Life with } z_{(j)} \text{ dies in } [t_{(j)}, t_{(j)}+\delta) \mid \text{One death from } R(t_{(j)}) \text{ at } t_{(j)}) \\
= \; & \frac{\Pr(\text{Life with } z_{(j)} \text{ dies in } [t_{(j)}, t_{(j)}+\delta) \mid \text{Alive at } t_{(j)})}{\Pr(\text{One death from } R(t_{(j)}) \text{ occurs in } [t_{(j)}, t_{(j)}+\delta) \mid R(t_{(j)}) \text{ alive at } t_{(j)})} \\
= \; & \frac{\Pr(\text{Life with } z_{(j)} \text{ dies in } [t_{(j)}, t_{(j)}+\delta) \mid \text{Alive at } t_{(j)})}{\sum_{\ell \epsilon R(t_{(j)})} \Pr(\text{Life with } z_\ell \text{ dies in } [t_{(j)}, t_{(j)}+\delta) \mid \text{Alive at } t_{(j)})}.
\end{aligned}$$

From the definition of the hazard function we have

$$\Pr(t \le T < t+\delta \mid T \ge t) \approx \delta\, \lambda(t)$$

and so

$$\Pr(\text{Life with } z_\ell \text{ dies in } [t_{(j)}, t_{(j)}+\delta) \mid \text{Alive at } t_{(j)}) \approx \delta\, \lambda_\ell(t_{(j)})$$

where $\lambda_\ell(t)$ is the hazard function of the life with z_ℓ. Substituting we find the required conditional probability can be written

$$\frac{\lambda_i(t_{(j)})}{\sum_{\ell \epsilon R(t_{(j)})} \lambda_\ell(t_{(j)})}$$

where we have assumed that the individual who actually dies is the ith. Substitute $\lambda_i(t) = \lambda_0(t)\exp(\beta' z_i)$. The baseline hazard function in the numerator and denominator cancels out, and we are left with

$$\frac{\exp(\beta' z_{(j)})}{\sum_{\ell \epsilon R(t_{(j)})} \exp(\beta' z_\ell)}.$$

Finally we take the product of these conditional probabilities over the r death times and we obtain the result. ∎

Comments

(a) The partial likelihood behaves very like an ordinary likelihood function. Suppose $\hat{\beta}$ is the estimate of β found by maximising $L(\beta)$, or equivalently $\ell(\beta)$, the log partial likelihood, and β_0 is the true unknown value of β. Then, viewed as a random variable, we have asymptotically

- $\mathrm{E}(\hat{\beta}) = \beta_0$;
- $\mathrm{Var}(\hat{\beta}) = -\,1/\mathrm{E}\left(\frac{\partial^2 \ell}{\partial \beta^2}\right)\Big]_{\beta=\beta_0} \approx -\,1/\left(\frac{\partial^2 \ell}{\partial \beta^2}\right)\Big]_{\beta=\hat{\beta}}$.

(b) The derivation of $L(\beta)$ makes no direct use of the actual censored and uncensored survival times. The censored and uncensored survival times only enter into the summation over the risk sets at the death times. For this reason $L(\beta)$ is not a true likelihood and it is usually referred to as a *partial likelihood*.

(c) The likelihood function depends only on the ranking of the death and censoring times, since this determines the risk set at each death time. Thus inferences about the β's depend only on the *rank order of the survival times*.

Example 1: Suppose the lifetime, the age and the sex of a number of lives are recorded and the Cox model is fitted by maximising the partial likelihood. The fitted parameters are: $\beta_f = -0.5$ where f denotes a female life and $\beta_a = 0.01$ is the effect of age in years. Taking a male aged 40 as the reference life, find the ratio of the hazards for (a) a male aged 20 (b) a male aged 60 (c) a female aged 20 (d) a female aged 60.

We compute

$$\frac{\exp(\beta' z_1)}{\exp(\beta' z_2)}$$

where $\beta' = (-0.5, 0.01)$, $z' = (0, age)$ for a male, and $z' = (1, age)$ for a female. We have

(a) $\exp(20 \times 0.01)/\exp(40 \times 0.01) = \exp(-0.2) = 0.82$
(b) $\exp(60 \times 0.01)/\exp(40 \times 0.01) = \exp(0.2) = 1.22$
(c) $\exp(-0.5 + 20 \times 0.01)/\exp(40 \times 0.01) = \exp(-0.7) = 0.50$
(d) $\exp(-0.5 + 60 \times 0.01)/\exp(40 \times 0.01) = \exp(-0.3) = 0.74$

■

Before giving our next example we give two definitions.

Definition: We define the **score function** $U(\beta)$ by

$$U(\beta) = \frac{\partial \ell}{\partial \beta}$$

where $\ell = \ell(\beta) = \log(L(\beta))$ is the log-likelihood function.

Definition: We define the **observed information function** $I(\beta)$ by

$$I(\beta) = -\frac{\partial^2 \ell}{\partial \beta^2}.$$

Example 2: The table gives the data on times to claim for permanent health insurance (PHI) policies for two groups of lives.

Male	2	5+	9	11+	14
Female	4	7+	10+	12+	15

where the + indicates that the observation was censored.
The partial likelihood in the Cox proportional hazards model is

$$\begin{aligned} L(\beta) &= \prod_{j=1}^{r} \frac{\exp(\beta' z_{(j)})}{\sum_{\ell \epsilon R(t_{(j)})} \exp(\beta' z_\ell)} \\ &= \frac{1}{5+5e^\beta} \times \frac{e^\beta}{4+5e^\beta} \times \frac{1}{3+3e^\beta} \times \frac{1}{1+e^\beta} \times 1 \\ &\propto \frac{e^\beta}{(1+e^\beta)^3(4+5e^\beta)} \\ \Rightarrow \ell(\beta) &= \beta - 3\log(1+e^\beta) - \log(4+5e^\beta). \end{aligned}$$

Estimation: The partial likelihood estimate of β satisfies

$$U(\beta) = \frac{\partial \ell}{\partial \beta} = 1 - \frac{3e^\beta}{1+e^\beta} - \frac{5e^\beta}{4+5e^\beta} = 0.$$

Putting $x = e^{\hat{\beta}}$ we find $15x^2 + 8x - 4 = 0$ so $x = (\sqrt{76} - 4)/15 = 0.3145$ (since $x > 0$) and hence $\hat{\beta} = -1.1567$.

What does $\hat{\beta}$ mean? We have

$$\frac{\lambda_f(t)}{\lambda_m(t)} = \frac{\lambda_0(t)e^{\beta}}{\lambda_0(t)} = e^{\beta} = e^{-1.1567} = 0.3145$$

and so $\lambda_f(t) \approx 0.3\lambda_m(t)$ for all t.

Standard error of $\hat{\beta}$: We use

$$\mathrm{Var}(\hat{\beta}) \approx 1/I(\hat{\beta}) = -\,1/\left(\frac{\partial^2 \ell}{\partial \beta^2}\right)\Big]_{\beta=\hat{\beta}}.$$

It is convenient to use the chain rule with $v = e^{\beta}$ to evaluate this expression. We find

$$\begin{aligned} -\frac{\partial^2 \ell}{\partial \beta^2} &= \left[\frac{d}{dv}\left(\frac{3v}{1+v} + \frac{5v}{4+5v}\right)\right] v \\ &= \left[\frac{d}{dv}\left(3 - \frac{3}{1+v} + 1 - \frac{4}{4+5v}\right)\right] v \\ &= \left(\frac{3}{(1+v)^2} + \frac{20}{(4+5v)^2}\right) v \\ &\approx 0.7486 \end{aligned}$$

since we know that $v = (\sqrt{76} - 4)/15 = 0.31452$. Thus

$$\mathrm{Var}(\hat{\beta}) \approx 1/0.7486 \Rightarrow \mathrm{StErr}(\hat{\beta}) = 1.1558.$$

Hypothesis testing: How can we test $H_0 : \beta = \beta_0$ *v* $H_1 : \beta \neq \beta_0$, where β_0 is the hypothesised value of β? In particular, we often want to test $H_0 : \beta = 0$ *v* $H_1 : \beta \neq 0$, the hypothesis of no difference between the groups (since $e^{\beta} = e^0 = 1$). We describe three approaches:

(a) **Wald test**. Compute $\hat{\beta}$ by solving $U(\beta) = 0$ and its variance by evaluating $V = 1/I(\hat{\beta})$. Compute $z^2 = (\hat{\beta} - \beta_0)^2/V$ and refer z^2 to χ^2_1.

(b) **LRT**. Compute $-2\log\Lambda = -2(\ell_0 - \ell_1)$ where ℓ_i is the maximised value of ℓ under H_i, $i = 0, 1$. Refer to χ^2_1.

(c) **Score test**. This has the advantage that it is not necessary to compute $\hat{\beta}$. The score test is

$$S = \frac{U(\beta_0)^2}{I(\beta_0)}$$

and has a χ^2 distribution with 1 degree of freedom if β is a scalar.

NB: All three tests are asymptotically equivalent.

The Wald test

We solve $U(\beta) = 0$ (graphical, Newton-Raphson, trial and error, or exactly (as in this example)). We know that $\hat{\beta} = -1.1567$ and $V = 1/0.7486$. Hence (with $\beta_0 = 0$)

$$z^2 = \frac{(\hat{\beta} - \beta_0)^2}{V} = (-1.1567)^2 \times 0.7486 = 1.00$$

and refer z^2 to the χ^2_1 tables.

The LRT test

We know

$$\ell(\beta) = \beta - 3\log(1+e^{\beta}) - \log(4+5e^{\beta})$$

so we easily compute

$$\begin{aligned} \ell_0 &= \ell(0) = -3\log 2 - \log 9 = -4.276666 \\ \ell_1 &= \ell(\hat{\beta}) = -3.694984 \\ -2\log\Lambda &= -2(-4.276666 + 3.694984) = 1.1634 \end{aligned}$$

The score test

We have done most of the work for the score test of $H_0 : \beta = 0$ *v* $H_1 : \beta \neq 0$ already. We have with $v = \exp(\beta) = 1$ at $\beta = 0$

$$\begin{aligned} U(\beta) &= 1 - \frac{3v}{1+v} - \frac{5v}{4+5v} \\ \Rightarrow U(0) &= 1 - \frac{3}{2} - \frac{5}{9} = -\frac{19}{18} \\ I(\beta) = -\frac{\partial^2 \ell}{\partial \beta^2} &= \left(\frac{3}{(1+v)^2} + \frac{20}{(4+5v)^2} \right) v \\ \Rightarrow I(0) &= \frac{3}{4} + \frac{20}{81} = \frac{323}{324}. \end{aligned}$$

The score test of $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$ is

$$S = \frac{U(0)^2}{I(0)} = \left(\frac{19}{18}\right)^2 \Big/ \frac{323}{324} = 1.12.$$

Comment: Of course, the Wald test, the score test and the LRT are all consistent; we have $z^2 = 1.00$, $S = 1.12$ and $-2\log\Lambda = 1.16$. None of these tests provides any sort of evidence that the mortality of the two groups differ - but then the sample size is very small.

Using `coxph( )` in R to fit the Cox model

The R-code, `Cox.r`, for a full analysis is on the web in the section to fit Kaplan-Meier and Cox models. The `summary( )` function gives the values of the Wald test, the LRT and the score test. The outline code follows:

```
library(survival)
Time <- c(2,5,9,11,14,4,7,10,12,15)
Censor <- c(1,0,1,0,1,1,0,0,0,1)
Sex <- factor(c(rep(1,5), rep(2,5)))
Cox.fit <- coxph(Surv(Time, Censor) ~ Sex)
summary(Cox.fit)
```

Tutorial 2: The Kaplan-Meier Estimate and the Cox Model

1. The table gives the time (in days) until the recurrence of symptoms in an exploratory drug trial with ten (10) patients; a `+` indicates that the observation was censored.

Time to recurrence					
	2	5	6+	8	9
	10+	12	14+	16+	20

Obtain the Kaplan-Meier estimate of the survivor function, $S(t)$, and plot the estimated survival curve. Confirm your solution with R.

2. The table gives the data from a small medical trial on the time (in hours) until monitoring indicates that further treatment is necessary:

Drug A	3	6+	8+	10
Drug B	5	7+	9+	12

where the `+` indicates that the observation was censored.

(i) Show that the partial likelihood in the Cox proportional hazards model for comparing the lifetimes of the two drugs is

$$L(\beta) \propto \frac{e^{\beta}}{(1+e^{\beta})^2(3+4e^{\beta})}.$$

and plot the loglikelihood function over an appropriate range.

(ii) Let $\ell = \ell(\beta) = \log L(\beta)$. Show that

$$\frac{d\ell}{d\beta} = 0 \Rightarrow \hat{\beta} = -0.792$$

and that

$$-\left[\frac{d^2\ell}{d\beta^2}\right]_{\beta=\hat{\beta}} = 0.664.$$

Hence, conduct the Wald test of $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$.

(iii) Conduct the score test of H_0 *v* H_1.

(iv) Conduct the likelihood ratio test of H_0 *v* H_1.

(v) Confirm your solution with `coxph( )`.

Comment on the consistency of the tests in (ii), (iii) and (iv).

3. Over...

3. The table gives the survival time (in weeks) for fourteen (14) patients who received a steroid treatment for severe viral hepatitis, and fifteen (15) patients who received a control treatment; a `+` indicates that the observation was censored, and the figure in brackets after the censored observation of 16 indicates the number of patients with a survival time of at least 16 weeks.

Survival time								
	Steroid	1	1+	4	4+	5	5+	7
		8	10	10+	12+	16+(3)		
	Control	1+	2	3	3+	5+	5+	5+	16+(8)

   (a) Find the partial log likelihood, $\ell(\beta)$, for the Cox proportional hazards model for these data.

   (b) Define an R function to evaluate $\ell(\beta)$.

   (c) Use R to plot $\ell(\beta)$ *v* β over an appropriate range and by refining the plot find the maximum partial likelihood estimate of β to 2 d.p.

   (d) Use the `coxph( )` function to confirm your estimate of β and to conduct the Wald test, the score test and the LRT of $H_0 : \beta = 0$ *v* $H_1 : \beta \neq 0$.

4. (Based on HW, 2008) The table gives the data from a small portfolio of employment protection policies, and shows, for two groups of lives, the time in months until a claim is made; the `+` indicates that the observation was censored.

Group A	3	6+	8+	10	15
Group B	5	7+	9	12+	16

   Suppose the Cox proportional hazards model applies to these data and that `Group A` is taken as the reference category, i.e., $\lambda_B(t) = \lambda_A(t)\exp(\beta)$ in the standard notation.

   (a) Find the partial loglikelihood $\ell(\beta)$ and deduce the score function $U(\beta)$.

   (b) Define two R functions, one to evaluate $\ell(\beta)$ and one to evaluate $U(\beta)$.

   (c) Use R to obtain a plot of the score function $U(\beta)$ against β. Refine the plot and hence estimate the maximum partial likelihood estimate of β.

   (d) Let $\phi = \exp(\beta)$. Find the partial loglikelihood $\ell(\phi)$ and show that the score function with respect to ϕ, $U(\phi)$, can be written

$$U(\phi) = \frac{2}{\phi} - \frac{3}{1+\phi} - \frac{3}{2+3\phi} - \frac{5}{4+5\phi}.$$

   (e) Use R to obtain a plot of the score function $U(\phi)$ against ϕ, $0.5 \leq \phi \leq 0.7$. Use the plot to estimate the maximum partial likelihood estimate of ϕ together with an estimate of its standard error.

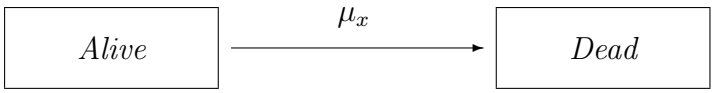
   (f) What is the connection between the estimate of β found in (b) and the estimate of ϕ found in (d)?

   (g) Use `coxph( )` in R to check that the LRT of $H_0 : \beta = 0$ *v* $H_1 : \beta \neq 0$ is equivalent to the LRT of $H_0 : \phi = 1$ *v* $H_1 : \phi \neq 1$.

3 Markov Models

3.1 Introduction

Up till now our approach has been to specify a future lifetime in terms of a random variable T or T_x. In this chapter we look at things rather differently, and use a **Markov model** of transfers between states. In the simplest case, a life can be alive or dead, and this gives a two-state model of mortality. We often represent this in a diagram like:



The probability that a life alive at any age x should be dead at some future age is determined by an age dependent force of mortality μ_{x+t}, $t \geq 0$, or **transition intensity**. The main advantages of the Markov model approach to modelling mortality over the random variable approach are that

- it can be generalised easily to multiple decrements or multiple state models, e.g. the three state model {*Well, Ill, Dead*}, and
- it deals easily with censoring, a common feature of mortality data.

3.2 Fundamental assumptions of the 2-state model

We make two fundamental assumptions in the 2-state model above.

1. **Markov assumption.** The probability that a life now aged x will be in either state {*Alive, Dead*} at any future time $x+t$ depends only on the age x and the state currently occupied.

2. The probability ${}_{dt}q_{x+t}$ is given by

$$ {}_{dt}q_{x+t} = \mu_{x+t}\, dt + o(dt),\ t \geq 0. $$

Informally, assumption 1 says that the future depends only on the present and not on the past, while assumption 2 says that the probability of death in a short interval of time is approximately proportional to the force of mortality at that time.
Reminder: The function $f(h)$ is said to be $o(h)$ or "little o of h" if

$$ \lim_{h \to 0} \frac{f(h)}{h} = 0. $$

In other words, $f(h)$ is $o(h)$ if $f(h) \to 0$ faster than h. In the next example, we use a "little o" argument to show that the random variable approach to modelling future lifetimes implies the second of our fundamental assumptions above.

Example 3.1: First, we recall that the Mean Value Theorem (MVT) says

$$ f(b) - f(a) = (b-a) f'(c) $$

for some $c \in (a, b)$ (subject to continuity and differentiability conditions). We apply this result as follows.

$$\begin{aligned}
{}_hq_x &= 1 - {}_hp_x \\
&= 1 - \exp\left(-\int_0^h \mu_{x+t}\, dt\right) \\
&= 1 - \exp\{-[F(h) - F(0)]\} \quad \text{for some function } F(\cdot) \\
&= 1 - \exp[-hF'(s)] \quad \text{for some } s \in (0, h) \text{ by MVT} \\
&= 1 - \exp(-h\mu_{x+s}) \\
&= 1 - \left(1 - h\mu_{x+s} + \frac{h^2\mu_{x+s}^2}{2!} \ldots\right) \\
&= h\mu_{x+s} + o(h) \\
&= h\mu_x + h(\mu_{x+s} - \mu_x) + o(h).
\end{aligned}$$

Now apply the MVT again to $\mu_{x+s} - \mu_x$ to find, for some $r \in (0, s)$

$$\begin{aligned}
{}_hq_x &= h\mu_x + hs\mu'_{x+r} + o(h) \\
&= h\mu_x + o(h) \quad \text{since } s \in (0, h).
\end{aligned}$$

Hence the random variable approach of chapter 1 $\Rightarrow$ assumption 2 in the Markov approach. ■

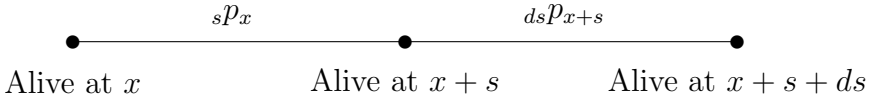
3.3 Computation of ${}_tp_x$

Our model is defined in terms of the transition intensities μ_x. How can we compute probabilities like ${}_tp_x$, ${}_tq_x$, etc? Notice that we already have formulae for these quantities from the random future lifetime approach in Chapter 1. We had better get the same answers with the Markov approach!

Lemma: Assumptions 1 and 2 imply that

$${}_tp_x = \exp\left(-\int_0^t \mu_{x+s}\, ds\right).$$

Proof: Let $s < t$. Notice first that ${}_sp_x$ is well defined by the Markov property, i.e. how the life got to age x is irrelevant. We consider the small interval of time **immediately after** $x + s$, and ask: how can a life aged x become a life aged $x + s + ds$? The diagram may help:



Hence

$$\begin{aligned}
{}_{s+ds}p_x &= {}_sp_x \times {}_{ds}p_{x+s} \\
&= {}_sp_x \times (1 - {}_{ds}q_{x+s}) \\
&= {}_sp_x \times (1 - \mu_{x+s}\, ds + o(ds)) \quad \text{by Assumption 2.}
\end{aligned}$$

Thus,

$${}_{s+ds}p_x - {}_sp_x = -{}_sp_x\, \mu_{x+s}\, ds + o(ds).$$

Now divide both sides by ds and let $ds \to 0$. We find

$$\frac{{}_{s+ds}p_x - {}_sp_x}{ds} = -{}_sp_x\,\mu_{x+s} + \frac{o(ds)}{ds} \quad \Rightarrow \quad \frac{\partial}{\partial s}{}_sp_x = -{}_sp_x\,\mu_{x+s} \;\Rightarrow\; \frac{\partial}{\partial s}\log\,{}_sp_x = -\mu_{x+s} \quad (3)$$

$$\Rightarrow {}_tp_x \;=\; \exp\left(-\int_0^t \mu_{x+s}\,ds\right)$$

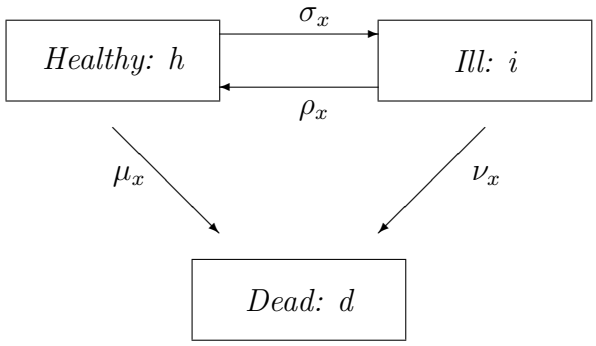
on integrating (3) from 0 to t. This agrees with the well known result obtained with the future lifetime approach; see Tutorial 1, Question 3. ■

Note: The partial differential equation above for ${}_tp_x$ is a simple example of a **Kolmogorov forward equation**. As an aside I recommend that you have a look in the St Andrews University history of mathematics web site for a biography of Kolmogorov. This is found at

`http://www-history.mcs.st-and.ac.uk/~history/`

3.4 Multi-state Markov models

The 2-state model of mortality can be extended to any number of states. Many insurance products, e.g. Disability Insurance (also known as Permanent Health Insurance (PHI) or Income Protection Insurance (IPI)), Joint Life Annuities, can be modelled by a multi-state model. Here is a 3-state model for IPI



The set $J = \{Healthy,\ Ill,\ Dead\} = \{h,i,d\}$ is called the **state space**.

Let g and h be any two states. We need to extend the assumptions in section 3.2 for the 2-state model to cope with a multi-state model. We do this in terms of the **transition probability** ${}_tp_x^{gh}$ (analogous to ${}_tq_x$) and the **transition intensity** μ_x^{gh} (analogous to μ_x). We define the transition probability

$${}_tp_x^{gh} = \Pr(\text{In state } h \text{ at time } x+t \mid \text{In state } g \text{ at time } x)$$

for any two states g and h.

The assumptions which govern the simple 2-state model of mortality extend to the multi-state model as follows:

1. **Markov assumption.** The probability that a life now aged x will be in a particular state at any future time $x + t$ depends only on the age x and the state currently occupied.

2. For any two **distinct** states g and h the transition probability ${}_{dt}p_{x+t}^{gh}$ is given by

$${}_{dt}p_{x+t}^{gh} = \mu_{x+t}^{gh}\,dt + o(dt),\ t \geq 0.$$

3. The probability that a life makes two or more transitions in time dt is $o(dt)$.

Notice that assumptions 1 and 2 are essentially the same as in the 2-state model but applied in the multi-state context. Assumption 3 is new and says, in effect, that only one transfer can take place at one time.

Letting $X(t)$ denote the state of someone's existence at time t, assuming that $t = 0$ is when they were born, it follows that in the language and notation of Chapter 8 of F79SP Stochastic Processes, we have the following

$$ {}_tp_x^{gh} = \Pr(X(x+t) = h \mid X(x) = g) = p_{gh}(x, x+t) $$

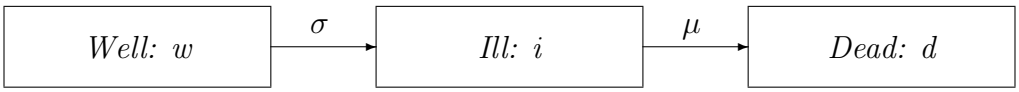
and

$$ \mu_x^{gh} = \frac{\partial}{\partial u} p_{gh}(x,u)\Big|_{u=x} = q_{gh}(x). $$

What does ${}_tp_x^{gg}$ mean? This is the probability that we are in state g at time $x+t$, given that we are in state g at time x. This **does not imply** that we have been in state g for the whole of the time interval from x to $x+t$, and for this we define

$$ {}_tp_x^{\overline{gg}} = \Pr(\text{In state } g \text{ from } x \text{ to } x+t \mid \text{In state } g \text{ at time } x). $$

However ${}_tp_x^{\overline{gg}}$ will equal ${}_tp_x^{gg}$ in one common situation, namely when return to state g is impossible. For example, in this model of terminal illness



we have ${}_tp_x^{\overline{ww}} = {}_tp_x^{ww}$ since return to the well state is impossible. In a similar fashion, we also have ${}_tp_x^{\overline{ii}} = {}_tp_x^{ii}$.

Again, relating this to Stochastic Processes, the quantity ${}_tp_x^{\overline{gg}}$ is what was previously denoted $p_{\overline{gg}}(x, x+t)$.

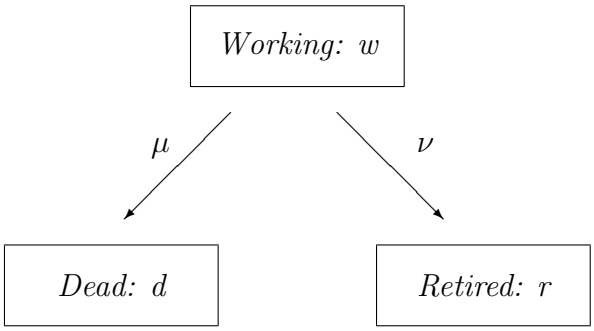
3.5 Kolmogorov forward equations

What can we say about the relationship between the transition intensities μ_{x+t}^{gh}, $g \neq h$, and the transition probabilities ${}_tp_x^{gh}$? We have the following result, proven in Stochastic Processes:

Theorem: For any states g and h, and times x and t, we have

$$ \frac{\partial}{\partial t} {}_tp_x^{gh} = \sum_{j \neq h} \left({}_tp_x^{gj} \mu_{x+t}^{jh} - {}_tp_x^{gh} \mu_{x+t}^{hj} \right), \; g \neq h. $$

Example 3.2: Consider the 3-state model for working, retiring and dying.



In this simple model we have assumed **constant** transition intensities. There are **three transition probabilities** which correspond to the events (a) *Working* to *Dead* (b) *Working* to *Retired* (c) *Working* to *Working*.
In this example, we have the following Kolmogorov differential equations:

$$\frac{\partial}{\partial t}{}_tp_x^{wd} = \mu \cdot {}_tp_x^{ww}; \tag{4}$$

$$\frac{\partial}{\partial t}{}_tp_x^{wr} = \nu \cdot {}_tp_x^{ww}; \tag{5}$$

$$\frac{\partial}{\partial t}{}_tp_x^{ww} = -(\mu + \nu){}_tp_x^{ww}. \tag{6}$$

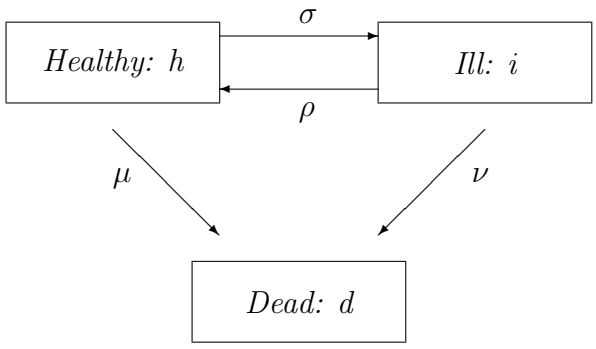
These can be solved using standard methods to find ${}_tp_x^{wd}$, ${}_tp_x^{wr}$ and ${}_tp_x^{ww}$, making use of the boundary conditions

$${}_0p_x^{wd} = {}_0p_x^{wr} = 0$$

and

$${}_0p_x^{ww} = 1.$$

Example 3.3: For our second example we return to the 3-state model for PHI



where we have assumed that the transition intensities are constant. There are six transition probabilities ${}_tp_x^{hh}$, ${}_tp_x^{hi}$, ${}_tp_x^{hd}$, ${}_tp_x^{ii}$, ${}_tp_x^{ih}$ and ${}_tp_x^{id}$.

We have the following differential equations for the probabilities of leaving (and remaining in) state h:

$$\frac{\partial}{\partial t}{}_tp_x^{hi} = \sigma \cdot {}_tp_x^{hh} - (\rho + \nu){}_tp_x^{hi}; \tag{7}$$

$$\frac{\partial}{\partial t}{}_tp_x^{hh} = \rho \cdot {}_tp_x^{hi} - (\sigma + \mu){}_tp_x^{hh}; \tag{8}$$

$$\frac{\partial}{\partial t}{}_tp_x^{hd} = \mu \cdot {}_tp_x^{hh} + \nu \cdot {}_tp_x^{hi}. \tag{9}$$

Again, these can be solved using standard methods. Similar differential equations can written down for the probabilities of leaving (and remaining in) state s.

Back to the general case, we also have the following equation for ${}_tp_x^{\overline{gg}}$, the probability that the state g is occupied continuously from time x to time $x + t$:

$${}_tp_x^{\overline{gg}} = \exp\left(-\int_0^t \sum_{j \neq g} \mu_{x+s}^{gj}\, ds\right). \tag{10}$$

Comment: This formula generalises a number of familiar formulae. In the simplest case, suppose that a lifetime has an exponential distribution with mean $1/\mu$. Then the probability that the life survives for at least t is $\exp(-\mu t)$; this corresponds to (10) with two states, *Alive* and *Dead*, and a single (constant) transition intensity μ. If the transition intensity, μ_x, is age dependent then (10) reduces to another well-known formula:

$${}_tp_x = \exp\left(-\int_0^t \mu_{x+s}\, ds\right).$$

3.6 Introduction to Data and Estimation

We now have a mathematical model for various situations that arise in insurance, e.g. a model of mortality with states {*Alive, Dead*}, a model for IPI with states {*Well, Ill, Dead*}. These models depend on transition intensities μ_x^{gh}, $g \neq h$, between states g and h. The values of these transition intensities are unknown - they are parameters in the models. Thus, we will need some data with which to estimate the values of the μ_x^{gh}. We will start with the 2-state model of mortality with states {*Alive, Dead*}, and work through this example in detail. It turns out that the answers for the 2-state model give a very good guide to what happens in the multi-state case.

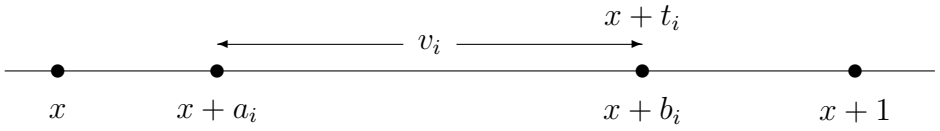
3.7 Data and notation for the 2-state model

Suppose we have data on n independent lives all aged between x and $x + 1$. We suppose for life i

$$\begin{aligned}
x + a_i &= \text{age at which observation begins, } 0 \leq a_i < 1 \\
x + b_i &= \text{age at which observation ends, if life does not die, } 0 < b_i \leq 1 \\
x + t_i &= \text{age at which observation stops, by death or censoring} \\
v_i &= t_i - a_i, \text{ the observed } \textbf{waiting time} \\
d_i &= 1, \text{ if life } i \text{ dies} \\
d_i &= 0, \text{ if life } i \text{ is censored}
\end{aligned}$$

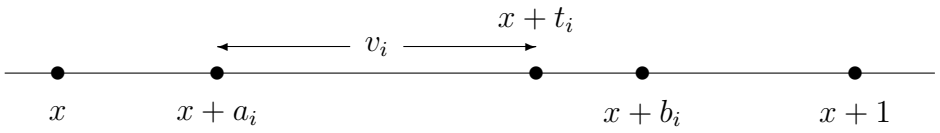
The following diagrams may be helpful.

Case 1: The ith life does not die, i.e. reaches the end of the observation period at $x + b_i$.



In this case, $d_i = 0$, $t_i = b_i$, $v_i = b_i - a_i$.

Case 2: The ith life dies at $x + t_i$ before the end of the observation period.



In this case, $d_i = 1$, $a_i < t_i < b_i$, $v_i = t_i - a_i$.

3.8 The random variables D_i and V_i

The observations on the ith life are: whether the life died or not, i.e. the value d_i, and the observed waiting time, i.e. the value v_i. We can think of the observations d_i and v_i as realisations of random variables D_i and V_i. Inference for the transition intensity μ_x will depend on the joint distribution of D_i and V_i. Before deriving this distribution we make two important points:

- D_i and V_i are **not independent**.
- V_i has a **mixed distribution**, i.e. the probability distribution has a **continuous density** for values of v_i which correspond to death, i.e. for $v_i < b_i - a_i$, and a **mass of probability** for the value of v_i which corresponds to survival, i.e. for $v_i = b_i - a_i$.

Assumption: Before deriving the joint distribution of D_i and V_i we will make a further assumption: we assume that $\mu_{x+t} = \mu_x = \mu$ for $0 \leq t \leq 1$, i.e. the transition intensity μ_{x+t} is constant from age x to age $x + 1$. We can now prove:

Result: The joint probability function of D_i and V_i can be written

$$f_i(d_i, v_i) = e^{-\mu v_i} \mu^{d_i} \tag{11}$$

where $\mu_{x+t} = \mu$ for $0 \leq t \leq 1$.

Proof: We consider the two cases: $D_i = 0$ and $D_i = 1$.

Case 1: $D_i = 0$. We need the probability the life survives from $x + a_i$ to $x + b_i$. We have

$$\begin{aligned}
& \Pr(D_i = 0, V_i = b_i - a_i) \\
= \; & \Pr(\text{Life } i \text{ survives from } x + a_i \text{ to } x + b_i) \\
= \; & \exp\left(- \int_0^{v_i} \mu_{x+a_i+s} \, ds \right), \text{ where } v_i = b_i - a_i \\
= \; & \exp(-\mu v_i), \text{ since } \mu_{x+a_i+s} \text{ is constant} \\
= \; & e^{-\mu v_i} \mu^{d_i}, \text{ since } d_i = 0.
\end{aligned}$$

Case 2: $D_i = 1$. We need the probability density function that a life aged $x + a_i$ dies at time $x + a_i + v_i$. We have

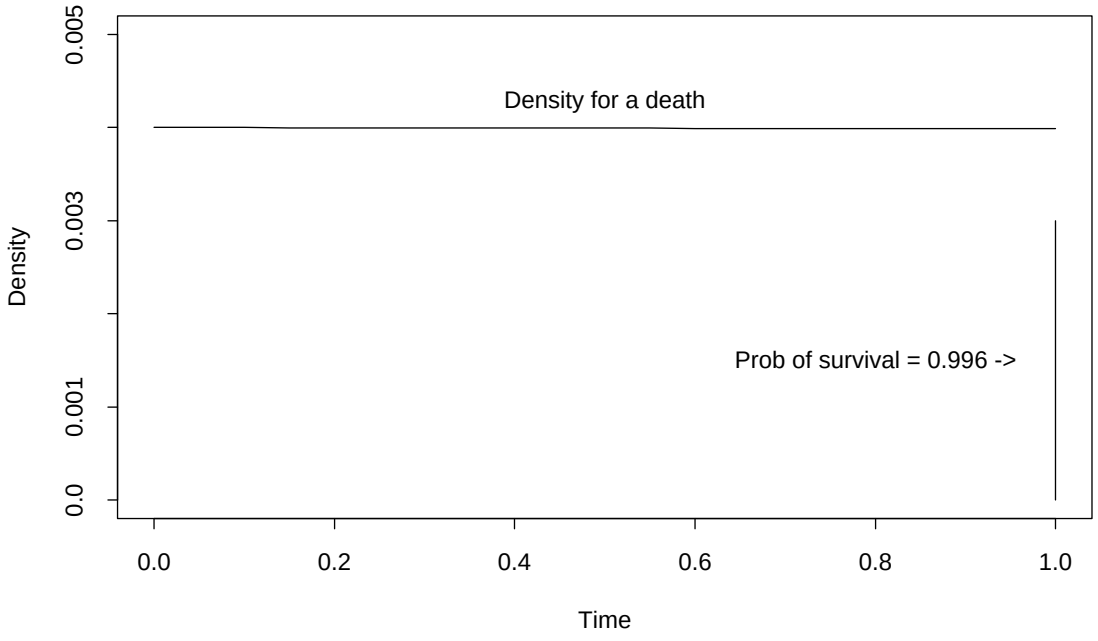
$$\begin{aligned} & \Pr(D_i = 1, v_i \le V_i < v_i + dv_i) \\ = \; & {}_{v_i}p_{x+a_i} \times \left(\mu_{x+a_i+v_i}\, dv_i + o(dv_i)\right) \\ = \; & \exp(-\mu v_i) \times \left(\mu\, dv_i + o(dv_i)\right) \text{ since } \mu_{x+t} \text{ is constant} \\ = \; & e^{-\mu v_i}\mu^{d_i}\, dv_i + o(dv_i), \text{ since } d_i = 1. \end{aligned}$$

Dividing by dv_i and taking the limit as $dv_i \to 0$, we get the pdf as $e^{-\mu v_i}\mu^{d_i}$ in the case of a death. Thus, in **both cases** we can write the probability function as

$$f_i(d_i, v_i) = e^{-\mu v_i}\mu^{d_i}$$

and this completes the proof. ■

What does the mixed distribution of D_i and V_i look like? Here is a simple example: take $a_i = 0$, $b_i = 1$, i.e. the observation period is the full year of life, unless death occurs. Suppose $\mu = 0.004$, the approximate force of mortality for a male life aged 50 (ELT 15). The probability of survival is $e^{-0.004} = 0.996$. The density function for a death is $\mu e^{-\mu t} = 0.004e^{-0.004t}$, $0 \le t \le 1$. In the plot of the mixed distribution below note that the probability mass for survival at 1 is drawn on a different scale from the density function for death.



3.9 The maximum likelihood estimate of μ

Suppose that the waiting times of the n lives are $\boldsymbol{v}' = (v_1, v_2, \ldots, v_n)$ and the record of deaths is $\boldsymbol{d}' = (d_1, d_2, \ldots, d_n)$. The n lives are independent so, from (11), the total likelihood is

$$L(\mu; \boldsymbol{d}, \boldsymbol{v}) \propto \prod_{i=1}^{n} e^{-\mu v_i} \mu^{d_i} = e^{-\mu \sum v_i} \mu^{\sum d_i}.$$

We can write this in more compact form as

$$L(\mu) \propto e^{-\mu v} \mu^{d} \tag{12}$$

where

$$\begin{aligned} v &= \sum v_i = \text{Total waiting time}, \\ d &= \sum d_i = \text{Total number of deaths}. \end{aligned}$$

The maximum likelihood estimate of μ follows from (12) as

$$\hat{\mu} = \frac{d}{v} = \frac{\text{Total number of deaths}}{\text{Total waiting time}}. \tag{13}$$

We are interested in the statistical properties of $\hat{\mu}$. For this purpose, we think of (12) and (13) as random variables. Corresponding to (12) we define the random variable

$$L(\mu; D, V) \propto e^{-\mu V} \mu^{D}$$

where $D = \sum D_i$ is the random total number of deaths and $V = \sum V_i$ is the random total waiting time; corresponding to (13) we define the estimator

$$\tilde{\mu} = \frac{D}{V}.$$

Comment: These are very simple answers, but an obvious question is "How do we extend the estimation process to a wider range of ages, e.g. from 20 to 70?" One way of doing this is to split the age range into a number of short age ranges in such a way that the μ_x can be assumed constant within each sub-range. Unfortunately, this solution creates another problem: there is no reason to expect that the large number of estimates of μ_x for different x-values will be consistent. For example, we expect mortality to increase, at least from age 20, say. We will be forced to smooth or **graduate** these raw rates; see Chapters 5 and 6.

3.10 A statistical interlude

We will make use of two important general results from mathematical statistics:

- properties of the **score function**, and
- the **maximum likelihood theorem**.

These topics are covered in your statistics modules. Here we give a brief review of this material. First, we consider the one parameter case.

Definition: The **score function**. Suppose $\boldsymbol{X}' = (X_1, X_2, \ldots, X_n)$ is a vector of random variables where the X_i are independent random variables with probability function $f_i(x, \theta)$ which depends on a parameter θ (which does not determine the range of X_i, $i = 1, 2, \ldots, n$). Let

$$\ell = \ell(\theta) = \ell(\theta, \boldsymbol{X}) = \sum_1^n \log f_i(X_i, \theta)$$

be the log likelihood function (viewed as a random variable). Then

$$U = U(\theta) = \frac{\partial \ell}{\partial \theta}$$

is called the **score function**.

Note: The score function is **additive** as follows: let

$$U_i = U_i(\theta) = \frac{\partial \ell_i}{\partial \theta} = \frac{\partial}{\partial \theta} \log f_i(X_i, \theta)$$

be the score function for the ith observation, $i = 1, \ldots, n$. Then

$$U = \sum_1^n U_i.$$

Result: The score function satisfies:

(a) $\mathrm{E}(U) = 0$, and

(b) $\mathrm{Var}(U) = -\mathrm{E}\left(\dfrac{\partial U}{\partial \theta}\right) = -\mathrm{E}\left(\dfrac{\partial^2 \ell}{\partial \theta^2}\right)$.

Proof: First, take $n = 1$ and without loss of generality $f_i = f$.

(a) Consider the identity

$$\int f(x, \theta)\, dx = 1.$$

Differentiate with respect to θ. We get

$$\int \frac{\partial f}{\partial \theta}\, dx = 0.$$

Rewrite this equation as

$$\int \left(\frac{\partial}{\partial \theta} \log f\right) f\, dx = 0. \tag{14}$$

The left hand side of this equation is $\mathrm{E}(U)$ so $\mathrm{E}(U) = 0$ for $n = 1$. Hence $\mathrm{E}(U) = 0$ for all n by the additivity property.

(b) The proof uses the same trick. Differentiate (14) with respect to θ and use the product rule. We get

$$\begin{aligned}
& \int \left\{ \frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}\log f\right) f \;+ \left(\frac{\partial}{\partial\theta}\log f\right)\frac{\partial f}{\partial\theta}\right\} dx = 0 \\
\Rightarrow \quad & \int \frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}\log f\right) f\, dx + \int \left(\frac{\partial}{\partial\theta}\log f\right)\left(\frac{\partial}{\partial\theta}\log f\right) f\, dx = 0 \\
\Rightarrow \quad & \mathrm{E}\left(\frac{\partial U}{\partial\theta}\right) + \mathrm{E}(U^2) = 0 \\
\Rightarrow \quad & \mathrm{E}(U^2) = -\mathrm{E}\left(\frac{\partial U}{\partial\theta}\right) \\
\Rightarrow \quad & \mathrm{Var}(U) = -\mathrm{E}\left(\frac{\partial U}{\partial\theta}\right) \quad \text{since } \mathrm{E}(U) = 0 \\
\text{or} \quad & \mathrm{Var}(U) = -\mathrm{E}\left(\frac{\partial^2 \ell}{\partial\theta^2}\right)
\end{aligned}$$

and (b) is proved for $n = 1$. The general case follows by the additivity of the score function and the independence of the observations. ■

The score function U is a fundamental quantity in statistics. It has already appeared in the Cramer-Rao Lower Bound Theorem (CRLB). Now it turns up again in connection with the properties of the maximum likelihood estimate of θ.

- The MLE satisfies $\partial\ell/\partial\theta = 0$ i.e. $U = 0$.

- The asymptotic variance of the MLE is found from $\mathrm{Var}(U)$.

Definition: Fisher's **information function** $I(\theta)$ is

$$I(\theta) = -\mathrm{E}\left(\frac{\partial^2 \ell}{\partial\theta^2}\right) = -\mathrm{E}\left(\frac{\partial U}{\partial\theta}\right) = \mathrm{Var}(U).$$

We usually just call $I(\theta)$ the information function. We are now ready to state (without proof) the **Maximum Likelihood Theorem**.

Theorem: Suppose $\hat{\theta}$ satisfies $U = 0$ and that θ_0 is the true (but unknown) value of θ. If $\tilde{\theta}$ is the random variable corresponding to $\hat{\theta}$, then $\tilde{\theta}$ is

(i) asymptotically unbiased, i.e. $\mathrm{E}(\tilde{\theta}) \to \theta_0$,

(ii) asymptotically efficient, i.e. achieves the Cramer-Rao Lower Bound, i.e. $\mathrm{Var}(\tilde{\theta}) \to I(\theta_0)^{-1}$,

(iii) asymptotically normal.

We can summarise this in

$$\tilde{\theta} \approx \mathcal{N}(\theta_0, I(\theta_0)^{-1}).$$

Of course, θ_0 is unknown so how do we calculate $I(\theta_0)^{-1}$? There are two main approximations.

- Replace θ_0 with $\hat{\theta}$, i.e.
$$I(\theta_0) \approx I(\hat{\theta}).$$

- If we can't work out the expectation in $\mathrm{E}(\partial U/\partial\theta)$ then try
$$I(\theta_0) \approx -\left(\frac{\partial^2 \ell}{\partial \theta^2}\right)\Bigg]_{\theta=\hat{\theta}}.$$

The maximum likelihood theorem is a most powerful and general result and we apply it now to the problem of determining the properties of the MLE of the force of mortality in the 2-state model.

3.11 Properties of $\tilde{\mu} = D/V$

We know from section 3.9 that

$$\begin{aligned} L(\mu) &= e^{-\mu V}\mu^D \\ \Rightarrow \ell(\mu) &= -\mu V + D\log\mu \\ \Rightarrow U(\mu) &= -V + \frac{D}{\mu} \\ \Rightarrow \tilde{\mu} &= \frac{D}{V} \end{aligned} \tag{15}$$

Now apply the Maximum Likelihood Theorem. Suppose that μ_0 is the true, but unknown, value of the unknown force of mortality.

(i) First, $\tilde{\mu}$ is asymptotically unbiased so

$$\mathrm{E}(\tilde{\mu}) \to \mu_0.$$

(ii) Second, $\tilde{\mu}$ is asymptotically efficient so

$$\begin{aligned}
\mathrm{Var}(\tilde{\mu}) &\to I(\mu_0)^{-1} \\
&= \left\{-\mathrm{E}\left(\frac{\partial U}{\partial \mu}\right)\right\}^{-1}\Bigg]_{\mu=\mu_0} \\
&= -\left\{\mathrm{E}\left(-\frac{D}{\mu^2}\right)\right\}^{-1}\Bigg]_{\mu=\mu_0} \quad \text{by differentiating (15)} \\
&= \left(\frac{\mathrm{E}(D)}{\mu^2}\right)^{-1}\Bigg]_{\mu=\mu_0} \\
&= \frac{\mu^2}{\mathrm{E}(D)}\Bigg]_{\mu=\mu_0}.
\end{aligned}$$

Now $\mathrm{E}(U) = 0$ so from (15) we get

$$\mathrm{E}\left(\frac{D}{\mu}\right) = \mathrm{E}(V) \Rightarrow \mathrm{E}(D) = \mu \mathrm{E}(V).$$

Hence

$$\mathrm{Var}(\tilde{\mu}) \to \frac{\mu}{\mathrm{E}(V)}\Bigg]_{\mu=\mu_0} = \frac{\mu_0}{\mathrm{E}(V)}.$$

(iii) Third, $\tilde{\mu}$ is asymptotically normal.

Our final result is

$$\tilde{\mu} \approx \mathcal{N}\left(\mu_0, \frac{\mu_0}{\mathrm{E}(V)}\right). \tag{16}$$

■

Comments: We note the following

- The observed waiting time $v = \sum v_i$ is usually called the **central exposed to risk** E_x^c in actuarial language.

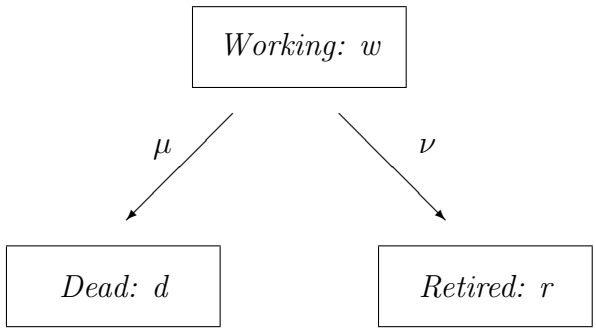
- In practice, we evaluate $\mathrm{Var}(\tilde{\mu}) = \mu_0/\mathrm{E}(V)$ by

$$\mathrm{Var}(\tilde{\mu}) = \frac{\mu_0}{\mathrm{E}(V)} \approx \frac{\hat{\mu}}{v} = \frac{d}{v^2}.$$

How does the estimation result (16) for the 2-state model extend to more complicated multi-state models such as Illness/Death models?

3.12 Estimation in the 3-state model

In section 3.5 we considered the following 3-state model for working, retiring and dying.



Suppose we have data on n lives and for the ith life we observe

$$\begin{aligned} d_i &= \begin{cases} 0 & \text{when } i\text{th life does not die} \\ 1 & \text{when } i\text{th life does die} \end{cases} \\ r_i &= \begin{cases} 0 & \text{when } i\text{th life does not retire} \\ 1 & \text{when } i\text{th life does retire} \end{cases} \\ v_i &= t_i - a_i, \quad \text{the } i\text{th waiting time.} \end{aligned}$$

Our observation is the triple (v_i, d_i, r_i) with corresponding underlying random variable (V_i, D_i, R_i). The distribution of (V_i, D_i, R_i) is given by

$$\Pr(D_i = d_i, R_i = r_i, v_i < V_i < v_i + dv_i).$$

Notice that once more this is a mixed distribution with a mass of probability corresponding to remaining in the working state and a continuous density function corresponding to retirement or death. The key result is

$${}_tp_x^{ww} = e^{-(\mu+\nu)t},$$

as we showed when we discussed this model in section 3.5.

We discuss the three cases: (i) remaining working, (ii) dying, (iii) retiring.

Case 1: Remaining working. In this case $d_i = 0, r_i = 0$ and the life remains in the working state from $x + a_i$ to $x + b_i$ so we have

$$\begin{aligned} & \Pr(D_i = 0, R_i = 0, V_i = b_i - a_i) \\ = \ & \Pr(\text{Neither died nor retired during } x + a_i \to x + b_i) \\ = \ & e^{-(\mu+\nu)(b_i - a_i)} \\ = \ & e^{-(\mu+\nu)v_i} \end{aligned}$$

since $v_i = b_i - a_i$ when $d_i = 0, r_i = 0$.

Case 2: Dying. In this case $d_i = 1, r_i = 0, v_i < V_i < v_i + dv_i$ so we have

$$\begin{aligned} & \Pr(D_i = 1, R_i = 0, v_i < V_i < v_i + dv_i) \\ = \ & \Pr(\text{Neither died nor retired during } (x + a_i, x + a_i + v_i) \ldots \\ & \qquad \ldots \text{ and then died in } (x + a_i + v_i, x + a_i + v_i + dv_i)) \\ = \ & e^{-(\mu+\nu)v_i} \times (\mu \, dv_i + o(dv_i)). \end{aligned}$$

Dividing by dv_i and taking the limit we find the pdf is

$$f_i(1,0,v_i) = e^{-(\mu+\nu)v_i}\mu.$$

Case 3: Retiring. In this case $d_i = 0, r_i = 1, v_i < V_i < v_i + dv_i$ so by symmetry with case 2 we have

$$\Pr(D_i = 0, R_i = 1, v_i < V_i < v_i + dv_i) = e^{-(\mu+\nu)v_i} \times (\nu\, dv_i + o(dv_i))$$

and the pdf in this case is

$$f_i(0,1,v_i) = e^{-(\mu+\nu)v_i}\nu.$$

In summary we have

- Case 1: $d_i = 0, r_i = 0 \Rightarrow f_i(0,0,v_i) = e^{-(\mu+\nu)v_i}$,
- Case 2: $d_i = 1, r_i = 0 \Rightarrow f_i(1,0,v_i) = e^{-(\mu+\nu)v_i}\mu$,
- Case 3: $d_i = 0, r_i = 1 \Rightarrow f_i(0,1,v_i) = e^{-(\mu+\nu)v_i}\nu$.

All three cases can conveniently be packaged as

$$f_i(d_i, r_i, v_i) = e^{-(\mu+\nu)v_i} \times \mu^{d_i} \times \nu^{r_i}$$

and so, assuming that all n lives are independent, we obtain the likelihood as

$$L(\mu,\nu) = \prod_1^n e^{-(\mu+\nu)v_i} \times \mu^{d_i} \times \nu^{r_i}$$

or

$$L(\mu,\nu) = e^{-\mu v}\mu^d \times e^{-\nu v}\nu^r \tag{17}$$

where $v = \sum v_i$ is the total waiting time, $d = \sum d_i$ is the total number of deaths and $r = \sum r_i$ is the total number of retirals.

Notice that $L(\mu,\nu)$ **factorises** into a product of two functions, one of μ and one of ν. As a result the log likelihood has the particularly simple form

$$\ell(\mu,\nu) = -\mu v + d\log\mu - \nu v + r\log\nu$$

and the estimators of μ and ν are easily obtained:

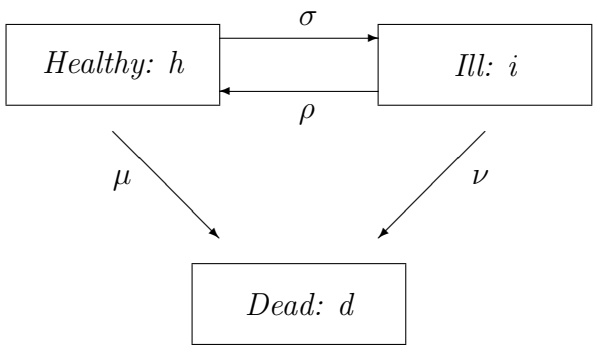
$$\tilde{\mu} = \frac{D}{V}, \quad \tilde{\nu} = \frac{R}{V}. \tag{18}$$

Comment: There are obvious parallels between

- (17) and the likelihood $L(\mu) = e^{-\mu v}\mu^d$ in the 2-state model, and
- (18) and the estimator $\tilde{\mu} = D/V$ in the 2-state model.

3.13 More estimation in the 3-state model

Our second (and final) example of the 3-state model is the model for IPI discussed in section 3.5. Consider the diagram

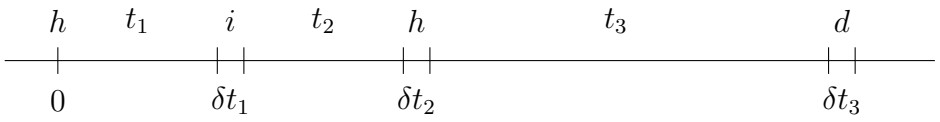


where we suppose that the transition intensities are constant for the n lives in the investigation. Suppose that the observation on the ith life is

$$\begin{aligned} v_i &= \text{total waiting time in state } h \\ w_i &= \text{total waiting time in state } i \\ s_i &= \text{number of transitions from } h \text{ to } i \\ r_i &= \text{number of transitions from } i \text{ to } h \\ d_i &= 1 \quad \text{if } h \to d, \text{ 0 otherwise} \\ u_i &= 1 \quad \text{if } i \to d, \text{ 0 otherwise.} \end{aligned}$$

A straightforward extension of the argument in the previous example for the ith life gives the likelihood. An example may help.

Example 3.4: Consider a life that is initially healthy, falls ill after time t_1, recovers after a further time t_2 and then dies after a further time t_3. As always we consider the events falling ill, recovering and dying as occurring in a short interval of time, as in the diagram



The probability of observing this sequence of events or **life history** is

$$e^{-(\mu+\sigma)t_1}[\sigma\delta t_1 + o(\delta t_1)] \times e^{-(\rho+\nu)t_2}[\rho\delta t_2 + o(\delta t_2)] \times e^{-(\mu+\sigma)t_3}[\mu\delta t_3 + o(\delta t_3)].$$

Dividing by $\delta t_1 \delta t_2 \delta t_3$ and taking the limit as $\delta t_i \to 0,\ i = 1, 2, 3$ we find the contribution to the likelihood from this life is

$$e^{-(\mu+\sigma)(t_1+t_3)} e^{-(\rho+\nu)t_2} \mu\, \sigma\, \rho\, \nu^0.$$

The general formula follows by multiplying up over all lives:

$$\begin{aligned} & \prod_1^n e^{-(\sigma+\mu)v_i} \times e^{-(\rho+\nu)w_i} \times \sigma^{s_i} \times \rho^{r_i} \times \mu^{d_i} \times \nu^{u_i} \\ = \ & e^{-(\sigma+\mu)v} e^{-(\rho+\nu)w} \sigma^s \rho^r \mu^d \nu^u. \end{aligned} \tag{19}$$

Once more the likelihood factorises into separate functions and so the estimators are easily obtained. We find

$$\tilde{\mu} = \frac{D}{V}, \quad \tilde{\nu} = \frac{U}{W}, \quad \tilde{\sigma} = \frac{S}{V}, \quad \tilde{\rho} = \frac{R}{W}. \tag{20}$$

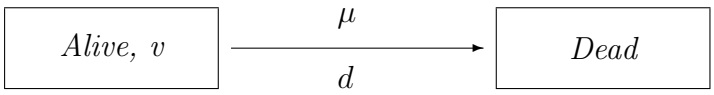
The estimates in all these examples have the form

$$\begin{matrix}\text{MLE of transition} \\ \text{intensity } g \to h\end{matrix} = \frac{\text{No. transfers from } g \text{ to } h}{\text{Total waiting time in } g}.$$

3.14 The likelihood in multi-state models

It is very important that you can write down the likelihood for a multi-state model. The rule is very simple and to emphasise this we collect the results of all our examples into one place.

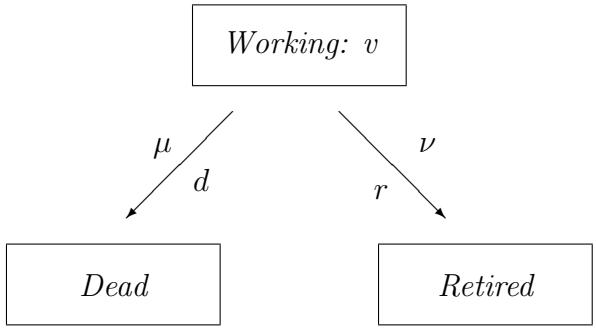
Example 1: Simple mortality; see 3.9.



where v is the total waiting time in the alive state and d is the total number of deaths. Then

$$L(\mu) = e^{-\mu v}\mu^d.$$

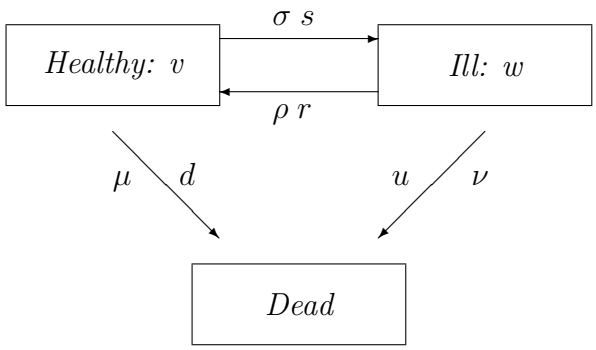
Example 2: Retiral; see 3.12.



where v is the total waiting time in the working state, d is the total number of deaths and r is the total number of retirals. Then

$$L(\mu, \nu) = e^{-(\mu+\nu)v}\mu^d\nu^r.$$

Example 3: IPI; see 3.13.



where v and w are total waiting times and r, s, d and u are total numbers of transfers between states as shown. Then

$$L(\mu, \nu, \sigma, \rho) = e^{-(\mu+\sigma)v} \times e^{-(\nu+\rho)w} \times \mu^d \times \nu^u \times \sigma^s \times \rho^r.$$

3.15 The likelihood rule

These examples allow us to formulate a general rule as follows:

(a) Each **waiting time** w_i in state i contributes a factor

$$e^{-(\sum_j \mu_{ij})w_i}$$

where $\sum_j \mu_{ij}$ is the sum of the transition intensities **out of state** i.

(b) Each **transfer number** t_{ij} from state i to state j contributes a factor

$$\mu_{ij}^{t_{ij}}$$

where μ_{ij} is the transition intensity from state i to state j.

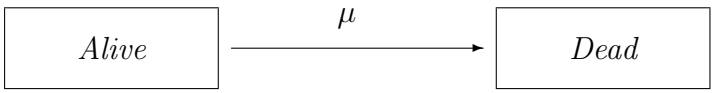
(c) The total likelihood is the product of all the above contributions.

Exam tip: You must be able to

- apply this rule to any multi-state model, and
- derive *from first principles* the likelihood for simple examples, as done in lectures.

3.16 Properties of the MLE's

We saw in the simple



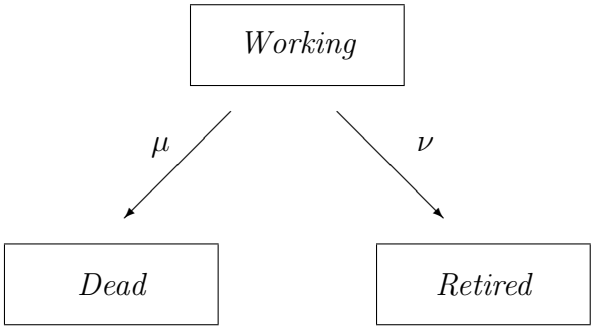
model that

$$\tilde{\mu} = \frac{D}{V} = \frac{\text{No. deaths}}{\text{Total waiting time}}$$

and that asymptotically

$$\tilde{\mu} \approx \mathcal{N}\left(\mu_0, \frac{\mu_0}{\mathrm{E}(V)}\right).$$

The proof consisted of applying the Maximum Likelihood Theorem (MLT) together with some properties of the score function. For the model for retiral



we have the likelihood

$$L(\mu, \nu) = e^{-\mu v}\mu^d \times e^{-\nu v}\nu^r$$

where v is the total waiting time in the working state, d is the number of deaths and r is the number of retirals. What is the joint distribution of $\tilde{\mu}$ and $\tilde{\nu}$? We need to extend the MLT to the multiparameter case. The following parallels the one parameter case dealt with in 3.10.

Definition: The **score function**. Suppose $\boldsymbol{X}' = (X_1, X_2, \ldots, X_n)$ is a vector of random variables where the X_i are independent random variables with probability function $f_i(x, \boldsymbol{\theta})$ which depends on a parameter $\boldsymbol{\theta}' = (\theta_1, \ldots, \theta_r)$ with r components (which do not determine the range of X_i, $i = 1, 2, \ldots, n$). Let

$$\ell = \ell(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}, \boldsymbol{X}) = \sum_1^n \log f_i(X_i, \boldsymbol{\theta})$$

be the log likelihood function (viewed as a random variable). Let

$$U_i = U_i(\boldsymbol{\theta}) = \frac{\partial \ell}{\partial \theta_i},\ i = 1, \ldots r.$$

Then $\boldsymbol{U}' = (U_1, \ldots, U_r)$ is called the **score function**.

Result: The score function satisfies:

(a) $\mathrm{E}(\boldsymbol{U}) = 0$, and

(b) $\mathrm{Var}(\boldsymbol{U}) = -\mathrm{E}\left(\dfrac{\partial \boldsymbol{U}}{\partial \boldsymbol{\theta}}\right) = (v_{ij})_{r\times r}$ where $v_{ij} = -\mathrm{E}\left(\dfrac{\partial^2 \ell}{\partial\theta_i \partial\theta_j}\right)$.

Proof: The proof is pretty much the same as the one-parameter case. First, we take $n = 1$ and write $f_i = f$.

(a) Consider the identity

$$\int f(x, \boldsymbol{\theta})\, dx = 1.$$

Differentiate with respect to θ_i. We get

$$\int \frac{\partial f}{\partial \theta_i}\, dx = 0.$$

Rewrite this equation as

$$\int \left(\frac{\partial}{\partial \theta_i} \log f\right) f\, dx = 0. \tag{21}$$

The left hand side of this equation is $\mathrm{E}(U_i)$, so $\mathrm{E}(\boldsymbol{U}) = 0$ as required.

(b) The proof uses the same trick. Differentiate (21) with respect to θ_j and use the product

rule. We get

$$\begin{aligned}
& \int \left\{ \frac{\partial}{\partial\theta_j}\left(\frac{\partial}{\partial\theta_i}\log f\right) f + \left(\frac{\partial}{\partial\theta_i}\log f\right)\frac{\partial f}{\partial\theta_j} \right\} dx = 0 \\
\Rightarrow \quad & \int \frac{\partial}{\partial\theta_j}\left(\frac{\partial}{\partial\theta_i}\log f\right) f\, dx + \int \left(\frac{\partial}{\partial\theta_i}\log f\right)\left(\frac{\partial}{\partial\theta_j}\log f\right) f\, dx = 0 \\
\Rightarrow \quad & \int \frac{\partial U_i}{\partial\theta_j} f\, dx + \int U_i\, U_j\, f\, dx = 0 \\
\Rightarrow \quad & \mathrm{E}(U_i\, U_j) = -\mathrm{E}\left(\frac{\partial U_i}{\partial\theta_j}\right) \\
\Rightarrow \quad & \mathrm{Cov}(U_i, U_j) = -\mathrm{E}\left(\frac{\partial^2\ell}{\partial\theta_i\theta_j}\right) \quad \text{since } \mathrm{E}(U_i) = 0
\end{aligned}$$

and (b) is proved.

The case of general n follows by the additivity of the score function and the independence of the X_i. ■

The information function $I(\theta)$ extends as follows:

Definition: Fisher's **information function** $I(\boldsymbol{\theta})$ is

$$I(\boldsymbol{\theta}) = -\mathrm{E}\left(\frac{\partial^2\ell}{\partial\theta_i\theta_j}\right) = -\mathrm{E}\left(\frac{\partial \boldsymbol{U}}{\partial\boldsymbol{\theta}}\right) = \mathrm{Var}(\boldsymbol{U}).$$

As before we usually call $I(\boldsymbol{\theta})$ the information function. We are now ready the state (without proof) the multi-parameter version of the **Maximum Likelihood Theorem**.
Theorem: Suppose $\hat{\boldsymbol{\theta}}$ satisfies $\boldsymbol{U} = \boldsymbol{0}$ and that $\boldsymbol{\theta}_0$ is the true (but unknown) value of $\boldsymbol{\theta}$. If $\tilde{\boldsymbol{\theta}}$ is the random variable corresponding to $\hat{\boldsymbol{\theta}}$, then $\tilde{\boldsymbol{\theta}}$ is

(i) asymptotically unbiased, i.e. $\mathrm{E}(\tilde{\boldsymbol{\theta}}) \to \boldsymbol{\theta}_0$,

(ii) asymptotically efficient, i.e. achieves the Cramer-Rao Lower Bound, i.e. $\mathrm{Var}(\tilde{\boldsymbol{\theta}}) \to I(\boldsymbol{\theta}_0)^{-1}$,

(iii) asymptotically normal.

We can summarise this in

$$\tilde{\boldsymbol{\theta}} \approx \mathcal{N}(\boldsymbol{\theta}_0, I(\boldsymbol{\theta}_0)^{-1}).$$

■

3.17 Examples of MLE's

Example 3.5: Retiral/Death model. We found the likelihood for this model in 3.12 as

$$L(\mu, \nu) = e^{-\mu v}\mu^d \times e^{-\nu v}\nu^r$$

which gives

$$\ell(\mu, \nu) = -\mu v + d\log\mu - \nu v + r\log\nu \tag{22}$$

as the log likelihood and

$$\tilde{\mu} = \frac{D}{V}, \quad \tilde{\nu} = \frac{R}{V}$$

as the estimators of μ and ν. Now the asymptotic distribution of $\tilde{\mu}$ and $\tilde{\nu}$ is determined by the information matrix $I(\mu, \nu)$. Observe that

$$\begin{aligned} U_\mu &= \frac{\partial \ell}{\partial \mu} = -v + \frac{d}{\mu} \\ \Rightarrow U_{\mu\nu} &= \frac{\partial^2 \ell}{\partial \mu \partial \nu} = 0. \end{aligned}$$

Thus $I(\mu, \nu)$ is **diagonal** and so $\tilde{\mu}$ and $\tilde{\nu}$ are **asymptotically independent**.

N.B. The estimators will always be asymptotically independent whenever the log likelihood can be expressed as the sum of separate functions of the parameters as in (22).

We can now apply the one parameter result for the 2-state model separately to the μ-part and the ν-part of $\ell(\mu, \nu)$. We find

$$I(\mu, \nu) = \begin{bmatrix} \dfrac{\mathrm{E}(V)}{\mu} & 0 \\ 0 & \dfrac{\mathrm{E}(V)}{\nu} \end{bmatrix}$$

and so

$$\begin{aligned} &\tilde{\mu} \approx \mathcal{N}\left(\mu_0, \frac{\mu_0}{\mathrm{E}(V)}\right), \\ &\tilde{\nu} \approx \mathcal{N}\left(\nu_0, \frac{\nu_0}{\mathrm{E}(V)}\right), \\ &\tilde{\mu} \text{ and } \tilde{\nu} \text{ are asymptotically independent} \end{aligned}$$

■

Example 3.6: PHI model. We found the likelihood for this model in 3.13 as

$$L(\mu, \nu, \rho, \sigma) = e^{-(\sigma+\mu)v} e^{-(\rho+\nu)w} \sigma^s \rho^r \mu^d \nu^u$$

which gives

$$\tilde{\mu} = \frac{D}{V}, \quad \tilde{\nu} = \frac{U}{W}, \quad \tilde{\sigma} = \frac{S}{V}, \quad \tilde{\rho} = \frac{R}{W}.$$

as the estimators of μ, ν, σ and ρ.
Once more the likelihood factorises into separate functions and so the asymptotic distribution of the estimators is easily obtained. First, since $I(\mu, \nu, \sigma, \rho)$ is diagonal, $\tilde{\mu}$, $\tilde{\nu}$, $\tilde{\sigma}$ and $\tilde{\rho}$ are asymptotically independent. Moreover

$$\tilde{\mu} \approx \mathcal{N}\left(\mu_0, \frac{\mu_0}{\mathrm{E}(V)}\right)$$

with corresponding results for $\tilde{\nu}$, $\tilde{\sigma}$ and $\tilde{\rho}$. ■

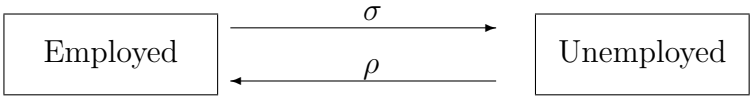
Conclusion: The Markov model gives simple estimates of the transition intensities with simple properties. We need to keep track of only two things:

- the number of transfers between states, and
- the total time spent in each state.

This simplicity is part of the appeal of this class of models.

Tutorial 3: Markov Models

1. Unemployment at ages x and over can be modelled by a multiple state model with two states, e = employed and u=unemployed, with constant forces of transition σ and ρ between them. Deaths are ignored.



   (a) State the standard Markov model assumptions for this setup and find ${}_tp_x^{\overline{ee}}$. Hence write down ${}_tp_x^{\overline{uu}}$.

   (b) Define: W_e to be the total waiting time spent in the employed state, W_u to be the total waiting time spent in the unemployed state, T_e to be the total number of transfers out of the employed state, and T_u to be the total number of transfers out of the unemployed state. Write down the likelihood function, $L(\sigma, \rho)$, and find the MLE's of σ and ρ.

   (c) Use the properties of the score function to prove that

      (i) $\mathrm{E}(T_e) = \sigma \mathrm{E}(W_e)$;

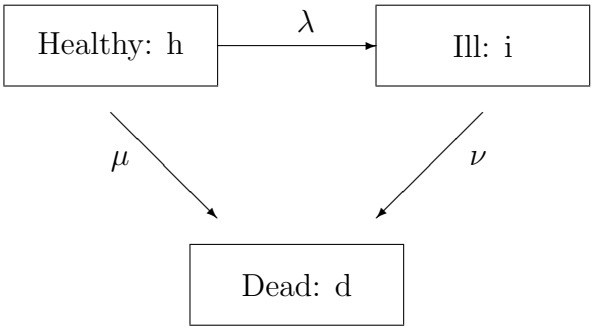
      (ii) $\mathrm{E}(T_u) = \rho \mathrm{E}(W_u)$;

      (iii) the information function $\mathrm{I}(\sigma, \rho)$ is given by

$$\begin{pmatrix} \dfrac{\mathrm{E}(W_e)}{\sigma} & 0 \\ 0 & \dfrac{\mathrm{E}(W_u)}{\rho} \end{pmatrix}.$$

   (d) What is the asymptotic distribution of the MLE's of σ and ρ?

2. Consider the following three-state model for terminal illness where the transition intensities can be considered constant for the ages concerned.

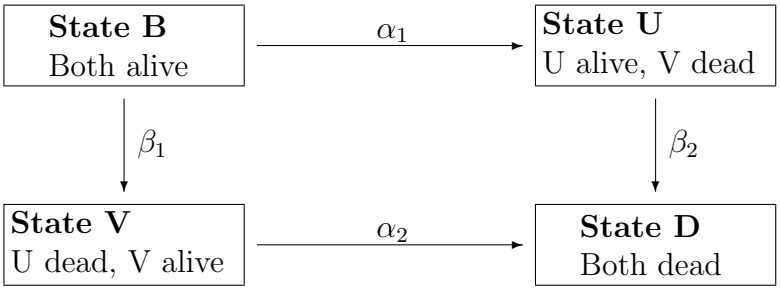


   (a) Find expressions for ${}_tp_x^{\overline{hh}}$ and ${}_tp_x^{\overline{ii}}$ in terms of μ, λ and ν. What can you say about ${}_tp_x^{\overline{hh}}$ and ${}_tp_x^{hh}$, and about ${}_tp_x^{\overline{ii}}$ and ${}_tp_x^{ii}$?

   (b) Obtain the likelihood function $L(\mu, \lambda, \nu)$; define carefully any symbols that you use. Find the MLE's of μ, λ and ν.

   (c) Derive the asymptotic distribution of the MLE's of μ, λ and ν.

3. The following has been suggested as a model of the mortality of two lives U and V:

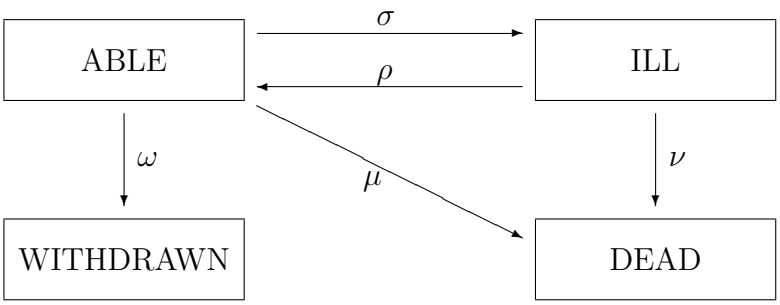


in which the transition intensities $\alpha_1, \alpha_2, \beta_1$ and β_2 are constants.

(a) Write down the likelihood function for α_1, α_2, β_1 and β_2 in terms of the observed numbers of transitions and the waiting times in each state. Find the MLE's of the parameters in the model, and derive their asymptotic distribution.

(b) Suppose you are told that $\alpha_1 = \alpha_2$, and $\beta_1 = \beta_2$. What now are the MLE's of the parameters, and what is their asymptotic distribution?

4. A group of lives who hold sickness insurance policies can be in one of four states : $a =$ able, $i =$ ill, $d =$ dead, $w =$ withdrawn. The lives can move between states according to the following diagram:



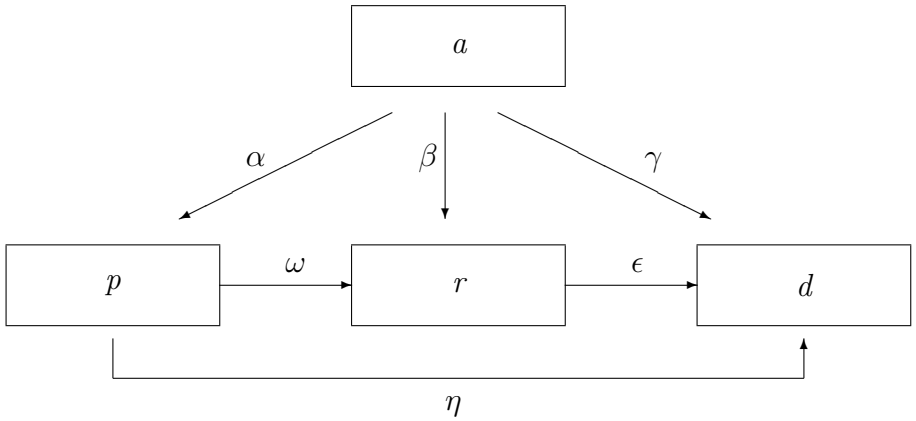
We assume that the forces μ, ν, ω, σ and ρ are constant, will make appropriate assumptions about the probabilities of multiple transitions in short times.

(a) Write down the likelihood function for an arbitrary set of observed transitions made by a single life who enters observation at time $t = 0$ and is observed for one year, or until death or withdrawal. Define any symbols which you use. Hence find the MLE's of the parameters in the model.

(b) Derive the asymptotic distribution of the MLE's.

5. In Tutorial 2, Question 2, we found $\hat{\beta} = -0.792$. What is the ratio of the hazards under the Cox model? Now assume a 2-state model with constant forces of transition μ_A and μ_B. Find the ratio of the hazards under this model and comment.

6. HW, 2008. The following 4-state Markov model with constant transition intensities α, β, γ, ω, ϵ and η is used to model a pension scheme for male members aged 50. Age is defined as age last birthday on January 1st. The states are a: Active, p: Preserved benefit, r: Drawing pension, and d: Dead.



(a) (i) By assuming the appropriate result for the 2-state model of mortality with constant force of mortality, write down expressions for ${}_tp_x^{aa}$, ${}_tp_x^{pp}$ and ${}_tp_x^{rr}$. Explain your reasoning.

(ii) From first principles, show that the Kolmogorov forward equation for ${}_tp_x^{ap}$ can be written

$$\frac{du}{dt} + \psi\, u = \alpha\, e^{-\phi t}$$

where $u = u(t) = {}_tp_x^{ap}$, $\phi = \alpha + \beta + \gamma$ and $\psi = \omega + \eta$.

(iii) Verify that the solution to the equation in (ii) is

$$\frac{\alpha}{\phi - \psi}\left(e^{-\psi t} - e^{-\phi t}\right)$$

and that this solution satisfies an appropriate boundary condition.

(b) A life, subject to the above 4-state model, is observed from January 1, 1980. Initially the life is in the active state. On 1st January 1985 the life moves to the preserved benefit state and then on 1st January 1990 the life dies. Find the contribution to the likelihood function of this life history.

(c) Part of the data from a study of lives subject to the above 4-state model was: waiting time in state p, $W_p = 1250$ years and the number of transfers from state p to state d, $T_{pd} = 24$. What parameter in the above 4-state model does this information allow you to estimate, what is the value of the estimate and what is its standard error?

4 Binomial and Poisson Models of Mortality

4.1 Motivation

Consider n independent identical lives **aged** x **exactly**. Observe these n lives for **one year exactly**. Suppose d die and that $q_x = \Pr(\text{Die in } (x, x+1))$. Let D be the random variable corresponding to d. Then

$$D \sim \mathcal{B}(n, q_x),$$

i.e. D has the binomial distribution and we observe $D = d$. The likelihood is

$$\begin{aligned} L(q_x) &\propto q_x^d (1-q_x)^{n-d} \\ \Rightarrow \ell &= d \log q_x + (n-d)\log(1-q_x) \\ \Rightarrow \frac{d\ell}{dq_x} &= \frac{d}{q_x} - \frac{n-d}{1-q_x} \\ \Rightarrow \hat{q}_x &= \frac{d}{n}, \end{aligned}$$

as is well known. Furthermore,

$$\mathrm{E}(\tilde{q}_x) = \mathrm{E}\left(\frac{D}{n}\right) = \frac{nq_x}{n} = q_x$$

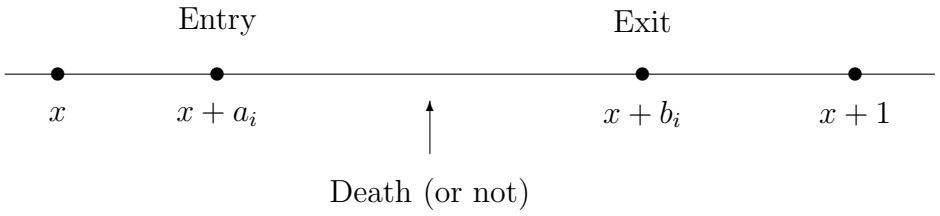
and

$$\mathrm{Var}(\tilde{q}_x) = \mathrm{Var}\left(\frac{D}{n}\right) = \frac{nq_x(1-q_x)}{n^2} = \frac{q_x(1-q_x)}{n}.$$

This is a simple version of the **binomial model of mortality**. The model might be appropriate if the exact times of death were not known in which case the waiting time (or central exposed to risk) could not be computed.

4.2 The binomial model

Consider now a more **realistic observation plan** for the ith life:



We observe whether death occurs or not in the observation period $x + a_i$ to $x + b_i$. Thus

$$D_i \sim \mathcal{B}(1, {}_{b_i-a_i}q_{x+a_i}).$$

Let

$$q_i = {}_{b_i-a_i}q_{x+a_i}$$

and then

$$D_i \sim \mathcal{B}(1, q_i)$$

and we can write

$$\Pr(D_i = d_i) = q_i^{d_i} \times (1-q_i)^{1-d_i}, \ d_i = 0, 1.$$

Hence the total likelihood over all n lives is

$$L(\boldsymbol{q};\boldsymbol{d}) \propto \prod_{i=1}^{n} q_i^{d_i} \times (1-q_i)^{1-d_i}$$

where

- $\boldsymbol{q}' = (q_1, \ldots, q_n) = ({}_{b_1-a_1}q_{x+a_1}, \ldots, {}_{b_n-a_n}q_{x+a_n})$ is the parameter set, and
- $\boldsymbol{d}' = (d_1, \ldots, d_n)$ is the data.

Now $\boldsymbol{q}$ is a vector with n-components so with n pieces of data we cannot make sensible progress without some further assumptions. The unrestricted MLE of ${}_{b_i-a_i}q_{x+a_i}$ is d_i which equals 0 or 1 depending on the value of d_i. Actually, $\boldsymbol{q}$ may not have dimension n if **some** of the $q_i = {}_{b_i-a_i}q_{x+a_i}$ are equal. However, in the great majority of cases we will need to put some restrictions on $\boldsymbol{q}$ in order to obtain **smooth estimates** of mortality. There are three common ways to reduce $L(\boldsymbol{q})$ to a function of a **single parameter**.

(a) Uniform distribution of deaths (UDD)

(b) Constant force of mortality (CFM)

(c) Balducci assumption (BA)

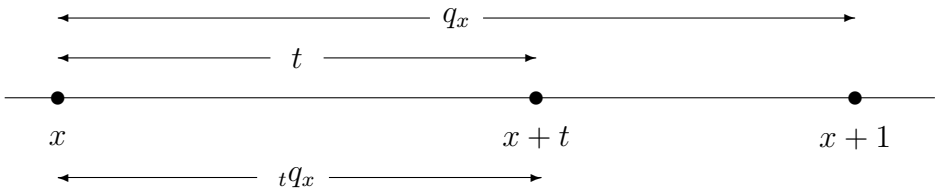
We discuss these in turn.

4.3 Uniform distribution of deaths

We suppose that

$${}_tq_x = t \cdot q_x, \ 0 \le t \le 1. \tag{23}$$

The following diagram might be helpful:



More precisely, under UDD, we will have

$$\Pr(\text{Die in } (x+t, x+t+\delta t) \mid \text{die in } (x, x+1)) = \delta t + o(\delta t) \tag{24}$$

Hence

$$\begin{aligned}
\frac{{}_tq_x}{q_x} &= \frac{\Pr(\text{Die in } (x, x+t) \mid \text{alive at } x)}{\Pr(\text{Die in } (x, x+1) \mid \text{alive at } x)} \\
&= \Pr(\text{Die in } (x, x+t) \mid \text{die in } (x, x+1)) \\
&= \int_0^t ds \quad \text{by (24)} \\
&= t, \quad \text{which is (23), as required.}
\end{aligned}$$

We can show (see Tut4, Q1) that under UDD

$$\mu_{x+t} = \frac{q_x}{1 - t \cdot q_x}, \ 0 \le t \le 1, \tag{25}$$

so in general μ_{x+t} is an **increasing function** of t for $0 \le t \le 1$.

4.4 Constant force of mortality

The constant force of mortality assumption is that

$$\mu_{x+t} = \mu_x, \ 0 \le t < 1.$$

We can show (see Tut4, Q2) that

$${}_tq_x = 1 - e^{-\mu_x t}, \quad 0 \le t < 1,$$

and in particular

$$q_x = 1 - e^{-\mu_x}$$

and

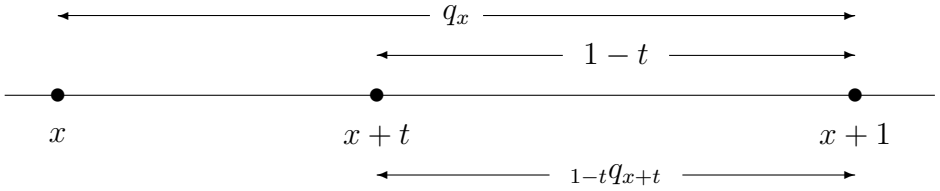
$$\mu_x = -\log(1 - q_x).$$

4.5 Balducci assumption

The Balducci assumption (BA) is that

$${}_{1-t}q_{x+t} = (1-t) \cdot q_x, \ 0 \le t \le 1. \tag{26}$$

The following diagram might be helpful:



Corresponding to (25), we can show that

$$\mu_{x+t} = \frac{q_x}{1-(1-t)q_x}, \ 0 \le t \le 1,$$

so BA has the highly unsatisfactory feature that μ_{x+t} is a **decreasing function** of t for $0 \le t \le 1$. However, BA still appears widely in the actuarial literature. We will make little further reference to BA and it is not examinable.

4.6 Reduction of the likelihood

In section 4.2 we showed that with $q_i = {}_{b_i-a_i}q_{x+a_i}$ we have

$$L(\boldsymbol{q}; \boldsymbol{d}) \propto \prod_{i=1}^{n} q_i^{d_i} \times (1-q_i)^{1-d_i}$$

and noted that, in general, this was function of the n parameters in $\boldsymbol{q}$. We can reduce $L(\boldsymbol{q}; \boldsymbol{d})$ to a function of a single parameter by any of the above three assumptions. For example, under UDD we know from Tut4, Q1 that

$$q_i = {}_{b_i-a_i}q_{x+a_i} = \frac{(b_i - a_i)q_x}{1 - a_i \cdot q_x}$$

and so $L(\boldsymbol{q};\boldsymbol{d})$ reduces to

$$L(q_x;\boldsymbol{d}) = \prod_{i=1}^{n}\left[\frac{(b_i-a_i)q_x}{1-a_i\cdot q_x}\right]^{d_i}\left[1-\frac{(b_i-a_i)q_x}{1-a_i\cdot q_x}\right]^{1-d_i}, \tag{27}$$

which is a function of q_x only. This function is messy to handle analytically but we can easily work out $\hat{q}_x$ and

$$-\frac{d^2\ell}{dq_x^2}\Big]_{q_x=\hat{q}_x}$$

with R so the MLE and its SE can be found.

Under the constant force of mortality, we have

$$\begin{aligned} q_i = {}_{b_i-a_i}q_{x+a_i} &= 1-{}_{b_i-a_i}p_{x+a_i} \\ &= 1-\exp\left\{-\int_0^{b_i-a_i}\mu_{x+a_i+s}\,ds\right\} \\ \Rightarrow q_i &= 1-\exp\{-(b_i-a_i)\mu_x\} \end{aligned}$$

and again $L(\boldsymbol{q};\boldsymbol{d})$ reduces to a 1-parameter function.

4.7 Example of the binomial model

Example 4.1: Here is some data from a small mortality investigation on six (6) lives all aged between x and $x+1$. We ignore for the moment the information on the exact times of death in the column t_i.

Table 3: Data from a mortality investigation on six lives

Life	a_i	b_i	d_i	t_i
1	0	1	0	1
2	0	1	1	0.5
3	0	0.5	0	0.5
4	0.25	0.75	1	0.5
5	0.5	1	1	0.75
6	0.75	1	0	1

In the table, for the ith life, we have

- $x+a_i$ is the exact age of entry,
- $x+b_i$ is exact age when life may be censored,
- $x+t_i$ is exact age of exit by death or censoring, and
- $d_i=0$ if life is censored, $d_i=1$ if life dies.

Suppose that deaths in $[x,x+1]$ obey the UDD assumption. Then

$${}_{b-a}q_{x+a} = \frac{(b-a)q_x}{1-a\cdot q_x}.$$

Table 4: Contributions to $L(q)$

Life	${}_{b-a}q_{x+a}$	d_i	Contribution
1	${}_1q_x = q$	0	$1-q$
2	${}_1q_x = q$	1	q
3	${}_{0.5}q_x = 0.5q$	0	$1-0.5q$
4	${}_{0.5}q_{x+0.25} = \dfrac{0.5q}{1-0.25q}$	1	$\dfrac{0.5q}{1-0.25q}$
5	${}_{0.5}q_{x+0.5} = \dfrac{0.5q}{1-0.5q}$	1	$\dfrac{0.5q}{1-0.5q}$
6	${}_{0.25}q_{x+0.75} = \dfrac{0.25q}{1-0.75q}$	0	$1-\dfrac{0.25q}{1-0.75q}$

For convenience we put $q_x = q$. We obtain the contribution to $L(q)$ from each life in the above table.

N.B. The binomial model discards the information on the exact times of death which are available in Table 3. The Markov model would make **full use of all** available information and should be used if exact times of death are known.

The full likelihood under the binomial model is

$$\begin{aligned} L(q) &\propto (1-q)\times q\times\left(1-\frac{q}{2}\right)\times\frac{q}{1-\frac{q}{4}}\times\frac{q}{1-\frac{q}{2}}\times\frac{1-q}{1-\frac{3q}{4}} \\ &\propto \frac{q^3(1-q)^2}{(4-q)(4-3q)} \end{aligned}$$

and the log likelihood is

$$\ell(q) = 3\log q + 2\log(1-q) - \log(4-q) - \log(4-3q).$$

The MLE satisfies

$$U = \frac{d\ell}{dq} = \frac{3}{q} - \frac{2}{1-q} + \frac{1}{4-q} + \frac{3}{4-3q} = 0$$

which is easily solved numerically in R. The programs `Binomial_UDD.r` and `Binomial_CFM.r` give R-code for the binomial model analysis of these data under UDD and CFM respectively.

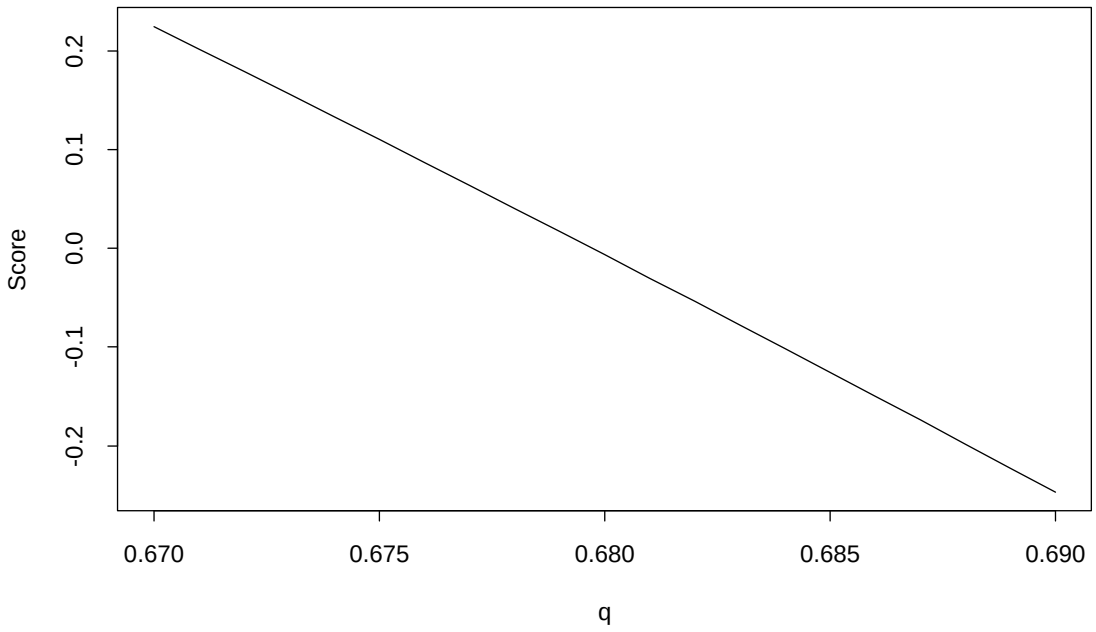
The **Newton-Raphson** algorithm is a very useful method for solving equations. Suppose x_0 is some estimate of the solution to the equation $f(x) = 0$ and that $x_1 = x_0 + h$ is an improved solution. Then by Taylor's Theorem

$$f(x_0+h) \approx 0 \Rightarrow f(x_0) + hf'(x_0) \approx 0 \Rightarrow h \approx -\frac{f(x_0)}{f'(x_0)}$$

and

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

is an improved solution. An important point is that often the derivative of $f(x)$ can be computed numerically from the function to evaluate $f(x)$. The program `NewtonRaphson.r` shows how this can be done.



The R code in `Binomial_UDD.r` in the section *Binomial models* on the web shows how to obtain the above plot and to find $\hat{q}$. In brief, we have

```
Score <- function(Q) 3/Q - 2/(1-Q) + 1/(4-Q) + 3/(4-3*Q)
Plot.Score <- function(Left, Right) {
  x <- seq(Left, Right, length = 1000)
  plot(x, Score(x), type = "l", col = "red", lwd = 2,
    main = "Score function for Q", ylab = "Score(Q)")
  grid( )
}
Plot.Score(0.1, 0.9)     # And we easily arrive at
Plot.Score(0.67, 0.69)
```

The MLE is $\hat{q} = 0.680$. The SE of $\hat{q}$ can be estimated in a number of ways

- the slope of U which we can estimate

$$\text{Slope of } U = \frac{U(0.685) - U(0.675)}{0.01} = -23.59$$

  or in R (better)

```
Slope = (Score(0.685) - Score(0.675))/0.01; Slope
```

  which gives an estimated standard error of $\sqrt{1/23.6} = 0.206$, or

- $d^2\ell/dq^2$ evaluated at $\hat{q}$. This gives

$$\frac{d^2\ell}{dq^2} = -\frac{3}{q^2} - \frac{2}{(1-q)^2} + \frac{1}{(4-q)^2} + \frac{9}{(4-3q)^2}\Bigg]_{q=\hat{q}} = -23.56$$

  which agrees which the previous answer.

Note: Under the constant force of mortality assumption we find

$$U = \frac{d\ell}{d\phi} = \frac{7}{\phi} - \frac{4\phi}{1-\phi^2} - \frac{4\phi^3}{1-\phi^4} = 0, \quad \phi = e^{-\mu/4}$$

from which it follows that $\hat{\phi} = 0.750$ and so $\hat{q} = 0.684 \approx 0.680$.

Markov model solution: In Table 3 we know the exact times of death so there is no need to use the binomial model. We use the Markov model and assume a constant force of mortality from x to $x+1$. From Table 3 we find

$$\begin{aligned} \hat{\mu}_x &= \frac{\text{No. deaths}}{\text{Total waiting time}} \\ &= \frac{3}{1+0.5+0.5+0.25+0.25+0.25} \\ &= \frac{3}{2.75} \\ &= 1.0909 \end{aligned}$$

from which we find

$$\hat{q}_x = 1 - e^{-\hat{\mu}_x} = 0.664 \approx 0.680$$

as found above.

The variance of $\hat{\mu}_x$ follows from

$$\text{Var}(\hat{\mu}_x) = \frac{\mu_x}{\text{E}(V)} \approx \frac{\hat{\mu}_x}{v} = \frac{1.0909}{2.75} = 0.3967.$$

But what about $\text{Var}(\hat{q}_x)$? We need the following very useful formula for computing $\text{Var}[f(X)]$ from $\text{Var}(X)$. This method is known as the **Delta method** or Δ-method.

Lemma: Let X be a random variable with mean μ, and let $f(\cdot)$ be some differentiable function. Then

$$\text{Var}[f(X)] \approx [f'(\mu)]^2 \text{Var}(X). \tag{28}$$

Proof: We use **Taylor's Theorem** and expand $f(X)$ about μ.

$$\begin{aligned} \text{Var}[f(X)] &\approx \text{Var}[f(\mu) + (X-\mu)f'(\mu) + \ldots] \\ &\approx \text{Var}[(X-\mu)f'(\mu)] \end{aligned}$$

and the answer follows. ■

We apply this formula as follows:

$$\begin{aligned} \text{Var}(\hat{q}_x) &= \text{Var}(1 - e^{-\hat{\mu}_x}) \\ &= \text{Var}(e^{-\hat{\mu}_x}) \\ &\approx [e^{-\mu_x}]^2 \text{Var}(\hat{\mu}_x) \\ &\approx e^{-2\hat{\mu}_x} \text{Var}(\hat{\mu}_x) \\ &= 0.04476 \end{aligned}$$

which gives a SE of 0.212 (compared to 0.206 above).

Comments: Of course, if we had the exact times of death we would **not** use the binomial model. We would estimate μ_x directly from the Markov model and then get q_x from that. We emphasise that we only use the binomial model when we do not have the exact times of death.

4.8 The actuarial estimate of q_x

The following formula is well-known to actuaries:

$$\hat{q}_x = \frac{d}{v + \frac{1}{2}d} \tag{29}$$

where v is the waiting time (central exposed to risk) and d is the number of deaths. This formula is known as *the actuarial estimate*. A heuristic argument is as follows: the denominator should be the waiting time **if no one dies**. So if we assume that

- a life is initially at risk of death throughout the whole year, and
- on average, a life that dies does so half way through the year

then on average the waiting time has lost $\frac{1}{2}d$ exposure to death.

Our attitude is: we have the 2-state model which gives $\hat{\mu} = d/v$ as the MLE of μ. Thus, we can estimate q by $\hat{q} = 1 - e^{-\hat{\mu}}$. Why bother with (29)? However, superficially a direct answer like (29) is attractive. Think of N lives all **initially exposed to risk** of death for the full year. Clearly, we have

$$\hat{q} = \frac{D}{N} \tag{30}$$

as the MLE in the binomial distribution. We might try to get a formula like

$$\hat{q} = \frac{D}{\text{something}},$$

to generalise (30). Here is one way to derive (29).

The 2-state model assumes

$$\mu_{x+t} = \mu_x, \quad 0 \le t < 1.$$

The constant force of mortality, μ_x, best represents the force of mortality mid-way through the interval $[x, x+1]$. Thus it would be more accurate to write

$$\mu_{x+t} = \mu_{x+\frac{1}{2}}, \quad 0 \le t < 1,$$

and so our 2-state model gives us

$$\hat{\mu}_{x+\frac{1}{2}} = \frac{d}{v}.$$

Now under UDD we know that

$$\mu_{x+\frac{1}{2}} = \frac{q_x}{1 - \frac{1}{2}q_x}$$

and it can be shown that this holds under BA too so we can estimate q_x by solving

$$\begin{aligned} \frac{d}{v} &= \frac{\hat{q}_x}{1 - \frac{1}{2}\hat{q}_x} \\ \Rightarrow \hat{q}_x &= \frac{d}{v + \frac{1}{2}d}, \end{aligned}$$

the actuarial estimate.

Note that (29) is not consistent with the 2-state model since under that model we have

$$\begin{aligned} q_x &= 1 - \exp[-\mu_{x+\frac{1}{2}}] \\ \Rightarrow \mu_{x+\frac{1}{2}} &= -\log(1-q_x) \end{aligned}$$

and this only approximately equals $q_x/(1-\frac{1}{2}q_x)$, as Taylor's Theorem shows. We have, with $q_x = q$,

$$\begin{aligned} & -\log(1-q) - \frac{q}{(1-\frac{1}{2}q)} \\ \approx\ & (q + \tfrac{1}{2}q^2 + \tfrac{1}{3}q^3) - q(1 + \tfrac{1}{2}q + \tfrac{1}{4}q^2) \\ =\ & \frac{1}{12}q^3, \end{aligned}$$

and this is a very small number.

Comments:

(a) In (29) we call $v + \frac{1}{2}d$ the **initial exposed to risk** and denote it by E_x. Thus

$$E_x = E_x^c + \tfrac{1}{2}d \Rightarrow \hat{q}_x = \frac{d}{E_x}. \tag{31}$$

We can also use the Δ-method to show (dropping the x-suffix) that

$$\mathrm{Var}(\hat{q}) \approx \frac{1}{(1+\frac{1}{2}\mu)^4}\mathrm{Var}(\hat{\mu}) \approx \frac{q(1-q)}{E}, \tag{32}$$

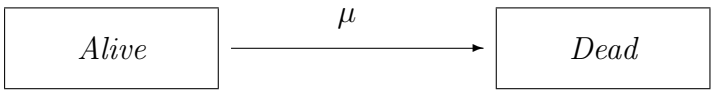
in line with the formula for the "binomial variance", pq/n.

(b) If we have census data, i.e., exact age of entry and exit (by death or censoring) the computation of an initial exposed to risk is a "pointless exercise". The 2-state model gives a simple solution.

(c) If we have more than one decrement then the binomial model becomes extremely difficult. Best stick with the multiple state framework.

4.9 The Poisson model

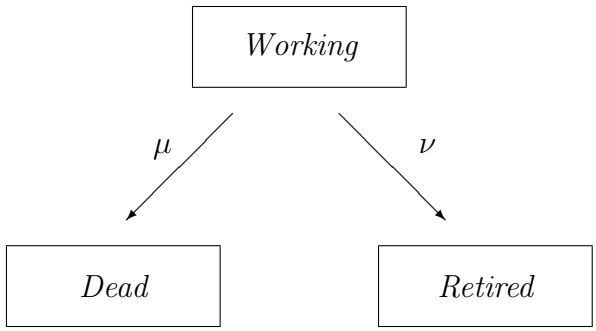
In our discussions of the Markov model we discussed the simple mortality model



which led to a likelihood of the form (see 3.9)

$$L(\mu) = e^{-\mu v}\mu^d.$$

For the retiral model



we obtained the likelihood (see 3.12)

$$L(\mu,\nu) = e^{-\mu v}\mu^d \times e^{-\nu v}\nu^r.$$

Alert readers will have recognised these as "Poisson Likelihoods". To be precise, if $X_1,\ldots,X_n$ are iid $\mathcal{P}(\lambda)$ then

$$L(\lambda) \propto \prod_{i=1}^{n} e^{-\lambda}\lambda^{x_i} = e^{-n\lambda}\lambda^{\sum x_i}$$

with the following correspondence

$$\begin{aligned} n, \text{ the number of observations} &\equiv v, \text{ the waiting time} \\ \textstyle\sum x_i, \text{ the total number of events} &\equiv d,\ r, \text{ the number of deaths, retirals.} \end{aligned}$$

This suggests that we could approach mortality with a Poisson model for the number of deaths in a group of lives, given the waiting time (or exposed to risk).

Let E_x^c be the total observed waiting time for n individuals all aged exactly x at the start of a mortality investigation. Thus $E_x^c = v$, the observed waiting time. Let us suppose that the period of the investigation is sufficiently short that we can assume that the force of mortality μ_x is constant, i.e. $\mu_x = \mu$. We have a Poisson model for mortality if D, the number of deaths, is s.t.

$$D \sim \mathcal{P}(\mu E_x^c) \tag{33}$$

$$\text{i.e. } \Pr(D=d) = \frac{e^{-\mu E_x^c}(\mu E_x^c)^d}{d!}$$

Comment: There is an obvious problem with this model: there is a non-zero probability of more than n deaths! In practice, unless n is small and q_x is large, this is not a serious problem.

Under (33) we easily find the MLE of μ:

$$\tilde{\mu} = \frac{D}{E_x^c}$$

with the properties

$$\begin{aligned} \mathrm{E}(\tilde{\mu}) &= \frac{\mu E_x^c}{E_x^c} = \mu, \text{ and} \\ \mathrm{Var}(\tilde{\mu}) &= \frac{\mu E_x^c}{(E_x^c)^2} = \frac{\mu}{E_x^c}. \end{aligned}$$

This is very like the two-state Markov model where we found $\mathrm{E}(\tilde{\mu}) = \mu$ and $\mathrm{Var}(\tilde{\mu}) = \mu/\mathrm{E}(V)$. In practice, $\mathrm{E}(V)$ is unknown so we replace it with $v = E_x^c$, and the two approaches give the same answer.

4.10 General comments

- If mortality rates are small and the exposure is large then there is little to choose between the three approaches (Markov, binomial, Poisson) for simple models.

- For a multi-state model the Markov approach gives a simple, efficient approach - not available for either the binomial or Poisson models.

Tutorial 4: Binomial and Poisson Models

1. This question looks at some consequences of the assumption that deaths in $[x, x+1]$ are uniformly distributed (the UDD assumption).

   (a) Prove that in general
   $$ {}_{b-a}q_{x+a} = 1 - \frac{{}_bp_x}{{}_ap_x}. $$
   Hint: Use $S_x(t) = S(x+t)/S(x)$.

   (b) Prove that in general
   $$ -\frac{\partial}{\partial t}{}_tp_x = {}_tp_x \cdot \mu_{x+t} $$
   Hint: Differentiate
   $$ {}_tp_x = \exp\left\{-\int_0^t \mu_{x+s}\,ds\right\} $$
   w.r.t. t.

   (c) Assume that the UDD assumption, ${}_tq_x = t \cdot q_x$, holds. Use (a) to show that
   $$ {}_{b-a}q_{x+a} = \frac{(b-a)q_x}{1 - a \cdot q_x}. $$
   Deduce an expression for ${}_tq_{x+a}$.

   (d) Assume that the UDD assumption, ${}_tq_x = t \cdot q_x$, holds. Use (b) to show that
   $$ \mu_{x+t} = \frac{q_x}{1 - t \cdot q_x}. $$
   Is μ_{x+t} an increasing or decreasing function of t?

2. If the force of mortality μ_{x+t} is constant for $0 \le t \le 1$ and equal to $\mu_{x+\frac{1}{2}}$, say, prove that
   $$ \mu_{x+\frac{1}{2}} = -\log(1 - q_x). $$
   English Life Table 15 (1990-92) for males gives $q_{70} = 0.03930$. Use the results in questions 1 and 2 to give a rough sketch of μ_{x+t}, for $0 \le t \le 1$, under the assumption (i) UDD (ii) constant μ_{x+t}. Comment.

   Repeat for females where $q_{70} = 0.02190$.

3. Over

3. In a mortality investigation, the following data have been recorded in respect of lives observed between the ages of x and $x+1$, where

   - a_i is the time after exact age x when the ith life came under observation;
   - b_i is the time after exact age x when the ith life will be censored;
   - $D_i = 1$ if the ith life dies before age $x + b_i$;
   - $D_i = 0$ if the ith life survives to age $x + b_i$.

   Suppose $x = 90$ and let $q_{90} = q$. The data are as follows:

Life i	a_i	b_i	d_i
1	0	1	0
2	0.25	1	0
3	0	1	1
4	0.25	0.75	1

   (a) Show that the log likelihood for q for the Binomial model for these data, assuming that deaths are uniformly distributed between integer ages, is

$$\ell(q) = 2\log q + 2\log(1-q) - 2\log(1-q/4).$$

   (b) Calculate $\hat{q}$, the Maximum Likelihood Estimator of $q = q_{90}$, together with an estimate of its standard error.

   Ans: $\hat{q} = 0.536$ with $SE = 0.249$.

4. You have collected the following data in respect of four lives aged between 99 and 100, where: a_i is the time (after age 99) at which observation begins, b_i is the time (after age 99) at which observation may be censored, and t_i is the time (after age 99) at which observation actually ended. You may assume that all uncensored observations were deaths.

Life i	a_i	b_i	t_i
1	0	0.7	0.7
2	0.3	0.7	0.7
3	0.3	0.6	0.5
4	0	1.0	0.5

   (a) Write down the likelihood for q_{99} for the Binomial model, assuming a uniform distribution of deaths between ages 99 and 100. Find the value of $\hat{q}_{99}$ which maximises this likelihood.

   (b) Write down the likelihood for μ, the constant force of mortality between ages 99 and 100, in terms of the two-state model. Hence find the maximum likelihood estimator of μ.

   (c) Compare the estimators of q_{99} obtained from (a) and (b).

5. Over

5. Based on HW, 2008.

   (a) A portfolio of n insured lives all exact age x is observed for 6 months and d_1 deaths are recorded. Let $q_x = q_1$. Show that the likelihood, $L_1(q_1)$, under the UDD assumption can be written

   $$L_1(q_1) \propto q_1^{d_1}(2-q_1)^{n-d_1}$$

   and find the maximum likelihood estimate, $\hat{q}_1$, of q_1 in terms of d_1 and n.

   (b) A second portfolio of n insured lives all exact age $x+\frac{1}{2}$ is observed for 6 months and d_2 deaths are recorded. Let $q_x = q_2$. Show that the likelihood, $L_2(q_2)$, for these lives under the UDD assumption can be written

   $$L_2(q_2) \propto \frac{q_2^{d_2}(1-q_2)^{n-d_2}}{(2-q_2)^n}$$

   and find the maximum likelihood estimate, $\hat{q}_2$, of q_2, in terms of d_2 and n.

   (c) Suppose now that $n = 100$, and that, by chance, $d_1 = d_2 = 20$.

      (i) Use the results in (a) and (b) to evaluate the maximum likelihood estimates of q_1 and q_2. *Ans:* $\hat{q}_1 = 2/5,\ \hat{q}_2 = 1/3$

      (ii) Define two R functions, one to evaluate the score function $U_1 = U_1(q)$ for the first set of lives and one to evaluate the score function $U_2 = U_2(q)$ for the second set of lives. Use these functions

         (α) to confirm your answers to (i), and

         (β) to estimate the standard errors of $\hat{q}_1$ and $\hat{q}_2$ by evaluating $U_1(q)$ and $U_2(q)$ at appropriate values of q. *Ans:* $SE(\hat{q}_1) = 0.080,\ SE(\hat{q}_2) = 0.056$

      (iii) An alternative approach is to work directly with the log likelihood function. Define two R functions, one to evaluate the log likelihood, $\ell_1 = \ell_1(q) = \log L_1(q)$ for the first set of lives and one to evaluate the log likelihood $\ell_2 = \ell_2(q) = \log L_2(q)$ for the second set of lives. Use these functions to plot ℓ_1 and ℓ_2 over suitably narrow ranges of values of q_1 and of q_2. Hence confirm your estimates found in (i).

      (iv) Recall that the second derivative of $y = f(x)$ wrt x is

      $$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

      for small h. Use this expression and your R functions defined in (iii) to estimate the standard errors of $\hat{q}_1$ and of $\hat{q}_2$, and confirm your answers found in (ii)(β).

   (d) Suppose instead that the force of mortality, $\mu_{x+t} = \mu$, is the same for both sets of lives and is constant over the full year x to $x+1$. Let $q = q_1 = q_2$. Suppose further that each of the $d_1 + d_2$ lives that dies does so midway through its period of exposure. Use the 2-state model of mortality to calculate the MLE of μ and deduce the MLE of q.

5 Methods of Graduation

5.1 Gompertz law

Benjamin Gompertz (1779-1865) was a pioneer in actuarial science. A biography can be found on the mathematical web site of the University of St Andrews (see page 25 for the web address). He is most famous for Gompertz's Law of Mortality (1825). Gompertz Law says that (St Andrews web site)

> "the mortality rate increases in a geometric progression. Hence, when death rates are plotted on a logarithmic scale, a straight line known as the Gompertz function is obtained. It is the most informative actuarial function for investigating the ageing process. The slope of the Gompertz function line indicates the rate of actuarial ageing. The differences in longevity between species are the result primarily of differences in the rate of ageing and are therefore expressed in differences in slope of the Gompertz function."

Example: Figure 3 illustrates the Gompertz Law for males ages 40 to 80 in the year 2000 for the CMI data set. Let d_i be the observed number of deaths at age i and E_i^c be the corresponding central exposed to risk. The *maximum likelihood estimate* of the force of mortality at age i is

$$\hat{\mu}_i = \frac{d_i}{E_i^c}$$

and the Gompertz Law says that $\log \mu_i$ is *linear* in age.

5.1.1 Gompertz law with normal errors

We can estimate the Gompertz line by fitting the linear regression of $\log \hat{\mu}_i$ on age, i.e., we take $\log \hat{\mu}_i$ as dependent variable and regress it on age. Suppose that `Obs` contains the values of $\log \hat{\mu}_i$ and the ages are in `AGE`, then the `lm( )` function in R does the work. We have

```
Gomp.fit <- lm(Obs ~ AGE)
names(Gomp.fit)
summary(Gomp.fit)
```

and the object `Gomp.fit` contains everything we want. The fitted Gompertz line is shown in Fig. 3.

We can improve the above solution by noting that $\text{Var}[\log(\hat{\mu}_i)]$ is not constant. Using the Δ-method (page 53) we have (with $d = d_i$ and $E = E_i^c$)

$$\begin{aligned}
\text{Var}\left[\log\left(\hat{\mu}\right)\right] &\approx \left(\frac{1}{\mu}\right)^2 \text{Var}(\hat{\mu}) \\
&= \left(\frac{1}{\mu}\right)^2 \text{Var}\left(\frac{d}{E}\right) \\
&= \left(\frac{1}{\mu}\right)^2 \left(\frac{1}{E^2}\right) E\mu \qquad \text{since } d \sim \mathcal{P}(E\mu) \\
&= \frac{1}{E\mu} \\
&\approx \frac{1}{d}.
\end{aligned}$$

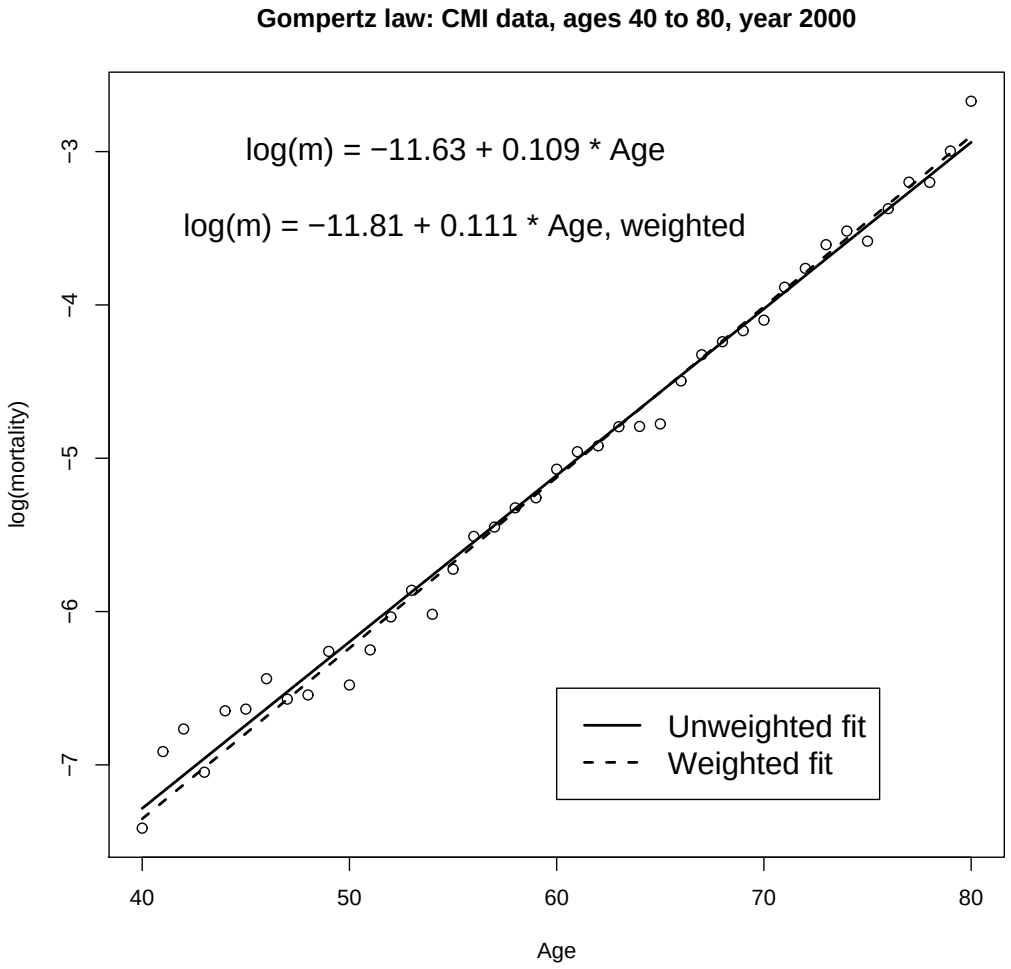


Figure 3: Log mortality for males in 2000 (CMI data).

This is an important result in its own right so we state

$$\text{Var}\left[\log\left(\hat{\mu}\right)\right] \approx \frac{1}{d}. \tag{34}$$

Notation: We denote by $\hat{\mu}_x = d_x/E_x^c$ the maximum likelihood estimate of μ_x. The smoothed or *graduated* value of μ_x will be denoted by $\dot{\mu}_x$.

We can perform weighted regression with the `weights` option to the `lm( )` function. With ordinary least squares we minimise

$$\text{RSS} = \sum (y_i - \dot{y}_i)^2$$

but with weighted least squares we minimise

$$\text{RSS(W)} = \sum w_i (y_i - \dot{y}_i)^2.$$

Thus the appropriate value of w_i is the reciprocal of the variance of y_i. In our example, this simply says that observations with more deaths get greater weight - which is entirely sensible. We modify our call of `lm( )` as follows:

```
Gomp.fit.w <- lm(Obs ~ AGE, weights = DTH)
```

where `DTH` contains the death counts. The weighted fit has been added to the plot. There is little difference between the two fits. See the R-code in the file `Gompertz.r` for details.

5.1.2 Gompertz law with Poisson errors

The above solution uses the Poisson assumption only to deduce the correct value of the weights in the linear regression. We can do much better since the Gompertz model is a simple example of a **generalised linear model** (GLM). We have for any age x

$$\begin{aligned} D_x &\sim \mathcal{P}(E_x^c \mu_x) \\ \Rightarrow \mathrm{E}(D_x) &= E_x^c \mu_x \\ \Rightarrow \log(\mathrm{E}(D_x)) &= \log(E_x^c) + \log \mu_x \\ &= \log(E_x^c) + a + bx \end{aligned}$$

by the Gompertz assumption. In conclusion, we have

$$\log(\mathrm{E}(D_x)) = \log(E_x^c) + a + bx. \tag{35}$$

We have a GLM with dependent variable D_x, Poisson error, log link, linear predictor $\log(E_x^c) + a + bx$ and offset $\log(E_x^c)$. We discuss each of these terms.

(a) The **dependent variable** is the left hand side of the regression, the 'y-value'. This is the number of deaths, D_x.

(b) The **error distribution** is the Poisson (unlike standard linear regression where the error distribution is the normal).

(c) The **linear predictor** is $\log(E_x^c) + a + bx$. This is the linear term on the right hand side of the regression.

(d) The **link function** is the log. The link function tells you how to go from the mean of the dependent variable (here $\mathrm{E}(D_x)$) to the explanatory variable (here age, x).

(e) The **offset** is the name given to a term in the linear predictor which appears with *a known coefficient*. Here, the offset is $\log(E_x^c)$; it is part of the linear predictor but its coefficient is *known* (and equal to one).

GLMs are fitted by maximum likelihood and the `glm( )` function in R enables easy fitting of the model. In the present case we have

```
Gomp.glm <- glm(DTH ~ offset(log(EXP)) + AGE, family = poisson)
summary(Gomp.glm)
```

where `EXP` contains the central exposures. The model is specified by the $\sim$ on the left of which we have the dependent variable, `DTH`. On the right of $\sim$ we have the linear predictor: the constant does not need specified since R assumes this will be fitted; the explanatory variable is `AGE`; the offset is included with the `offset` function. The error distribution is specified with the `family` option. With Poisson errors R assumes that the log link is used.

The model gives the fitted values of $\log \mu_i$ directly as

$$\log \dot{\mu}_i = \dot{a} + \dot{b} x_i$$

and this gives the Gompertz line as $\log \mu_x = -11.82 + 0.112x$, extremely close to the weighted fit we found above. We can add the fitted line to Figure 3 in R in a number of ways, all of which are instructive.

(a) Using the values of the *linear predictor* as follows:

```
names(Gomp.glm)
lines(AGE, Gomp.glm$lin - log(EXP))
```

where we must remember to subtract the offset from the linear predictor.

(b) Using the *estimated coefficients* to construct the values of the linear predictor:

```
lines(AGE, Gomp.glm$coef[1] + Gomp.glm$coef[2]*AGE)
```

(c) Using the *estimated number of deaths* to construct the values of $\log \dot{\mu}$:

```
lines(AGE, log(Gomp.glm$fit/EXP))
```

All three methods give the same result; see again the file `Gompertz.r`.

5.1.3 Gompertz law with binomial errors

The binomial model is an alternative to the Poisson model. Under the Poisson model we suppose that

$$D_x \sim \mathcal{P}(E_x^c \mu_x)$$

and that

$$\log \mathrm{E}(D_x) = \log(E_x^c) + a + bx.$$

Thus the Gompertz assumption with Poisson errors leads to a GLM with a log link. Under the binomial model we suppose that

$$D_x \sim \mathcal{B}(E_x, q_x)$$

where E_x is the *initial* exposed to risk and q_x is the probability a life with exact age x dies in the following year. The Gompertz assumption under the binomial model is

$$\operatorname{logit}(q_x) = \log \frac{q_x}{1 - q_x} = a + bx.$$

The maximum likelihood estimate of q_x is

$$\hat{q}_x = \frac{d_x}{E_x}$$

and Figure 4 shows a plot of $\operatorname{logit}(\hat{q}_x)$ against age. Once more we see the Gompertz line and the logit of q_x is linear in age.

We have a GLM with

(a) dependent variable $\hat{q}_x$, (**not** D_x as in the Poisson model),

(b) binomial error distribution,

(c) linear predictor $a + bx$,

(d) logit link function,

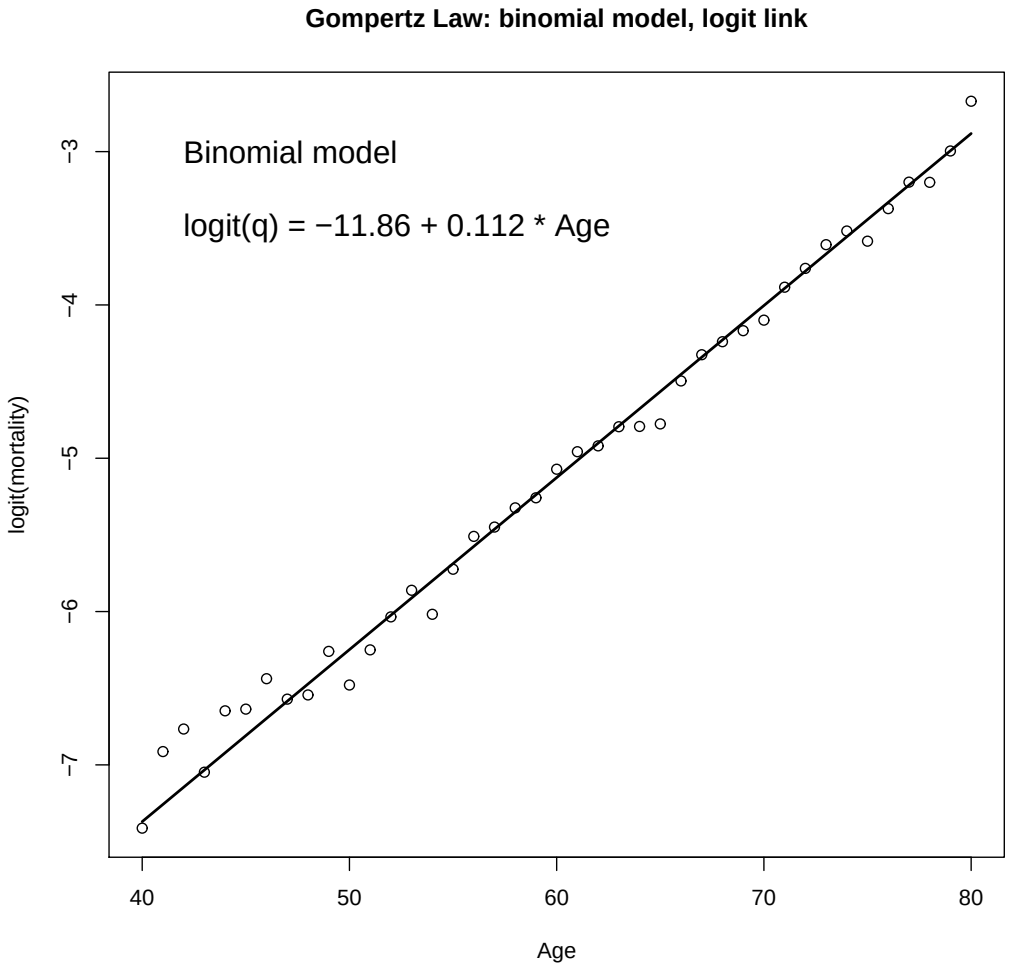


Figure 4: Logit mortality for males in 2000 (CMI data).

(e) number of trials, E_x.

Notice that since we use the probability of death as the dependent variable we must have some way of knowing the reliability of these proportions. In R we use the `offset( )` function with the Poisson model, but we use the number of trials with the binomial model.

We start by calculating the initial exposed to risk, E_x, and then use the `glm( )` function. Suppose `DTH` holds the death counts and `EXP` holds central exposures E_x^c. In brief,

```
E.init <- EXP + DTH/2              # Initial exposed to risk
Q.x <- DTH/E.init
Logit <- function(x) log(x/(1-x))   # Define logit function
Obs <- Logit(Q.x)
plot(AGE, Obs, xlab = "Age", ylab = "logit(mortality)",
    main = "Gompertz Law:  binomial model, logit link")
Gomp.bin <- glm(Q.x ~ AGE, weights = E.init, family = binomial)
lines(AGE, Gomp.bin$lin, lwd = 2)
```

As with the Poisson case, the model is specified by the $\sim$, on the left of which we have the dependent variable, `Q.x`, the maximum likelihood estimate of q_x. On the right of the $\sim$ we have the linear predictor: again, the constant does not need specified since R assumes this will be fitted; the explanatory variable is `AGE`. The 'Number of trials' is included with the

`weights` option. The error distribution is specified with the `family` option. With binomial errors R assumes the logit link will be used.

The model gives the fitted values of $\text{logit}\, q_i$ directly as

$$\text{logit}\, \dot{q}_i = \dot{a} + \dot{b}x_i$$

and this gives the Gompertz line as $\text{logit}\, q_x = -11.86 + 0.112x$, very close to the result with Poisson errors. We can add the fitted line to Figure 4 in R in the same ways as we did with Poisson errors. Poisson and binomials models are specified differently so note the differences in detail between the two cases.

(a) Using the values of the *linear predictor* as follows:

```
names(Gomp.bin)
lines(AGE, Gomp.bin$lin)
```

(b) Using the *estimated coefficients* to construct the values of the linear predictor:

```
lines(AGE, Gomp.bin$coef[1] + Gomp.bin$coef[2]*AGE)
```

(c) Using the *estimated probabilities of death* to construct the values of $\text{logit}\, \dot{q}_x$:

```
lines(AGE, Logit(Gomp.bin$fit))
```

All three methods give the same result; again, `Gompertz.r` gives full details.

5.1.4 Gompertz style models in time with Poisson errors

The Gompertz Law tells us how to model the force of mortality, μ_x, in terms of age, x. The GLM formulation for μ_x in terms of x (age) for a given year can also be used to give us a model for μ_t in terms of t (year) for a given age. Figure 5 gives a plot of the observed force of mortality by year for age 65. A Gompertz style model is fitted with the same code as above

```
Gomp.yr.glm <- glm(DTH ~ offset(log(EXP)) + Year, family = poisson)
```

where `DTH` and `EXP` contain the deaths and exposures for a particular age, say 65, and `Year` contains the years 1950 to 2005.

The fitted line is shown and clearly gives a very poor fit. Figure 5 suggests that we could try a quadratic model in year. The values of `Year^2` are very large and this will distort the fitted regression coefficients. We add a quadratic term to the regression in terms of `Yr2 <- (Year - mean(Year))^2` with

```
Gomp.yr2 <- glm(DTH ~ offset(log(EXP)) + Year + Yr2, family = poisson)
```

and the fitted quadratic has been added to plot.

5.2 Residuals

There are three cases: (a) the model for $\log \mu_x$ with normal errors (section 5.1.1), (b) the model for the death counts, d_x, with Poisson errors (section 5.1.2), and (c) the model for the probability of death, q_x, with binomial errors (section 5.1.3).

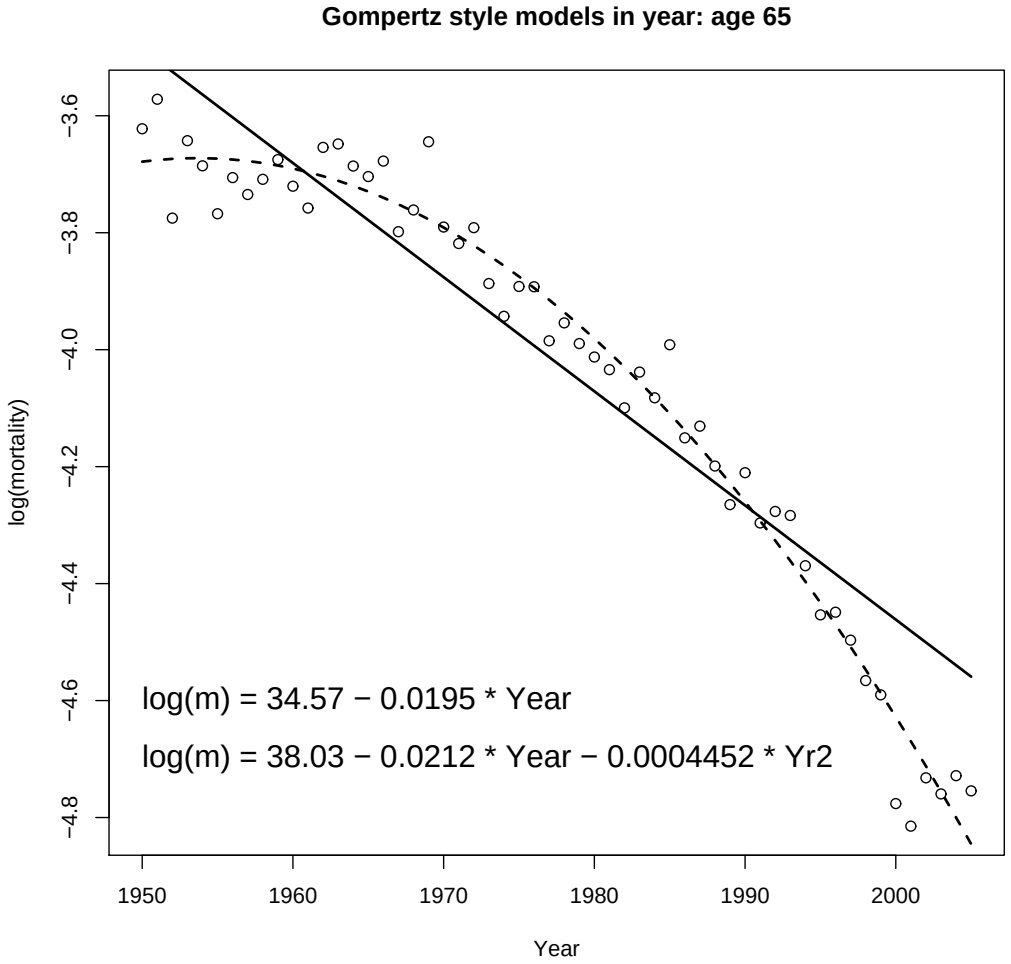


Figure 5: Log mortality for males age 65 (CMI data) (`Yr2 = (Year - mean(Year))^2.`)

5.2.1 Residuals with normal errors

We are familiar with the use of residuals as a tool to investigate the goodness-of-fit of a model. We first estimate the error variance σ^2 in the normal model by

$$s^2 = \frac{\sum(y_i - \dot{y}_i)^2}{n - \mathrm{df}} \tag{36}$$

where n is the number of observations and df is the degrees of freedom of the fitted model (in the Gompertz model this is 2). In R we can either pick out the estimate of σ (not σ^2) from the output of `summary`, as in

```
summary(Gomp.fit)
Sig <- 0.1293
```

or compute (36) directly with

```
Sig <- sqrt(sum((Obs - Gomp.fit$fit)^2)/Gomp.fit$df.res)
```

The standardised residual is then defined, using an obvious general notation, as

$$z_i = \frac{y_i - \dot{y}_i}{s}. \tag{37}$$

The `standardised residuals` are computed with

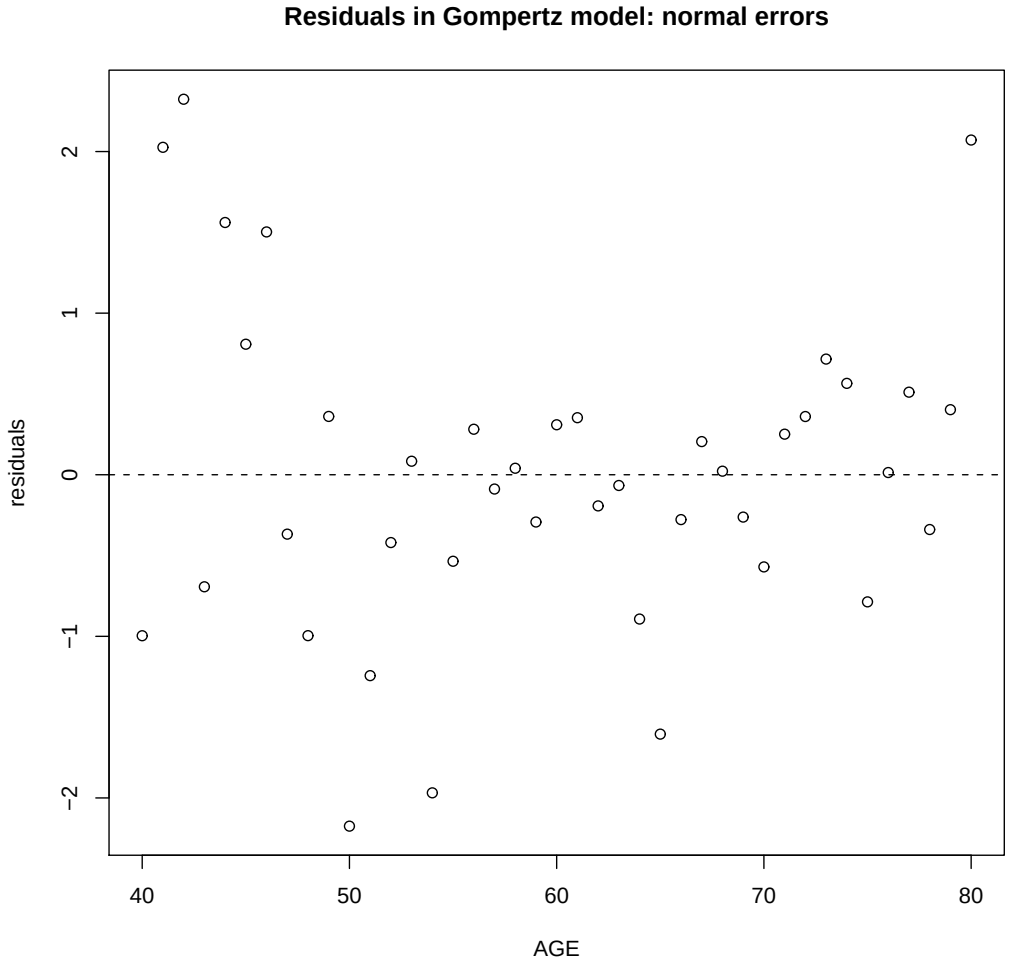


Figure 6: Standardised residuals in normal Gompertz model

```
Z.values <- (Obs - Gomp.fit$fit)/Sig
plot(AGE, Z.values, xlab = "Age", ylab = "Residuals",
    main = "Standardised residuals in Gompertz model")
```

and Figure 6 gives a plot of these residuals against age. There are no noticeably large values nor any pattern to these residuals so the Gompertz model provides a reasonable model for our data.

The raw residuals, $y_i - \dot{y}_i$, are also part of the output of `lm( )` and you can use the fitted object from the `lm( )` call to find them in two ways.

```
Raw.z <- Gomp.fit$res
Raw.z <- resid(Gomp.fit)
```

5.2.2 Residuals with Poisson errors

Residuals in Poisson models are based on the same principles as normal models but there are two differences.

(a) In the normal model the variance is determined by the scale parameter, σ^2; this parameter is quite separate from the mean, μ. In the Poisson model the variance is equal to the mean.

(b) The normal model has symmetric errors whereas the Poisson model has skew (long-tailed) errors. This means that a raw residual, r, $r > 0$, above the mean is less serious that a raw residual, $-r$, with the same absolute value, below the mean. Informally, this is because there is more probability available above the mean than is available the same distance below the mean. For example, we can compute lower and upper tails in the Poisson distribution with `ppois( )` as in

```
c(ppois(0.5, 4), 1 - ppois(7.5, 4))
[1] 0.01831564 0.05113362
```

which shows that $\Pr(X - \mathrm{E}(X) < -3.5) < \Pr(X - \mathrm{E}(X) > 3.5)$ where $X \sim \mathcal{P}(4)$. In a symmetric distribution these probabilities are the same.

We describe two kinds of residuals:

(a) Pearson residuals; these ignore the lack of symmetry in the error distribution.

(b) deviance residuals; these take account of the lack of symmetry (and are to be preferred).

Let d_i and $\dot{d}_i$ be the observed and fitted (graduated) deaths at the ith age. Then, corresponding to (37) we have the **Pearson residuals**

$$z_i = \frac{d_i - \dot{d}_i}{\sqrt{\mathrm{Var}(d_i)}} \approx \frac{d_i - \dot{d}_i}{\sqrt{\dot{d}_i}} \tag{38}$$

since $\mathrm{Var}(d_i) = E_i^c \mu_i \approx \dot{d}_i$.

The following code (which you can find in `Gompertz.r`) shows how to obtain the Pearson and deviance residuals. Suppose that `glm( )` has been used to fit the Gompertz model with Poisson errors and the results are in the object `Gomp.glm`. The code also shows how to calculate the Pearson residuals directly from (38).

```
cbind(resid(Gomp.glm, type = "pear"),
    (DTH - Gomp.glm$fit)/sqrt(Gomp.glm$fit))
plot(AGE, resid(Gomp.glm, type = "pear"), ylab = "residuals",
    xlab = "Age", main = "Residuals in Gompertz model")
points(AGE, resid(Gomp.glm), pch = 2)
legend(50, 4, legend = c("Pearson", "Deviance"), pch = c(1,2))
```

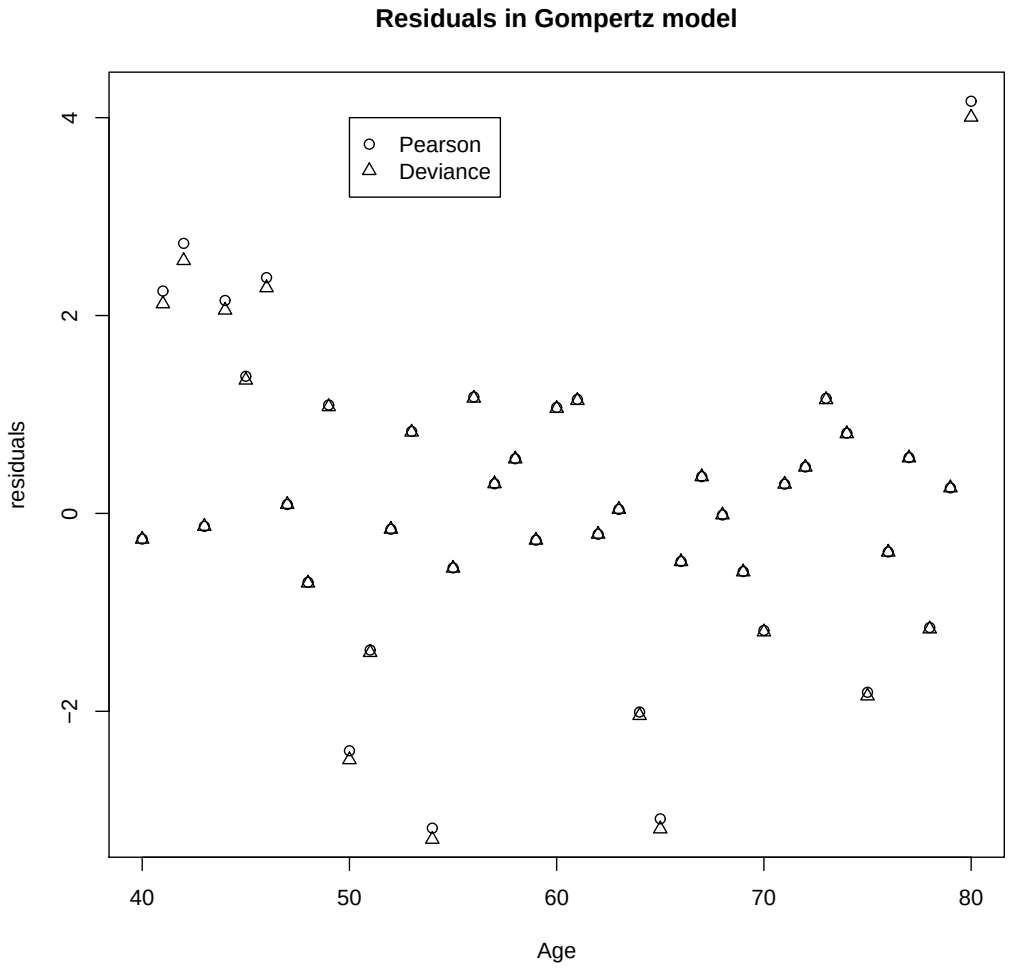


Figure 7: Pearson and deviance residuals in Poisson Gompertz model

Figure 7 shows the resulting plot. We make three remarks

(a) There is no pattern in the residuals so the Gompertz model for the mean is acceptable.

(b) There are now a few large residuals so it looks as though there is more variation in the data than is allowed for with the Poisson distribution. This is an example of *overdispersion* where the data show more variation than the error distribution allows. One possible explanation is that our data are counts of claims rather than actual deaths. Since some policyholders will hold more that one policy a single death can result in more than one claim. Such a death will inflate the variance in the data. Ideally, we should combine such policies into a single policy but until very recently this has not been possible. Certainly there are duplicate policies present in the CMI data used here. We discuss the *problem of duplicates* further in section 5.5.

(c) The deviance residuals can be accessed with `resid(Gomp.glm)`. These are the default residuals so do not need specified with the `type` option. The deviance residuals have been added to Fig. 7. Notice how the positive residuals have been moved towards the mean while the negative residuals have been moved away from the mean; this is in line with our discussion on symmetric and non-symmetric errors above.

5.2.3 Residuals with binomial errors

The treatment of residuals in the binomial model follows that of the Poisson model very closely. We start by showing that (38) holds (approximately) in the binomial case. The Pearson residual is

$$z_x = \frac{\hat{q}_x - \dot{q}_x}{\sqrt{\mathrm{Var}(\hat{q}_x)}}. \tag{39}$$

Now

$$\hat{q}_x - \dot{q}_x = \frac{d_x}{E_x} - \frac{\dot{d}_x}{\mathrm{E}_x} = \frac{d_x - \dot{d}_x}{E_x}$$

and

$$\mathrm{Var}(\hat{q}_x) \approx \frac{\dot{q}_x(1-\dot{q}_x)}{E_x} \approx \frac{\dot{q}_x}{E_x} = \frac{\dot{d}_x}{E_x^2}.$$

Putting this together we find that (39) reduces to

$$z_x = \frac{(d_x - \dot{d}_x)/E_x}{\sqrt{\dot{d}_x/E_x^2}} = \frac{(d_x - \dot{d}_x)}{\sqrt{\dot{d}_x}}$$

which is the same as (38). As in the Poisson case, we can access the Pearson and deviance residuals as follows

```
resid(Gomp.bin)     # deviance residuals
resid(Gomp.bin, type = "pear")     # Pearson residuals
```

and these can be plotted as required (the object `Gomp.bin` holds the output from a call of `glm( )` with binomial errors).

5.3 Graduation by reference to a standard table

5.3.1 What is graduation by a standard table?

Suppose we have two mortality experiences. The first experience has been graduated, probably with large amounts of data, and is an established graduation, a so-called **standard table**. We can use this graduation as a reference to graduate a second experience (provided of course we have checked that this is a sensible thing to do). There are two situations when this method might be appropriate: (a) a mortality study with relatively little data, (b) a mortality study where a standard table graduation provides a simple and simple-to-use graduation.

We suppose we have a standard table graduation of the force of mortality, μ, in the table $\{\mu_x^s\}$ and raw (ungraduated) rates $\{\hat{\mu}_x\}$ from our second experience. We attempt to graduate the $\hat{\mu}_x$ by finding a simple formula relating $\hat{\mu}_x$ to μ_x^s. Similar remarks apply to rates of mortality $\hat{q}_x$ and q_x^s. Possible functions which might be considered include the following

1. $\dot{\mu}_x = a\mu_x^s$
2. $\dot{\mu}_x = a\mu_x^s + b$
3. $\dot{\mu}_x = (ax+b)\mu_x^s$
4. $\dot{\mu}_x = \mu_{x+k}^s$, known as **age rating** - commonly used to adjust for mortality improvements with $k = \pm 1, \pm 2$ etc.

5. $\dot{\mu}_x = a\mu_x^{s_1} + b\mu_x^{s_2}$, a blend of two standard tables.

The first step is the choice of an **appropriate standard table**. This is guided by the type of lives under consideration.

1. For assured lives we might refer to A67/70, or a more recent table for this class of business.
2. For annuitants we might refer to a(90), or a similar experience.
3. For pensioners we might refer to PA(90), or a similar experience.
4. For a population based study we might choose a National Life Table (available online from the ONS).

Clearly where tables are published for both sexes, reference is made to males or females as appropriate. In summary we must consider:

1. The class of lives being graduated (assurances, sex, geography, etc.)
2. The period of time underlying the standard table.
3. The range of ages covered.
4. Unusual features of the standard table.
5. The method used to graduate the standard table.
6. Selection characteristics of the standard table (e.g. is the magnitude of the selection effect likely to be similar for our data).

After choosing a standard table, we must then determine a suitable relationship between $\hat{\mu}_x$ and μ_x^s. Various plots are possible.

1. $\hat{\mu}_x$ against μ_x^s to see if a linear relationship is suggested.
2. $\hat{\mu}_x/\mu_x^s$ against x to explore other formulae linking our data to the standard table.
3. Transformations of the data, e.g. $\log\left(\dfrac{1-\hat{q}_x}{1-q_x^s}\right)$ against x.

Note If $\hat{\mu}_x = \mu_x^s + c$ where c is a constant then all values of $\hat{p}_x/p_x^s$ will be constant. (Use $p_x^s = \exp(-\mu_{x+1/2}^s)$ and $\hat{p}_x = \exp(-\hat{\mu}_{x+1/2})$, etc.)

Advantages of Standard Table Graduation

- Information in a more extensive graduation of similar data is used.
- Can be used with little data, particularly at extreme ages.
- Since the standard table will be smooth, the graduated rates will also be smooth (as long as a simple formula has been used).
- Can incorporate prior expectations of the form of the curve.

Disadvantages of Standard Table Graduation

- It is not always obvious which standard table to use.
- Adherence to data can be a problem.
- Features in the standard table will be exhibited in the graduated rates.
- Not appropriate for producing new standard tables based on large amounts of data.

5.3.2 Methods of graduation by a standard table

We discuss two methods of graduation by a standard table (a) weighted least squares (b) GLMs.

Weighted least squares

Suppose we choose to graduate the data $\{\hat{\mu}_x\}$ with the standard table $\{\mu^s_x\}$. We have satisfied ourselves that a graduation of the form $\hat{\mu}_x = a + b\mu^s_x$ is suitable. We choose the parameters which minimise the weighted sum of squares

$$\sum_x w_x(\hat{\mu}_x - a - b\mu^s_x)^2.$$

The weights should be the reciprocal of the variance of $\hat{\mu}_x$, where $\text{Var}(\hat{\mu}_x) \approx \hat{\mu}_x/E^c_x$. Thus, we have a weighted linear model with dependent variable $\hat{\mu}_x$, explanatory variable μ^s_x and weight $E^c_x/\hat{\mu}_x$. This can be fitted with the `lm( )` function as in section 5.1.1.

Generalised linear model

Weighted least squares is a general method and can always be used. Sometimes we can write the graduation in the form of a GLM. Suppose we change the above example and assume that $\log(\hat{\mu}_x) = a + b\log(\mu^s_x)$. We can express this as a GLM as follows.

$$\begin{aligned} D_x &\sim \mathcal{P}(E^c_x\mu_x) \\ \Rightarrow \log[\text{E}(D_x)] &= \log E^c_x + \log\mu_x \\ &= \log E^c_x + a + b\log(\mu^s_x). \end{aligned}$$

We have a GLM with dependent variable D_x, explanatory variable $\log(\mu^s_x)$, Poisson errors, log link and offset $\log E^c_x$.

Example

We illustrate the ideas of graduation with reference to a standard table with an example. Suppose we use data from the year 2000 in the CMI data set to construct our standard table. We then use this graduation as the standard table in a graduation of the 2005 data. Panel A in Figure 8 is a plot of $\log\hat{\mu}_x$ against x for the year 2000 with the fitted Gompertz line (obtained with the GLM Poisson model). Suppose we save the standard table (on the log scale) as $\log\mu^s_x$; on the log scale, the standard table is the Gompertz line. Panel B in Figure 8 is a plot of $\log\hat{\mu}_x$ (data from 2005) against $\log\mu^s_x$ (standard table from 2000). This panel suggests a linear regression of $\log\hat{\mu}_x$ on $\log\mu^s_x$.

Suppose that the deaths and exposures from 2005 are in `Dth` and `Exp` respectively, and that the values of $\log\mu^s_x$ are in `log.mu.s`. We take a normal model with dependent variable

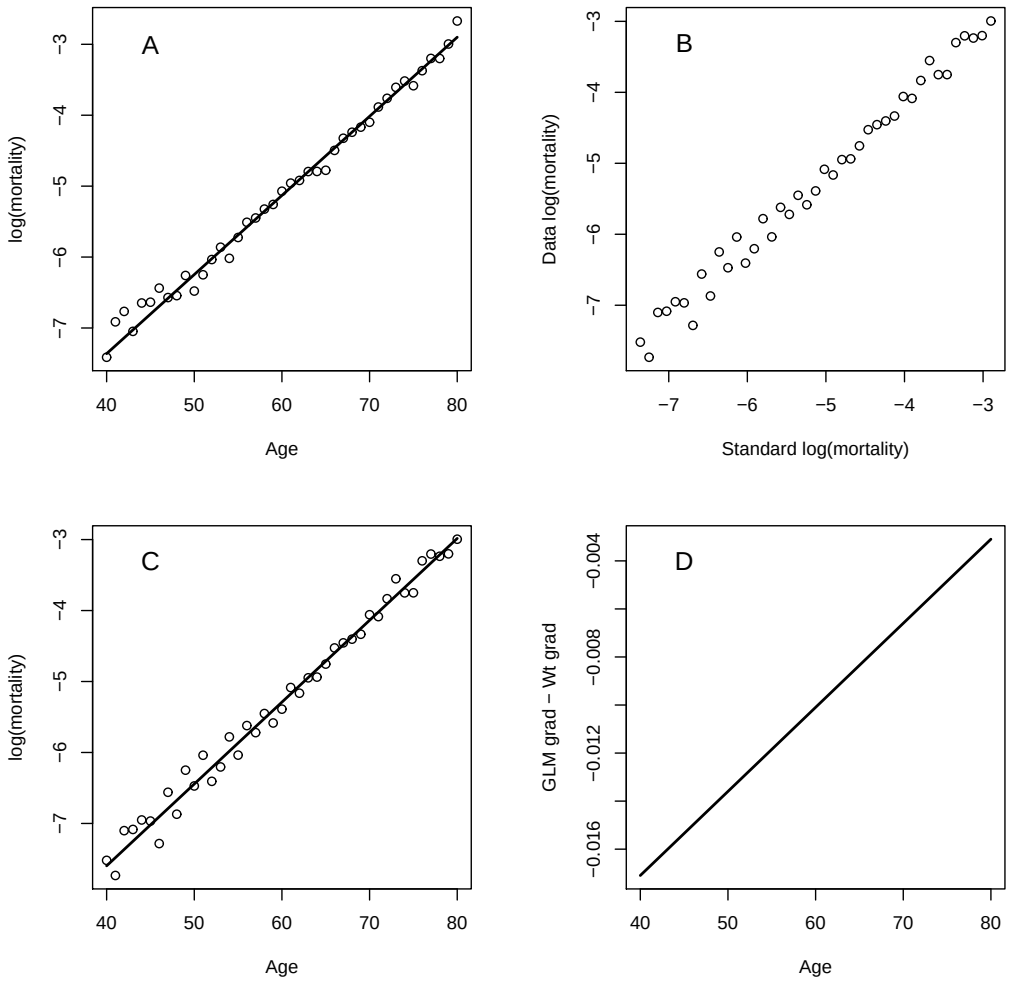


Figure 8: Panel A: 2000 data and Gompertz line; Panel B: $\log \mu_x$ from 2005 against standard table (Gompertz line from 2000); Panel C: 2005 data and standard table graduation (GLM); Panel D: difference in GLM and weighted graduation of 2005 data

$\log \hat{\mu}_x$, explanatory variable `log.mu.s` and weight `DTH` (from equation (34)). The R code for the weighted least squares fit is

```
Logm <- log(Dth/Exp)
Grad.wt <- lm(Logm ~ log.mu.s, weights = Dth)
```

For the GLM we have

$$\begin{aligned} D_x &\sim \mathcal{P}(E_x^c \mu_x) \\ \Rightarrow \log[\mathrm{E}(D_x)] &= \log E_x^c + \log \mu_x \\ &= \log E_x^c + a + b \log(\mu_x^s). \end{aligned}$$

We have a GLM with dependent variable D_x, explanatory variable $\log(\mu_x^s)$, Poisson errors, log link and offset $\log E_x^c$. The R code is

```
Grad.glm <- glm(Dth ~ log.mu.s + offset(log(Exp)), family = poisson)
```

Panel C shows the result of the graduation with the GLM fit while Panel D shows that the difference between the GLM and weighted fits is very small (as we would expect). See the

R file `Standard.r` for full details of the code.

5.4 Graduation by Mathematical Formula

In this section we review some of the graduation formulae that have been used by actuaries. Graduation by parametric formulae is the preferred method for large data sets but finding a formula which gives adequate adherence to data can be difficult. In some cases a single formula can be used for all ages but in other cases a blend of functions is used (which of course introduces additional problems). Other related graduation methods include use of spline functions. Splines join together low order polynomials to achieve a smooth continuous function across the age range; this method has been used for certain published life tables (e.g. ELT13) and has been used more widely by actuaries in recent years.

Some parametric formulae

- **Gompertz:** $\mu_x = Bc^x$ (1825)
- **Makeham:** $\mu_x = A + Bc^x$ (1860)
- **Perks:** $\mu_x = \dfrac{A + Bc^x}{Ec^{-x} + 1 + Dc^x}$ (1932) and $\mu_x = \dfrac{A + Bc^x}{1 + Dc^x}$ (1932)

The table A49/52 was graduated using

$$q_x = A + \frac{Bc^x}{Ec^{-2x} + 1 + Dc^x}.$$

The table A67/70 was graduated using

$$\frac{q_x}{p_x} = A + Hx + Bc^x.$$

More recently the Continuous Mortality Investigation Committee has used other mathematical formulae such as

$$\text{logit}(q_x) = \log\left(\frac{q_x}{1 - q_x}\right) = \beta_0 + \beta_1 x + \beta_2 x^2 + \ldots,$$

typically linear or quadratic. We recognise this as a standard binomial GLM with logit link.

Note that the transformed function $\text{logit}\,(q_x)$ is often easier to model than q_x itself as the range $[0, 1]$ is transformed to $(-\infty, \infty)$.

The main problem with a single formula is capturing all the features of a typical mortality experience. The Australian National 1960/62 table was graduated using a single formula:

$$\frac{q_x}{p_x} = A^{(x+b)^c} + De^{-E(\log(x) - \log(F))^2} + GH^x$$

where the first term corresponded to Infant Mortality, the second term to the Accident Hump and the third term to General Aging. This gave a good result in practice.

The Gompertz-Makeham Family

Let r and s be integers ≥ 0 which are not both zero, and let

$$\boldsymbol{\alpha}' = (\alpha_1, \alpha_2, \ldots, \alpha_{r+s})$$

be a vector of parameters. Define

$$GM^{r,s}_{\boldsymbol{\alpha}}(x) = \sum_{i=1}^{r} \alpha_i x^{i-1} + \exp\left\{\sum_{i=1}^{s} \alpha_{r+i} x^{i-1}\right\}$$

where GM stands for Gompertz-Makeham.

Comments

- Gompertz's formula is $GM^{0,2}_{\boldsymbol{\alpha}}(x)$.
- Makeham's formula is $GM^{1,2}_{\boldsymbol{\alpha}}(x)$.
- Observe that
$$\frac{q_x}{p_x} = GM^{r,s}_{\boldsymbol{\alpha}}(x) \Leftrightarrow q_x = \frac{GM^{r,s}_{\boldsymbol{\alpha}}(x)}{1 + GM^{r,s}_{\boldsymbol{\alpha}}(x)}$$
and so the equation used for graduating A67/70
$$\frac{q_x}{p_x} = A + Hx + Bc^x \Leftrightarrow q_x = \frac{GM^{2,2}_{\boldsymbol{\alpha}}(x)}{1 + GM^{2,2}_{\boldsymbol{\alpha}}(x)}$$
- We also have
$$\log\left(\frac{q_x}{p_x}\right) = \text{polynomial in } x \text{ of degree } n-1 \Leftrightarrow q_x = \frac{GM^{0,n}_{\boldsymbol{\alpha}}(x)}{1 + GM^{0,n}_{\boldsymbol{\alpha}}(x)}$$
- The Logit Gompertz-Makeham formula of type (r, s) is then defined more generally as
$$LGM^{r,s}_{\boldsymbol{\alpha}}(x) = \frac{GM^{r,s}_{\boldsymbol{\alpha}}(x)}{1 + GM^{r,s}_{\boldsymbol{\alpha}}(x)}$$

Model Selection

Assume we are comparing two models M_1 and M_2 where M_1 has n parameters and M_2 has an **additional** k parameters. Thus M_1 is a **sub-model** of M_2; we also say that M_1 is **nested** in M_2. The larger model will naturally adhere better to the data but the improvement in fit may not be statistically significant. We approach the selection of the most efficient number of parameters in a similar way to most multiple regression variable selection methods (see statistics courses). We compute the likelihood ratio test statistic

$$\Lambda = L_0/L_1 = L(n)/L(n+k)$$

where $L(n)$ and $L(n+k)$ are the maximised values of the likelihood with n and $n+k$ parameters respectively. We refer $-2\log\Lambda$ to the χ^2_k distribution.

5.5 The problem of duplicates

The investigations of life office mortality carried out in the UK by the CMI (Continuous Mortality Investigation) have one particular feature that affects the statistical properties of the resulting estimates: they are **based on policies and not lives**. That is, instead of observing persons and recording:

1. E_x^c = Number of person-years observed
2. d_x = Number of deaths

the CMI observes policies, and records

1. E_x^c = Number of policy-years observed
2. d_x = Number of policies becoming claims by death.

The reason for this is simple: life-office record keeping is based on policies. It can be very difficult to establish when two policies are, in fact, owned by the same person, especially if many life offices pool their data (as in the CMI investigations). The upshot is that we cannot be sure that we are observing **independent** claims, since it is quite possible that one death will result in more than one claim. This is called the problem of **duplicate policies**.

A Toy Example

Consider a tiny portfolio consisting of two policies. There are two possibilities:

(a) The policies are held by two independent lives.

(b) The policies are held by the same life.

Let C be the number of claims arising from the portfolio and let q be the probability that either life claims in (a), and the probability the sole life claims in (b). Then it is easy to see that $\mathrm{E}(C) = 2q$ in both cases. However, $\mathrm{Var}(C) = 2q(1-q)$ in (a) but $\mathrm{Var}(C) = 4q(1-q)$ in (b). (This follows since $C \sim \mathcal{B}(2, q)$ in case (a) but $C = 2X$ where $X \sim \mathcal{B}(1, q)$ in case (b).)

Conclusion: The non-independence of claims results in an **increase in the variability** of claim numbers.

A General Approach

We analyse the effect of duplicates in the simplest possible setting. Suppose we observe N lives from age x to $x+1$, with no censoring, new entrants, or other complications. The lives are statistically independent in the usual way, but a proportion π_i, $i = 1, 2, \ldots$ of the N lives each own i policies. Thus the total number of policies observed is $N \sum_i i\pi_i$. Suppose each life dies during the year with probability q_x.

The problem is this: we observe $N \sum_i i\pi_i$ policies but the proportions π_i are unknown. If we assume that the number of claims $C \sim \mathcal{B}(N \sum_i i\pi_i, q_x)$ then we will be in error because of the dependencies.

Let D_i be the number of **deaths** among the $N\pi_i$ lives with i policies, and let C_i be the number of **claims** arising from these same lives. Then

$$D_i \sim \mathcal{B}(N\pi_i, q_x)$$

because of the independence of deaths. Thus, we have

$$\begin{aligned} \mathrm{E}(C) &= \mathrm{E}\left(\sum_i C_i\right) = \mathrm{E}\left(\sum_i iD_i\right) = \sum_i i\mathrm{E}(D_i) = Nq_x \sum_i i\pi_i \\ \mathrm{Var}(C) &= \mathrm{Var}\left(\sum_i C_i\right) = \mathrm{Var}\left(\sum_i iD_i\right) = \sum_i i^2\mathrm{Var}(D_i) \\ &= Nq_x(1-q_x)\sum_i i^2\pi_i. \end{aligned}$$

On the other hand, if we observed $N\sum_i i\pi_i$ **independent** lives (or policies) we would have

$$\begin{aligned} \mathrm{E}(C) &= Nq_x \sum_i i\pi_i \\ \mathrm{Var}(C) &= Nq_x(1-q_x)\sum_i i\pi_i, \end{aligned}$$

so the effect of duplicate policies is to increase the variance of the number of claims by the ratio

$$r = \frac{\sum_i i^2\pi_i}{\sum_i i\pi_i}.$$

This is called the **variance ratio**. Notice that the proportions π_i will depend on age, so we have a different variance ratio r_x for each age x.

The Toy Example Again

We have one life with two policies, so $\pi_2 = 1$. Hence,

$$r = \frac{\sum_i i^2\pi_i}{\sum_i i\pi_i} = \frac{2^2}{2} = 2,$$

as found above.

Conclusion

If we knew the r_x we could allow for them in the graduations. The CMI has occasionally conducted special investigations to derive estimates of the r_x which are suitable for use.

Tutorial 5: Graduation I

1. Extract the data for females ages 40 to 80 for the year 2005 from the England and Wales (E&W) data set. Use R to fit Gompertz type models; i.e., for μ_x assume that $\log \mu_x$ is linear in age x while for q_x assume that $\text{logit}\, q_x$ is linear in x.

   (a) The model for μ_x using weighted least squares to estimate the parameters.

   (b) The model for μ_x using a GLM with Poisson errors to estimate the parameters.

   (c) The model for q_x using weighted least squares to estimate the parameters. (Hint: You need $\text{Var}[\text{logit}(\hat{q}_x)] \approx 1/d_x$).

   (d) The model for q_x using a GLM with binomial errors to estimate the parameters.

   (e) Compare the GLM graduations of μ_x and q_x.

   (f) Obtain the standardised residuals, z_x, in each of the four graduations and plot the z_x against x. Comment informally on what these plots tell you.

2. (a) Extract the male data for age 65 for all years from the E&W data set. Plot the maximum likelihood estimates of $\log \mu_t$ against year, t. Consider Gompertz style graduations of μ_t with

$$\log(\mu_t) = a_0 + a_1 t + a_2 t^2 \ldots + a_p t^p$$

   for $p = 1, 2, 3, 4$. Use the `glm( )` function and Poisson errors to estimate the regression coefficients. Compare the various fits with appropriate plots.

   Hint: You will do well to use powers of $t - \bar{t}$ in the calls of `glm( )`*; see section 5.1.4.*

   (b) Repeat (a) for age 20.

   (c) Note for interest the amazing changes in mortality that have taken place from 1841 to 2005.

6 Graduation and Statistical Tests

6.1 Introduction

In the previous chapter we discussed methods of graduation. Our discussion centred on the fitting of Gompertz style models although we referred to some other functions that can also be used. Once we have our graduated rates we have the following obvious question: does our graduation provide a reasonable description of our data? We present a number of statistical tests which can be used to examine this question. We conclude the chapter with a worked example.

Our tests will be based on the Pearson residuals as discussed in the previous chapter in sections 5.2.2 and 5.2.3. There, we derived basic formulae for the Pearson residuals in terms of the observed and graduated number of deaths, d_i and $\dot{d}_i$, respectively.

1. Poisson model
$$z_i = \frac{d_i - \dot{d}_i}{\sqrt{\dot{d}_i}}. \tag{40}$$

2. Binomial model
$$z_i = \frac{\hat{q}_i - \dot{q}_i}{\sqrt{\dot{q}_i(1-\dot{q}_i)/E_x}} \approx \frac{d_i - \dot{d}_i}{\sqrt{\dot{d}_i}}. \tag{41}$$

The following tests are all based round the idea that the z_i can be regarded as independent observations from the $\mathcal{N}(0,1)$ distribution. There are two important remarks about this assumption (a) the approximation works well provided n, the number of data points, is not too small, and (b) an adjustment is sometimes required to take account of the number of estimated parameters in the fitted model.
We have provided a set of R-functions to conduct these tests and these can be found in the file `Test_GoF.r`. A guide to the use of these functions is also provided. Both the functions and the guide can be found in *R Scripts* in the section *Testing graduations*. You must also be able to conduct the tests by hand.

6.2 Statistical tests

6.2.1 χ^2 test

This is a test for overall goodness of fit. There are two cases.

1. We can test whether our data are consistent with a **standard table** with
$$X = \sum \frac{(d_x - \dot{d}_x)^2}{\dot{d}_x} = \sum z_x^2$$
where d_x are the observed numbers of deaths and $\dot{d}_x$ are the expected number of deaths under the tested model. The higher the value of X the greater are the differences between our observed data and our graduated values, i.e., we have a **one-sided** test whose null distribution is approximately:
$$X = \sum_x Z_x^2 \sim \chi^2_N$$
where N is the number of age groups.

2. However, if a graduation has made use of the d_x data in the construction of $\dot{q}_x$ or $\dot{\mu}_x$ (as is usually the case) then the χ^2 statistic will have fewer than N degrees of freedom. The reduction in the degrees of freedom will depend on the number of estimated parameters in the fitted model. Again, we have a **one-sided** test whose null distribution is approximately:
$$X = \sum_x Z_x^2 \sim \chi^2_{N-r}$$
where N is the number of age groups and r is the number of estimated parameters. Large values of X indicate a poorly fitting model and lead to rejection of the graduation.

N.B. It is important in practical calculations to group data where expected numbers are low (say < 5) since $\dfrac{(d_x - \dot{d}_x)^2}{\dot{d}_x}$ is sensitive to small changes in $\dot{d}_x$. We often group ages until $\dot{d}_x > 5$ whilst accepting that this reduces the sensitivity of our test. We emphasise that we group ages until $\dot{d}_x > 5$, and **not** until $d_x > 5$.

6.2.2 The standardised deviations test

If the $\{z_i\}$ are a random sample from the $\mathcal{N}(0,1)$ distribution then their histogram should look like that obtained from a normal sample. We can conduct a test based on the observed and expected frequencies in the histogram. The test statistic is the familiar
$$\sum_1^k \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$
where k is the number of cells in the histogram. The null distribution of the test statistic is χ^2 with $k-1$ df and large values of the test statistic indicate the model is a poor fit. The test can be performed in two ways and Figure 9 shows the two cases: (a) equal width cells (b) equal area cells.

(a) **Equal width cells:** We must calculate the area of each cell and hence find the values of the expected frequencies. We must also count the values of the observed frequencies.

(b) **Equal area cells:** The expected frequencies are immediate. We must calculate the cell boundaries and hence count the values of the observed frequencies.

You can use the `hist( )` function to compute the observed frequencies in the cells. For example, with equal cell widths $-1.5, -1, \ldots, 1, 1.5$ and residuals in `Res` we have

```
Boundary <- seq(-1.5, 1.5, by = 0.5)
Observed.freq <- hist(Res, breaks = c(-10, Boundary, 10))$counts
```

The `hist( )` function normally draws a histogram but you can return the counts by referring to them directly with the above code. The R-functions `Standard.Area( )` and `Standard.Width( )` in the file `Test_GoF.r` can be used to perform these tests.

6.2.3 A test for bias - the sign test

Suppose that S denotes the number of positive z_i and that there are n ages. If there is no positive or negative bias in the $\{z_i\}$ then each z_i is equally likely to be positive or negative and so $S \sim \mathcal{B}(n, \frac{1}{2})$. Since we wish to identify positive or negative bias, a **two-sided test** is required. Suppose we observe $S = s$. There are two cases.

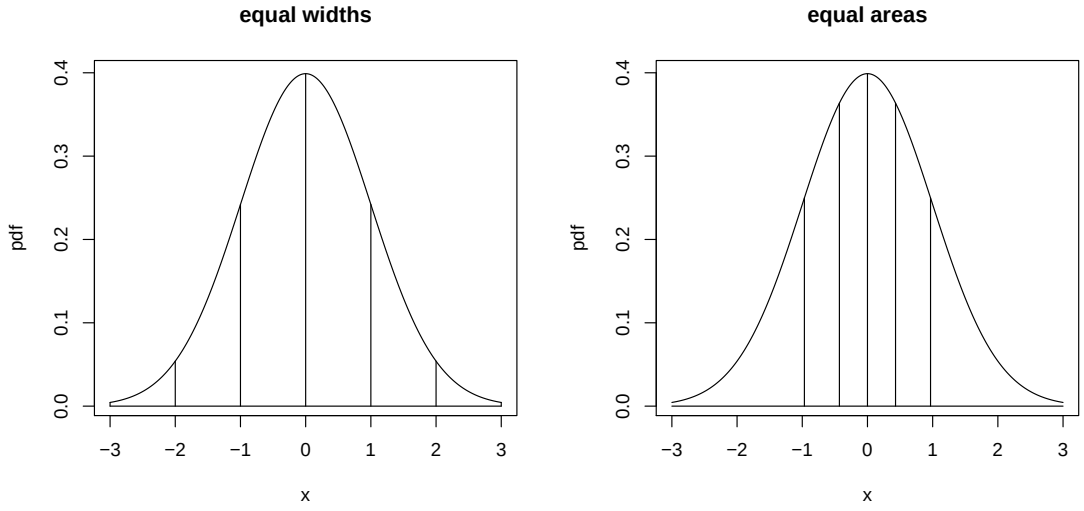


Figure 9: Proportions in a normal distribution: left panel shows **equal cell widths**, right panel shows **equal cell areas**

- If $s < n/2$ then the significance probability is $2\Pr(S \leq s)$ while if $s > n/2$ then the significance probability is $2\Pr(S \geq s)$. In R we use `2 * pbinom(s, n, 1/2)` in the first case and `2 * (1 - pbinom(s - 1, n, 1/2))` in the second.
- If R is not to hand and n is large then we can use the normal approximation to the binomial distribution with

$$\mathrm{E}(S) = \frac{n}{2}, \quad \mathrm{Var}(S) = n \times \frac{1}{2} \times \frac{1}{2} = \frac{n}{4}$$

or $S \sim \mathcal{N}(\frac{n}{2}, \frac{n}{4})$.

Example: We observe $s = 14$ positive values in $n = 20$ values of z_i.

$$\Pr(S \geq 14) = 1 - \Pr(S \leq 13) = 1 - 0.9423 = 0.0577$$

and the significance probability is 0.1154, as can be checked with the R-function `Sign( )`.

6.2.4 Change of sign test

This test examines non randomness of the signs of the residuals on a smaller scale. These might be 'runs' (i.e., a long sequence of positive Z_x) or 'groups' or 'clumps' (i.e., where sequences are shorter but not consistent with randomness.

If the z_x are independent and distributed $N(0,1)$ then with probability $\frac{1}{2}$ the sign of the $(x+1)$ th residual will be the same as the sign of the x th residual. If C denotes the total number of sign changes then $C \sim \mathcal{B}(n-1, \frac{1}{2})$. If c is the observed value of C then in most cases we are interested in small values of c, i.e., **too few sign changes**. We calculate $p = \Pr(C \leq c)$ and reject the graduation if $p \leq 0.05$, say. Again a normal approximation can be used if numbers are sufficiently large and R is not to hand.

Example: We observe $c = 5$ changes of sign with $n = 20$ values of z_i. We have $C \sim \mathcal{B}(19, \frac{1}{2})$ and the significance probability is `pbinom(5, 19, 1/2)` or 3.2% which suggests that all is not well with our graduation; this can be checked with the R-function `Change.Sign( )`

6.2.5 The grouping of signs test (or Stevens test)

This test looks at the number of *groups* or *runs* of positive signs amongst the residuals. We define the test statistic:

$$G = \text{Number of groups of positive residuals,}$$

and let g be the observed value of G. Suppose we have m residuals of which n_1 are positive and n_2 are negative. The null hypothesis is that the given n_1 positive residuals and n_2 negative residuals are in random order. We are on the look out for **too few groups** since this will indicate non-randomness. The test is **one-sided** and so we compute

$$\Pr(G \leq g)$$

and reject the null hypothesis if this is less than 5%, say.

Let $t \leq g$. We argue

- There are $\binom{n_1 - 1}{t - 1}$ ways to arrange n_1 positive signs into t positive groups.
- There are $\binom{n_2 + 1}{t}$ ways to arrange t positive groups among n_2 negative signs.
- There are $\binom{n_1 + n_2}{n_1}$ ways to arrange n_1 positive and n_2 negative signs.

Thus, the probability of obtaining exactly t positive groups is

$$p_t = \frac{\binom{n_1-1}{t-1}\binom{n_2+1}{t}}{\binom{n_1+n_2}{n_1}}.$$

The required significance probability is

$$\Pr(G \leq g) = \sum_{t=1}^{g} p_t.$$

If $m \leq 20$ we can use the Yellow Tables to conduct the test at the 5% level while if m is large, say $m > 20$, we can use a normal approximation

$$G \sim \mathcal{N}\left(\frac{n_1(n_2+1)}{n_1+n_2}, \frac{(n_1 n_2)^2}{(n_1+n_2)^3}\right).$$

With R you can conduct this test (and some of the other tests in this section) by using a **permutation test**. Permutation tests are an example of a **resampling method**. These are very powerful computer based methods. These tests are based on a very simple idea. We take a random permutation of the z_i and compute the value of our test statistic. We do this a large number of times and then by definition we have computed the null distribution of our test statistic. We then check to see where our observation lies in the null distribution.

Example: We have 20 observations and under some model we have 3 positive residuals, followed by 15 negative, followed by 2 positive. Clearly our model is very suspect! We conduct the runs test as follows. We have $n_1 = 5$, $n_2 = 15$ and $g = 2$

(a) **Formula:** If we expect to use the runs test on other data we can write the test as an R function. We show how to do this in two ways. In the first case we supply the values of n_1, n_2 and g as arguments

```
Runs <- function(n1, n2, g){
  Sig.Pr <- 0
  Denom <- choose(n1+n2, n1)
  for(i in 1:g) {
    Num <- choose(n1-1,i-1)*choose(n2+1,i)
    Sig.Pr <- Sig.Pr + Num/Denom
  }
Sig.Pr
}
n1 <- 5; n2 <- 15; g <- 2
Runs(n1, n2, g)
```

We find that 2 runs provides significant evidence against our model at 3.2%.

In the second case we supply the residuals, $\{z.x\}$. We write a function to calculate n_1, n_2 and g and then use our existing function `Runs( )`

```
Runs.test <- function(Z.x){
  Code <- Z.x; Code[ Z.x <= 0] = -1; Code[ Z.x > 0] = 1
  n1 <- sum(Code == 1); n2 <- sum(Code == -1)
  g <- 0; if(Code[1] > 0) g <- 1
  for(i in 1:(length(Code)-1)) if(Code[i] < Code[i+1]) g <- g + 1
  Sig.Pr <- Runs(n1, n2, g)
  return(list(n1 = n1, n2 = n2, g = g, Sig.Prob = Sig.Pr))
}
Z.x <- c(abs(rnorm(3)), -abs(rnorm(15)), abs(rnorm(2)))
Answer <- Runs.test(Z.x)
```

and we find the same answer as before.

(b) **Normal approximation:** We have

```
Mean <- n1*(n2+1)/(n1+n2)
SD <- sqrt((n1*n2)^2/(n1+n2)^3)
Sig.Norm <- pnorm(g+0.5, Mean, SD)
Sig.Norm
```

We find that 2 runs provides significant evidence against our model at 3.7%, in broad agreement with the exact calculation.

(c) **Permutation test:** The resampling is performed with the `sample( )` function. We have

```
Perm <- sample(Code)
```

gives a random permutation of the values in `Code`. We then calculate the number of runs in `Perm` using the above code. This gives us a single random observation from the null distribution of our test statistic. In this fashion, we build up the null distribution of our test statistic. The R-function `Runs.Test.Perm( )` lets you perform the permutation version of the runs test. Again we find a result significant around 3%.

(d) **Yellow tables:** The Yellow Tables tell us that 2 runs is significant evidence against our model at the 5% level, a somewhat weaker conclusion than obtained above.

6.2.6 Serial Correlation Test

This test again addresses the limitations of the χ^2 test in detecting groups of residuals of the same sign. It also takes into account the **size** of the residuals. We examine the tendency of residuals at adjacent ages to be **positively correlated**.

Consider the **first** $(n-1)$ residuals $\{z_1, \ldots, z_{n-1}\}$ as one set of data and the **last** $(n-1)$ residuals $\{z_2, \ldots, z_n\}$ as another set and define r_1 as the correlation between the two sets. Then R_1 has asymptotic distribution

$$R_1 \sim \mathcal{N}\left(0, \frac{1}{n-1}\right)$$

under the null distribution of randomness of the residuals. Students of time series will recognise the lag 1 serial correlation coefficient. We are concerned with the tendency of residuals at adjacent ages to be positively correlated, so the test is **one-sided** with large positive values of r_1 providing evidence against H_0. The significance probability is then $\Pr(R_1 > r_1)$. In R we have

```
Set.1 <- Res[-length(Res)]; Set.2 <- Res[-1]
Serial.Cor <- cor(Set.1, Set.2); Serial.Cor
Sig.Serial <- 1 - pnorm(Serial.Cor, 0, sqrt(1/(length(Res)-1)))
```

The lag 1 serial correlation coefficient can easily and obviously be generalised to a lag k correlation, with $\mathcal{N}(0, 1/(n-k))$ as its asymptotic null distribution.

6.3 Example: Testing a Graduation

We consider a small data set for illustration. The table gives the observed number of deaths, d_x, and initial exposures, E_x, for a set of lives. Note that the data are grouped into five year age bands for this example but we would normally avoid grouping since it results in a loss of information. We calculate the crude death rates, $\hat{q}_x = d_x/E_x$. The left panel of Figure 10 gives a plot of these death rates while the right panel, a plot of the logit $\hat{q}_x$, suggests that a Gompertz style linear, quadratic or cubic function will provide a suitable graduation.

We fit the GLM with binomial errors and logit link and quadratic linear predictor

$$\log\left(\frac{q_x}{1-q_x}\right) = a_0 + a_1 x + a_2 x^2$$

Age	E_x	d_x	$\dot{d}_x$	$\frac{d_x - \dot{d}_x}{\sqrt{\dot{d}_x}}$
12	8119	14	18.78	−1.104
17	7750	20	19.68	0.072
22	6525	22	18.83	0.730
27	5998	23	20.37	0.583
32	5586	26	23.10	0.603
37	5245	28	27.35	0.125
42	4659	32	31.70	0.054
47	4222	37	38.77	−0.285
52	3660	44	46.92	−0.427
57	3012	54	55.70	−0.228
62	2500	68	68.81	−0.097
67	2113	87	89.10	−0.222
72	1469	100	97.26	0.278
77	883	95	93.36	0.169
82	418	70	70.87	−0.103
87	181	49	48.40	0.086

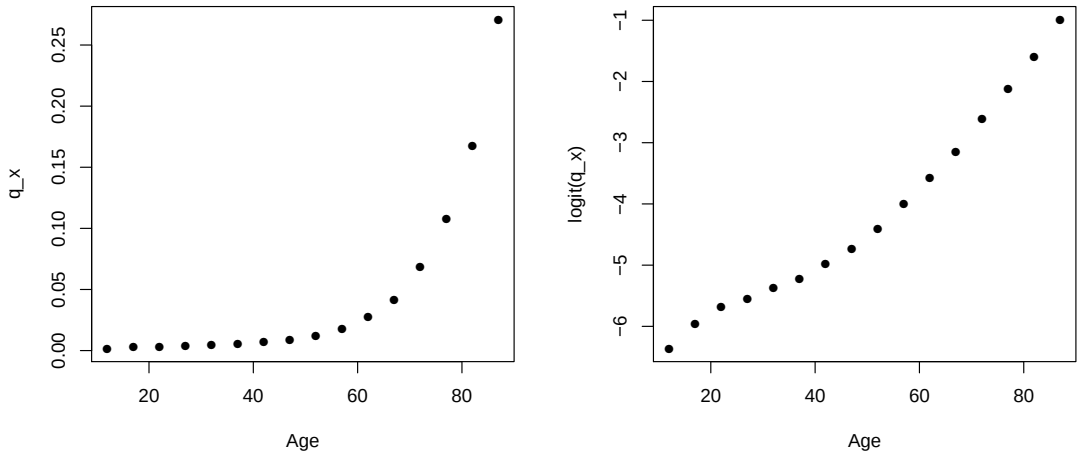


Figure 10: Left: Crude mortality rates $\hat{q}_x$ against age, x. Right: Logit of crude mortality rates $\hat{q}_x$ against age, x.

where x is age. We find $\hat{a}_0 = -6.149$, $\hat{a}_1 = -0.001511$ and $\hat{a}_2 = 0.0006966$. The data, fitted values and standardised residuals are in the table. We suppose the deaths, exposures and ages are in the objects `D.x`, `E.x` and `Age` respectively. Here is our R-code to produce the plots, fit the model and calculate the residuals.

```
Q.x <- D.x/E.x
Logit <- function(x) log(x/(1-x)) # Define logit function
Obs <- Logit(Q.x)
Age2 <- Age^2
Gomp.bin <- glm(Q.x ~ Age + Age2, weights = E.x, family = binomial)
```

```
Q.dot <- Gomp.bin$fit
Dth.dot <- E.x * Q.dot
Z.x <- (D.x - Dth.dot)/sqrt(Dth.dot)
cbind(Age, D.x, E.x, round(Dth.dot, digits = 2), round(Z.x, digits = 3))
```

The final line reproduces the table and it is now a straightforward matter to check our graduation by running through our battery of tests. The complete code for the example can be found in the R-file `Chap_6_Eg.r`.

6.3.1 χ^2 test

We compute `sum(Z.x^2)` and find $\chi^2 = 2.976$. We have $n - p = 16 - 3 = 13$ degrees of freedom and so there is no suggestion that the graduation is unsatisfactory. Indeed we have observed a closer fit than expected since $\mathrm{E}(\chi^2_r) = r$ and we have observed a very small value; this suggests that our model is a very close fit. We can also use `Chi.Square(Z.x, 3)` to obtain this answer.

6.3.2 Standardised Deviations Test

There are very few observations for this test. With four cells of equal width we calculate

Range of z	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
Observed	1	6	9	0
Expected	2.54	5.46	5.46	2.54

In R we have

```
Boundary <- c(-10, -1, 0, 1, 10)
Observed <- hist(Z.x, breaks = Boundary)$counts
Expected <- length(E.x) * diff(pnorm(Boundary))
Chis2 <- sum( (Observed - Expected)^2/Expected )
```

which gives

$$\chi^2 = \frac{1.54^2}{2.54} + \frac{0.54^2}{5.46} + \frac{3.54^2}{5.46} + \frac{2.54^2}{2.54} = 5.82$$

with 3 degrees of freedom. Now $\Pr(\chi^2 > 5.82) = 1 - \texttt{pchisq}(\texttt{Chis2}, 3) \approx 0.12$ when $\chi^2 \sim \chi^2_3$ so there is weak evidence that the graduated data do not fit the original data. If we are worried that the expected values (of 2.56) are rather small we can combine cells to give

$$\chi^2 = \frac{2 \times 1^2}{8} = 0.25$$

where $\chi^2 \sim \chi^2_1$. Any evidence of a poor fit has disappeared.
NB: The R-function call `Standard.Width(Z.x, 1)` automatically gives this second case.

A better approach is to use cells of equal area. With 16 observations we could use four (or three) cells. We find

Range of z	$(-\infty, -0.6745)$	$(-0.6745, 0)$	$(0, 0.6745)$	$(0.6745, \infty)$
Observed	1	6	8	1
Expected	4	4	4	4

In R we have

```
Expected <- rep(4, 4)
Posts <- c(-10, qnorm(c(0.25, 0.5, 0.75)), 10)
Observed <- hist(Z.x, breaks = Posts)$counts
Chis2 <- sum( (Observed - Expected)^2/Expected )
```

and $\chi^2 = 9.5$. Now $\Pr(\chi^2 > 9.5) = 0.023$ so we have quite strong evidence that the graduation should be rejected. The equal area approach has given a more powerful test, in this case at least. The R-call `Standard.Area(Z.x, 4)` also gives this result.

6.3.3 The Sign Test

We have 7 negative and 9 positive signs. This is clearly quite unexceptional. If S is the number of positive signs then $S \sim \mathcal{B}(16, \frac{1}{2})$, and, for example, $2 * \Pr(S \leq 3) = 0.021$ and $2 * \Pr(S \leq 4) = 0.077$ so we are nowhere near a significantly low (or high) value of S. The exact significance can be found with `Sign(Z.x)`.

6.3.4 The Change of Sign Test

We have 5 changes of sign. If C is the number of sign changes then $C \sim \mathcal{B}(15, \frac{1}{2})$, and $\Pr(C \leq 5) = 0.15$, so there is nothing to make us suspect non-randomness here. This agrees with the result given by `Change.Sign(Z.x)`.

6.3.5 The Grouping of Signs Test

We have $n_1 = 9$ positive signs, and $n_2 = 7$ negative signs. Let G denote the number of groups of positive signs. We observe $g = 3$. Then

$$p_t = \Pr(G = t) = \frac{\binom{n_1-1}{t-1}\binom{n_2+1}{t}}{\binom{n_1+n_2}{n_1}} = \frac{\binom{8}{t-1}\binom{8}{t}}{\binom{16}{9}} = \frac{\binom{8}{t-1}\binom{8}{t}}{11440}.$$

Thus,

$$p_1 = \frac{8}{11440}, \quad p_2 = \frac{224}{11440}, \quad p_3 = \frac{1568}{11440}$$

and so $\Pr(G \leq 3) = 0.16$. Once more there is nothing untoward here. We can confirm this calculation by using the function `Runs.test( )`, as in `Runs.test(Z.x)`.

We can also use the **normal approximation** with $\mathrm{E}(G) = n_1(n_2 + 1)/(n_1 + n_2) = 9 \times 8/16 = 4.5$ and $\mathrm{Var}(G) = (n_1 n_2)^2/(n_1 + n_2)^3 = 0.9689941$. Thus, $\Pr(G \leq 3) \approx \Pr[Z \leq (3.5 - 4.5)/\sqrt{0.9689941}] = 0.15$, in agreement with the exact calculation.

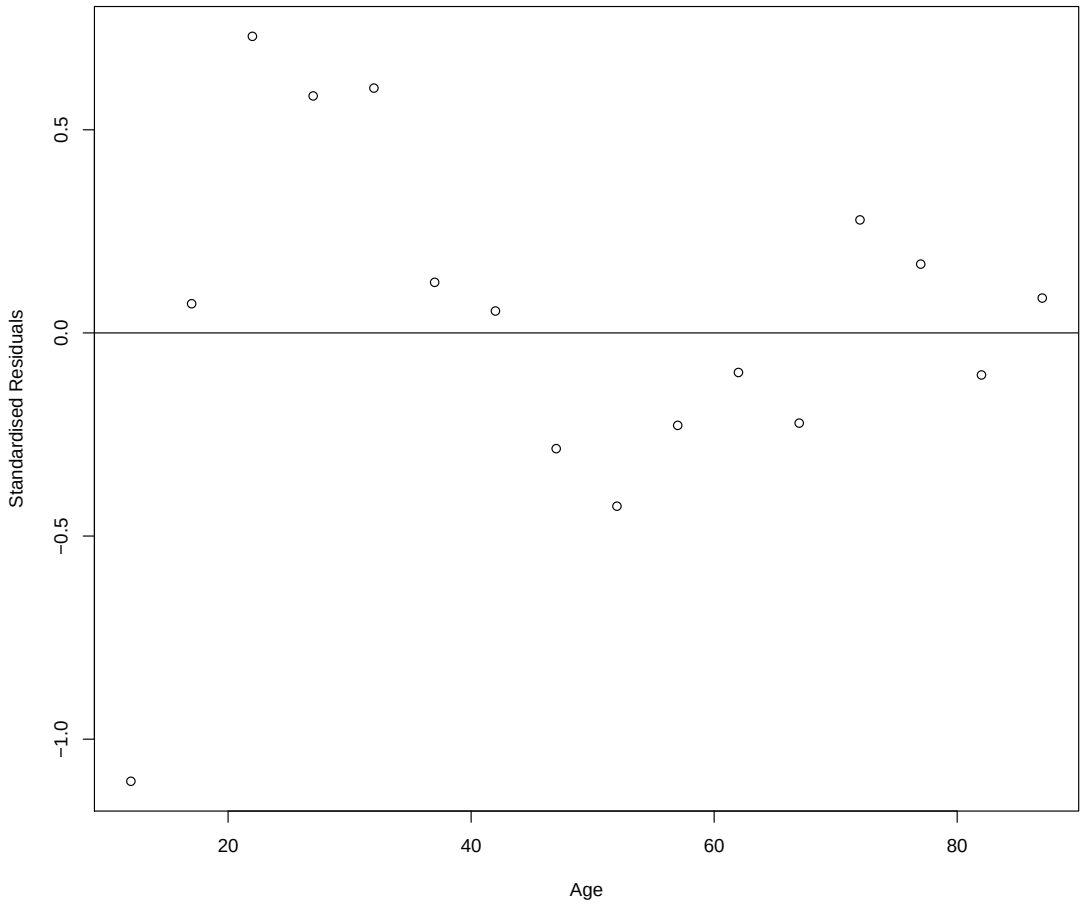


Figure 11: Pearson residuals against age, x

6.3.6 The Serial Correlation Test

We use the R-Code given earlier

```
Set.1 <- Z.x[-length(Z.x)]; Set.2 <- Z.x[-1]
Serial.Cor <- cor(Set.1, Set.2); Serial.Cor
Sig.Serial <- 1 - pnorm(Serial.Cor, 0, sqrt(1/(length(Z.x)-1)))
```

We find $r_1 = 0.461$, significant evidence against H_0 at 3.7%. Alternatively, `Serial(Z.x)` gives this result immediately.

Figure 11 is a plot of the Pearson residuals against age. The serial correlation coefficient test is probably picking up the (apparent) tendency for the deviations at young ages to be positive and at old ages to be negative.

Conclusion: The graduation passes some test but not others. We could fit a cubic logistic regression and see whether this gave an improved fit.

Functions for Testing Goodness-of-Fit

The file `Test_GoF.r` (available on the web site) contains a number of R functions to perform various goodness-of-fit tests. The table opposite summarizes the tests.

Test R-Function	Arguments	Output
χ^2-test `Chi.Square`	`Z`, residuals Z_x `Npar`, number of fitted parameters	`Chis2`, value of χ^2 `DF`, df `Sig.Pr`, significance probability
Standardized deviations `Standard.Area` (equal cell area form)	`Z`, residuals Z_x `N`, number of cells	`Boundary`, boundaries of cells `Obs`, O_i; `Exp`, E_i; `DF`, df `Chis2`, value of χ^2 `Sig.Pr`, significance probability
Standardized deviations `Standard.Width` (equal cell width form)	`Z`, residuals Z_x `W`, width of internal cells	`Boundary`, boundaries of cells `Obs`, O_i; `Exp`, E_i; `DF`, df `Chis2`, value of χ^2 `Sig.Pr`, significance probability
Sign test `Sign`	`Z`, residuals Z_x	`N.plus`, number of +'s `N.minus`, number of −'s `Sig.Prob`, significance probability
Change of sign test `Change.Sign`	`Z`, residuals Z_x	`N`, number of Z_x `Change`, number of changes `Sig.Pr`, significance probability
Runs test `Runs.test`	`Z`, residuals Z_x	`n1`, number of +'s `n2`, number of −'s `g`, number of groups of + signs `Sig.Pr`, significance probability
Runs test (permutation test) `Runs.Test.Perm`	`Z`, residuals Z_x	`Runs`, number of runs `Null.dist`, null distribution `Sig.Pr`, significance probability
Serial correlation test `Serial`	`Z`, residuals Z_x	`Serial`, correlation coef. `Sig.Pr`, significance probability

Tutorial 6: Graduation II

1. Based on HW, 2006. The table gives the central exposed to risk, E_x^c, the number of deaths, d_x, and $\log \hat{\mu}_x$ for a set of male assured lives, $\mathcal{S}$, for ages $x = 60, \ldots, 72$. Here $\hat{\mu}_x$ is the usual maximum likelihood estimate of the force of mortality at age $x + \frac{1}{2}$.

Age x	E_x^c	d_x	$\log \hat{\mu}_x$	$\dot{d}_x$	z_x
60	52250	314	−5.11	314.8	−0.044
61	47400	312	−5.02	321.1	−0.506
62	44860	352	−4.85	341.7	0.559
63	42920	334	−4.86	367.5	−1.749
64	38920	405	−4.57	374.7	1.564
65	24950	257	−4.58	270.1	−0.797
66	16900	206	−4.41	205.7	0.020
67	14990	212	−4.26	205.1	0.478
68	13550	235	−4.05	208.5	1.835
69	12160	223	−4.00	210.4	0.870
70	10950	200	−4.00	213.0	−0.891
71	9770	217	−3.81	213.7	0.226
72	8770	195	−3.81	215.7	−1.408

The table is graduated by fitting a Poisson generalised linear model with a graduation of the form $\log \dot{\mu}_x = a + bx$. The table also shows the fitted number of deaths, $\dot{d}_x$, and the Pearson residuals, z_x, from this model. You are to use R to answer this question.

(a) Enter the ages, deaths and central exposures into R and hence obtain a plot $\log \hat{\mu}_x$ against age.

(b) Use the `glm( )` function to fit the model and add the fitted log mortality to the plot.

(c) Verify that the graduated deaths, $\dot{d}_x$, and the Pearson residuals, z_x, are as shown in the table. Comment on the success of the graduation.

(d) Test the suitability of the graduation with (i) the χ^2-test (ii) the standardised deviations test (use four equal area cells for the test) (iii) the sign test (iv) the change of signs test (v) the runs test, and (vi) the serial correlation test. What is your conclusion?

(e) An actuary suggests graduating the table by minimising

$$S(\alpha, \beta) = \sum_x w_x (\log \hat{\mu}_x - \alpha - \beta x)^2.$$

Explain briefly the role of the weights w_x in the above expression and use the Δ-method to suggest suitable values for the weight function w_x. Fit the model in R and compare the fitted model to that obtained in (b).

2. Obtain the deaths, d_t, $t = 1950, \ldots, 2005$, and central exposures, E_t^c, for age 70 from the CMI data files. Convert the central exposures to initial exposures, E_t. Let $\hat{L}_t$ be the maximum likelihood estimates of $\operatorname{logit} q_t$.

   (a) Plot $\hat{L}_t$ against t.

   (b) Fit the generalised linear model for $\hat{q}_t$ on t with quadratic predictor, logit link and binomial error. Obtain the fitted logits, $\dot{L}_t$, add the fitted curve to your plot in (a) and comment informally on the quality of the fit.

   (c) Calculate the fitted deaths, $\dot{d}_t$.

   (d) Calculate the Pearson residuals, z_t, and plot them against t. Again, comment on the quality of the fit to the data.

   (e) Test the suitability of the graduation with (i) the χ^2-test (ii) the standardised deviations test (use ten equal area cells for the test) (iii) the sign test (iv) the change of signs test (v) the runs test, and (vi) the serial correlation test. What is your conclusion?

   (f) An actuary suggests graduating the table by minimising

$$S(\beta_0, \beta_1, \beta_2) = \sum_t w_t (\operatorname{logit} \hat{q}_t - \beta_0 - \beta_1 t - \beta_2 t^2)^2.$$

   Explain the role of the weights w_t in the above expression and use the Δ-method to suggest suitable values for the weight function w_t. Fit the model and compare the fitted model to that obtained in (b).

7 Mortality Projection

As a reference for this chapter, see Macdonald, A.S., Richards, S.J. and Currie, I.D. (2018). *Modelling Mortality with Actuarial Applications*, Cambridge University Press.

7.1 Introduction

There has been clear evidence, over many years, that human lifespan has been lengthening, under the influence of increasing affluence, improved healthcare and other reasons. Where such progress is halted or disrupted, there have been quite abrupt reverses in this trend, e.g. in the former USSR in the years after 1989.

For pension funds, and writers of annuities, this trend is financially very important. Such contracts are potentially loss-making if the mortality actually experienced in the future is *better* than was assumed in pricing the contract, or in reserving for future liabilities. If we expect the trend of increasing lifespan to continue in future, then basing annuity prices and reserves on the mortality we observe now will be inadequate. Mortality in future will be better than it is now, and we need to allow for that. Therefore, we need to *project* how mortality will improve in future. This chapter introduces some approaches to doing so.

Writers of life insurance contracts are not so troubled by this feature, as future mortality lower than that assumed in setting prices or reserves is a source of profit. However, competition may drive life insurers to use projected mortality improvements for this line of business as well (compare with the widespread use of select life tables for pricing life insurance contracts).

For example, Table 5 shows how life expectancy at age 65 has changed, in the UK and the USA, between 1970 and 2010.

Table 5: Life expectancy at age 65 in UK and US from the Human Mortality Database. Source: Macdonald, Richards and Currie (2018).

	UK		US	
Year	Male	Female	Male	Female
1970	12.1	16.0	13.0	16.9
1980	12.8	16.8	14.1	18.3
1990	14.0	17.9	15.1	19.0
2000	15.7	18.9	16.1	19.1
2010	18.0	20.6	17.9	20.5

These are examples of *period* life expectancy, meaning that they are based on an investigation of lives in a period around the calendar year concerned. For example, the figure of 12.1 years for Males in the UK in 1970 was based on an investigation made during a few years around 1970. If an actuary in 1970 expected mortality to improve in future, she would want to estimate the *cohort* life expectancy — that is, life expectancy based on the *future* mortality of a UK male now age 65. Of course, this is unknown and can only be estimated by modelling the dependence of mortality on calendar time, as well as age.

Instead of estimating μ_x or q_x only as a function of age x, we now must model $\mu_{x,y}$ or $q_{x,y}$ as a function of both age x and calendar year y. Generally, this entails three steps.

- Fit a model for μ_{x,y^p} or q_{x,y^p} for calendar years y^p in the past.

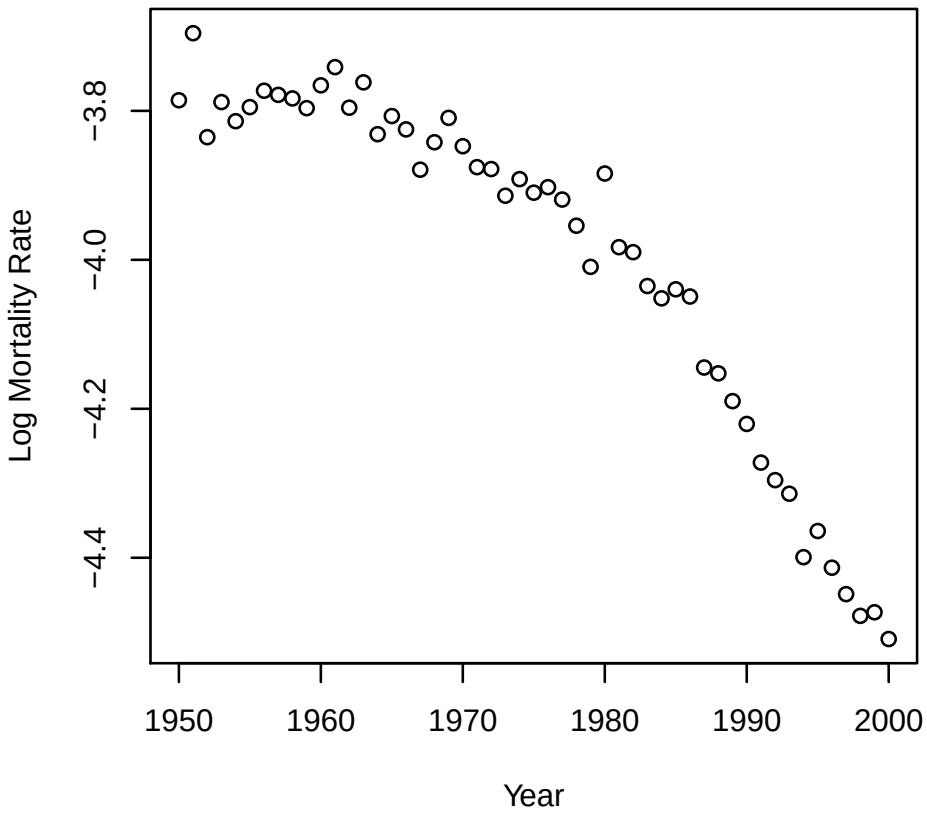


Figure 12: Log mortality for males, England & Wales, Age 60, from 1950 to 2000 (HMD data).

- Use this model to estimate how μ_{x,y^p} or q_{x,y^p} depended on calendar year y^p in the past.
- Project this form of dependence into calendar years y^f in the future, hence obtain estimates of μ_{x,y^f} or q_{x,y^f} in future calendar years y^f.

Some approaches may combine two, or even all three, of these steps into one procedure.

Suppose we fix a single age, $x = 60$ say, and carry out the above procedure. Then we get a function $\mu_{60,y}$ of all past and future years y. Then we repeat the process for ages 61, 62, ... and so on. However, doing so separately for each individual age will usually lead to problems, because there is no guarantee that the pattern of mortality rates in any future calendar year will make sense. For example, projecting forward from 2015 in this way, we might find that $\mu_{70,2035} > \mu_{80,2035}$, which we would not accept as sensible.

Hence all approaches to this problem, to a greater or lesser extent, attempt to model $\mu_{x,y}$ or $q_{x,y}$ as a *2-dimensional surface*, so that the results look sensible *both* as a function of calendar time for each age, and as a function of age in each calendar year. This is a formidable task, and is the subject of much current research in the actuarial literature.

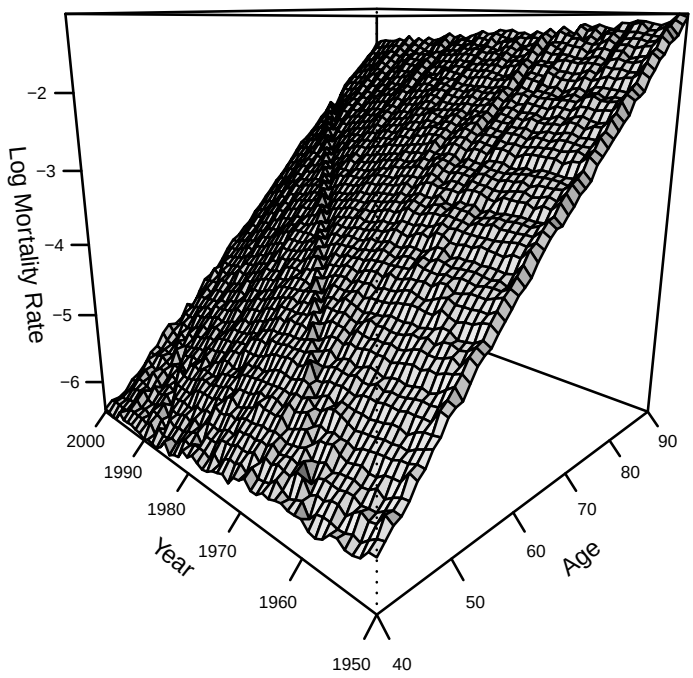


Figure 13: Log mortality for males, England & Wales, Ages 40–90 , from 1950 to 2000 (HMD data).

7.2 Data

Our examples will use the Human Mortality Database data (see the note on Data Files `Data_files.pdf` on Canvas). Figure 12 shows male mortality at age 60, in England and Wales, from 1950 to 2000 (log scale). This clearly shows the need to allow for further possible improvements in future. The R code to produce this plot is as follows.

```
source("EngWales_read.r")           # Reads HMD data
z <- Dth.M / Exp.M
plot( 1950:2000 , log( z[ Age==60 , Year>=1950 & Year <=2000] ) , xlab="Year"
, ylab = "Log Mortality Rate")
```

Figure 13 shows the surface of log mortality rates for males age 40 to 90, and from 1950 to 2000. The improvements over time are easy to see at the lowest and highest ages, less so in the middle of the surface, but note that Figure 12 is a cross-section through this surface, at age 60. The R code to produce this plot is as follows

```
source("EngWales_read.r")           # Reads HMD data
z <- Dth.M / Exp.M
persp( 40:90 , 1950:2000 , log( z[ Age>=40 & Age <= 90 , Year>=1950 & Year <=2000]
) , theta=-45 , xlab="Age" , ylab="Year" , zlab = "Log Mortality Rate" ,
ticktype="detailed" , cex.axis=0.5 , cex.lab=0.75 )
```

Note that projecting mortality requires data that have been collected on a consistent basis over a long period of time. In the UK, there are realistically only two source of data: the CMI and the ONS (data from the latter are the source for the UK data in the HMD). In many countries, official statistics corresponding to the ONS data will be the only possible source. This has the important consequence that projections of future trends in mortality may be based on an experience different from that that the actuary wishes to project. For example, the actuary may wish to project the future experience of annuitant mortality, but be forced to base the projections on national statistics of the whole population.

7.3 Approaches to Projecting Mortality

Many approaches to the problem of projecting mortality have been tried. We identify three: (a) methods based on deterministic projections (or expectations); (b) extrapolation of stochastic models; and (c) modelling underlying causes of death.

7.3.1 Deterministic Projections

Deterministic projections usually follow four steps.

- Choose a *base table*, which is a cross-sectional or secular life table based on the most recent calendar years. Examples would be the most recent graduations produced by the CMI (for insurance data) or the ONS (for national data).
- By studying the rate at which $\mu_{x,t}$ or $q_{x,t}$ has been changing in the recent past, extrapolate the rate of change in the near future.
- Choose a target for values of $\mu_{x,t}$ or $q_{x,t}$ in the distant future (possibly $\mu_{x,\infty}$ or $q_{x,\infty}$). This can be based on expert opinion, or by reference to countries that have had historically low mortality (Japan is often cited in this context).

- Choose a formula that blends the extrapolated changes in $\mu_{x,t}$ or $q_{x,t}$ in the near future into the target values of $\mu_{x,t}$ or $q_{x,t}$ in the distant future.

As an example, we show the method used by the CMI from its "80" Series tables until fairly recently (it now provides members with a spreadsheet-based projections toolkit that lets each insurer carry out its own projections).

The CMI chose to project $q_{x,t}$.

- Step 1: The "80" Series tables were graduated by the CMI based on the 1979–82 experience. These, and updated graduations published subsequently, were used as the base tables. The bases table(s) were denoted by $q_{x,0}$.

- Steps 2 and 3: At each age x, a 'target' ultimate reduction factor $\alpha(x)$ at age x was chosen as follows:

$$\alpha(x) = \begin{cases} 0.5 & x < 60 \\ \dfrac{x-10}{100} & 60 \le x \le 100 \\ 1 & x > 100. \end{cases}$$

  The projection model then took the form of a *reduction factor* applying in year t:

$$RF(x,t) = \alpha(x) + (1-\alpha(x))0.4^{t/20}$$

  and

$$q_{x,t} = q_{x,0} \times RF(x,t).$$

  Thus the fraction $(1-\alpha(x))$ of the base table which is to be 'lost' in the distant future is reduced by 60% every 20 years.

The advantage of deterministic projections is its simplicity — there is no need to fit a stochastic model. Its disadvantages are several.

- Setting the target level of mortality in the distant future relies on expert opinion. Experts disagree on fundamental questions, such as whether human mortality can improve without limit, or whether there is some biological barrier that can never be exceeded.

- A single projection is almost certain to be wrong. In the past actuaries have tended to *underestimate* future improvements in mortality, possibly through failing to foresee changes in lifestyle (especially smoking habits) and medical advances. (Underestimation leads to losses in annuity portfolios.) Today's actuaries are no more likely to get it right.

- Precisely because it is a deterministic projection, there is no information about any uncertainty around the projection, such as might be useful in setting up risk reserves in annuity portfolios.

7.3.2 Stochastic Models

Stochastic models attempt to fit a model to the two-dimensional mortality surface (see Figure 13) allowing for both trend and sampling uncertainty. Knowing something about the uncertainty of the fitted surface, we may hope to be able to say something about the uncertainty around the future projections. We will consider four methods that have been much used in recent years: the Lee-Carter model (Section 7.4); the Cairns-Blake-Dowd model (Section 7.5); the P-spline model (Section 7.6); and the age-period-cohort model (Section 7.7).

7.3.3 Cause-of-Death Models

We can observe death arising from several major causes (cancer, cardiovascular disease, pulmonary disease, kidney failure etc.) and other minor causes. An attractive idea, at first sight, is to model mortality by each cause separately, and then to combine these models into a model of aggregate mortality. We can then project how mortality from each cause might change in future, e.g. because of changing smoking habits and medical advances (or medical retreats, if antibiotic resistance is not defeated).

This idea goes back at least as far as Daniel Bernoulli, who in 1766 considered what the effect might be of eradicating smallpox in Europe.

In practice, cause-of-death modelling is very difficult, although some models exist in insurance practice.

- Cause of death is recorded on death certificates, but not always accurately. Apart from straightforward errors, there can be a reluctance to record certain causes for cultural reasons or out of consideration for family members; AIDS may be an example.

- Death certificates often allow more than one cause to be recorded, perhaps a primary cause and several underlying causes. It is not always clear that the primary cause has been correctly identified, nor what the contribution of underlying causes to a model should be. For example, the primary cause may be recorded as heart failure, while the underlying (and real) cause might be years of chronic pulmonary disease.

- Different causes may be strongly correlated because of lifestyle or genetic factors. For example, smoking is strongly associated with both lung cancer and heart disease. In such cases a joint model of the correlated causes is required. Modelling the correlation is difficult, both when fitting a model to data, and when projecting future mortality.

- As one cause of death recedes, perhaps because of medical advances, others take its place, in ways that would have been hard to predict. At the start of the 20th century, cancer was not the major cause of death it is today. Cancer is predominantly a disease of old age. As lifespans got longer during the 20th century, infectious diseases and much heart disease receded, and cancer emerged as a major cause.

- Data may not be available for the long periods of time needed to project mortality. Causes of death are recorded using the International Classification of Diseases (ICD) codes, set up in 1899 and revised in 1900, 1909, 1920, 1929, 1938, 1948, 1955, 1965, 1975, 1989 and 2018. Changes to ICD codes can disrupt the consistency of the data.

We do not consider cause-of-death models in more detail.

7.4 The Lee-Carter Model

7.4.1 Model Definition

The Lee-Carter model is defined as:

$$\log(\mu_{x+1/2,y+1/2}) = \alpha_x + \beta_x \kappa_y, \qquad x = x_1, x_2, \ldots, x_{n_x}, \quad y = y_1, y_2, \ldots, y_{n_y}. \tag{42}$$

- The range of ages is $x = x_1, x_2, \ldots, x_{n_x}$, so in the example from Section 7.2 above, $n_x = 51, x_1 = 40$ and $x_{n_x} = x_{51} = 90$. The range of years is $y = y_1, y_2, \ldots, y_{n_y}$, so in the example $y_1 = 1950, n_y = 51$ and $y_{n_y} = y_{51} = 2000$.
- The left-hand side is the log force of mortality at age x and in calendar year y. The notation reflects the fact that we assume an estimate of this quantity to apply at the mid-point of the year of age and the mid-point of the calendar year.
- The first term on the right-hand side, α_x, is a parameter for each age x. We often find in practice when we fit the Lee-Carter model that the function α_x is very close to a straight line αx for some constant α. In that case the first term is similar to the exponential term in a Gompertz model.
- The second term on the right-hand side is a product of two factors. One, κ_y, is the only term in the model which is a function of y. Moreover, for any fixed age x, the function $\log(\mu_{x+1/2,y+1/2})$ is just the function κ_y multiplied by a constant plus an additive constant. Therefore, the *shape* of the force of mortality as a function of time (see Figure 12) is the same for all ages.
- The term β_x 'modulates' the effect of κ_y at different ages. If β_x is large (small) in magnitude, mortality is changing more rapidly (less rapidly) at age x. If, for two ages x_i and x_j, $\beta_{x_i} > 0$ and $\beta_{x_j} < 0$, mortality would be moving in opposite directions at these two ages (which way would depend on the sign of κ_y). However, because of the property noted in the previous bullet point, mortality at these two ages would then *always* move in opposite directions, which would not often be realistic.

For convenience, define the vector quantities:

$$\begin{aligned} \boldsymbol{\alpha} &= (\alpha_1, \alpha_2, \ldots, \alpha_{x_{n_x}}) \\ \boldsymbol{\beta} &= (\beta_1, \beta_2, \ldots, \beta_{x_{n_x}}) \\ \boldsymbol{\kappa} &= (\kappa_1, \kappa_2, \ldots, \kappa_{y_{n_y}}). \end{aligned}$$

Compared to most models we are used to, the Lee-Carter model has a very large number of parameters, $2n_x + n_y$, or 153 in the example from Section 7.2 above. As a consequence, we cannot expect the Lee-Carter model to be smooth, either in age or in calendar time. If we want smoothness we will have to impose it some other way.

7.4.2 Model Identifiability

Consider the signs of β_x and κ_y. If we reverse them, and put $\beta_x^* = -\beta_x$ and $\kappa_y^* = -\kappa_y$, we get the same model:

$$\alpha_x + \beta_x \kappa_y = \alpha_x + \beta_x^* \kappa_y^*.$$

The same is true if we put $\beta_x^* = c\beta_x$ and $\kappa_y^* = \kappa_y/c$ for any $c \neq 0$. Certain other transformations of $\boldsymbol{\alpha}, \boldsymbol{\beta}$ and $\boldsymbol{\kappa}$ also leave the model unchanged. Since our aim is to estimate $\boldsymbol{\alpha}, \boldsymbol{\beta}$ and $\boldsymbol{\kappa}$ this is a problem; there is no unique solution. We say that the model is *unidentifiable*. The solution is to impose constraints on $\boldsymbol{\alpha}, \boldsymbol{\beta}$ and $\boldsymbol{\kappa}$ such that the model has a unique representation. The conventional choice is to require the following:

$$\sum_{x=1}^{n_x} \beta_x = 1 \qquad\qquad \sum_{y=1}^{n_y} \kappa_y = 0.$$

These constraints mean that $\boldsymbol{\alpha}$ has a simple interpretation. We have:

$$\sum_{y=1}^{n_y} \log(\hat{\mu}_{x+1/2,y+1/2}) = \sum_{y=1}^{n_y} (\hat{\alpha}_x + \hat{\beta}_x \hat{\kappa}_y) = n_y \hat{\alpha}_x$$

for each age x. That is, each $\hat{\alpha}_x$ is the average over time of the fitted values $\log(\hat{\mu}_{x+1/2,y+1/2})$.

7.4.3 Fitting the Model

This section follows closely the method in Chapter 12 of Macdonald, Richards and Currie (2018). The R programs `Lee_Carter.r` and `Forecast_LC.r` referred to there are freely available from the publisher's website: `www.cambridge.org/9781107045415`. We use slightly modified versions `LeeCarter.r` and `ForecastLC.r`.

The Lee-Carter model falls outside the framework of most standard statistics packages, in particular the cross-term $\beta_x \kappa_y$ means that it is not a GLM. However, it can be fitted using a specialised function `gnm()`, in a way that mirrors closely the call to `glm()`. The steps in fitting the model are as follows.

- If your local system does not have the `gnm` package installed, download it. On many systems typing `install.packages("gnm")` at R's command prompt will allow you to do this.
- Prepare your data. In keeping with the example in Section 7.2 we will use males age 40–90 in England & Wales during 1950–2000. First extract the required data in matrix form:

```
library("gnm")
source("EngWales_read.r")            # Reads HMD data
Dth <- Dth.M[ Age>=40 & Age <= 90 , Year>=1950 & Year <=2000 ]
Exp <- Exp.M[ Age>=40 & Age <= 90 , Year>=1950 & Year <=2000 ]
```

  Then convert these to vectors, which `gnm()` requires.

```
Dth.V <- c(Dth)
Exp.V <- c(Exp)
```

  Finally, age and year must be passed to `gnm()` as factors.

```
AGE <- 40:90
YEAR <- 1950:2000
Age.F <- factor(rep(AGE , ncol(Dth)))
Year.F <- factor(rep(YEAR , each=nrow(Dth)))
```

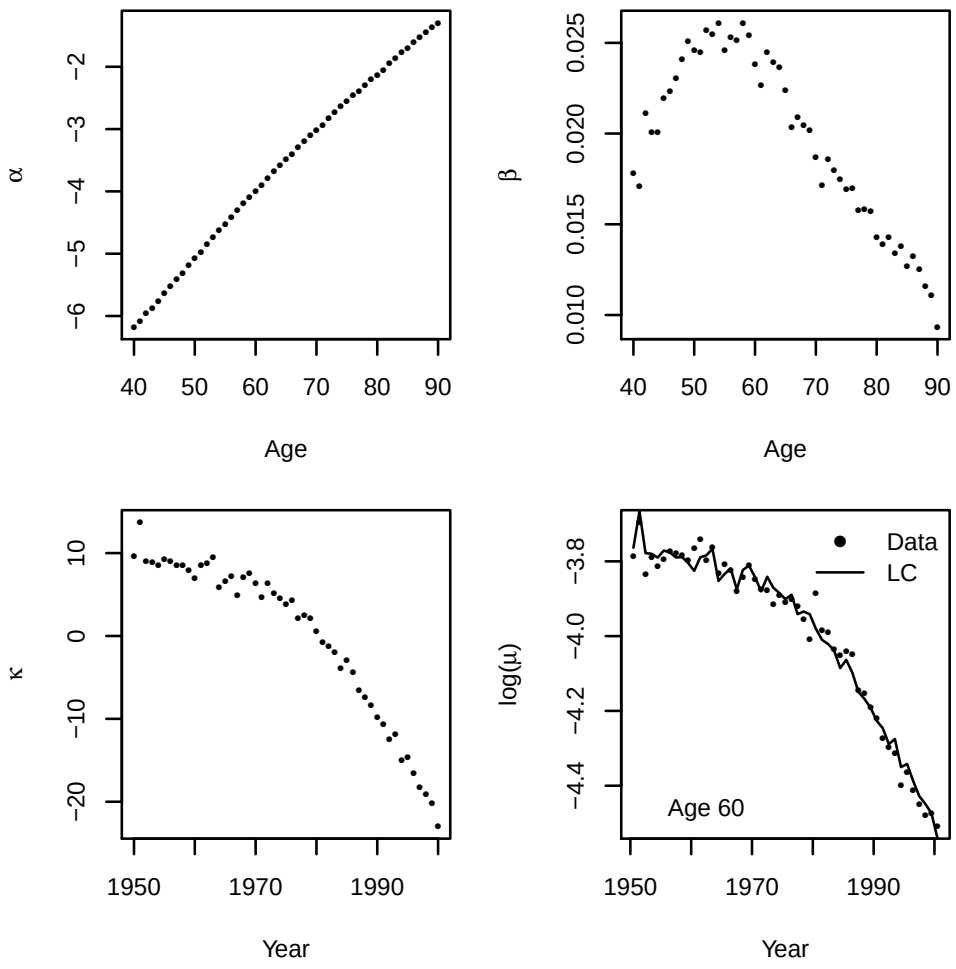


Figure 14: Plots of $\hat{\alpha}, \hat{\beta}$ and $\hat{\kappa}$ for a Lee-Carter model fitted to males, England & Wales, Ages 40–90, from 1950 to 2000 (HMD data). Also shown bottom right is a plot of the fitted force of mortality at age 60, from 1950 to 2000.

- Then we can make the function call to fit the model.

```
LC.Model <- gnm( Dth.V ~ -1 + Age.F + Mult(Age.F,Year.F) , offset=log(Exp.V)
, family=poisson)
```

`gnm()` returns the estimated values in a single vector, `LC.Model$coefficients`, which contains $\hat{\boldsymbol{\alpha}}_R$ followed by $\hat{\boldsymbol{\beta}}_R$ followed by $\hat{\boldsymbol{\kappa}}_R$, and from which they can be extracted.

```
Alpha.gnm <- LC.Model$coefficients[1:51]
Beta.gnm <- LC.Model$coefficients[52:102]
Kappa.gnm <- LC.Model$coefficients[103:153]
```

- However, `gnm()` does not implement the identifiability constraints defined in Section 7.4.2, so we have some further work to do.

```
Kappa.m <- mean(Kappa.gnm)
Beta.m <- mean(Beta.gnm)
Alpha.hat <- Alpha.gnm + Kappa.m * Beta.gnm
Beta.hat <- Beta.gnm / (nrow(Dth) * Beta.m)
Kappa.hat <- nrow(Dth) * Beta.m * (Kappa.gnm - Kappa.m)
sum(Kappa.hat); sum(Beta.hat)
```

It can be shown that `Alpha.hat`, `Beta.hat` and `Kappa.hat` do satisfy the required constraints.

Figure 14 shows the fitted α, β and κ (i.e. `Alpha.hat`, `Beta.hat` and `Kappa.hat`), and a plot of the fitted force of mortality at age 60, for 1950 to 2000. It is obvious that the fitted force of mortality is not a smooth function, the result of fitting so many parameters.

7.4.4 Projecting Future Mortality

Projecting future mortality in the Lee-Carter model is done by projecting κ_y to future years y. Most common is to model κ_y as a *time series* and use standard forecasting techniques from that discipline. No prior knowledge of time series is assumed here so we will consider just the simplest example. This assumes that the difference between successive values of κ_y consists of a constant δ (the trend) plus a zero-mean stochastic error term:

$$\kappa_{y+1} - \kappa_y = \delta + \epsilon_{y+1},$$

and more generally,

$$\kappa_{y+m} - \kappa_y = m\delta + \sum_{i=1}^{m} \epsilon_{y+i}.$$

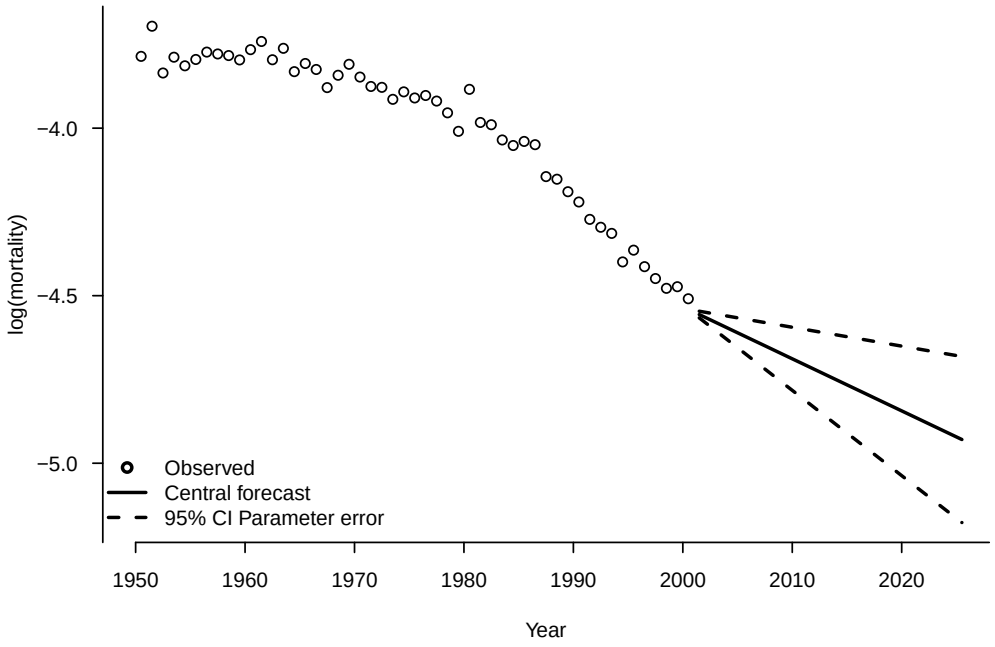


Figure 15: Plot of the Lee-Carter model fitted to males, England & Wales, ages 50–90, from 1950 to 2000 (HMD data) and projected to 2025 for age 60. Also shown are 95% intervals allowing for parameter error (equation (43)).

Fitting this model to the estimated $\hat{\boldsymbol{\kappa}}$ means estimating δ and the variance σ^2 of the error terms (assumed to be the same for all y). Standard estimates assuming normality of the error terms are:

$$\hat{\delta} = \frac{1}{n_y - 1} \sum_{y=1}^{n_y - 1} (\kappa_{y+1} - \kappa_y) = \frac{\kappa_{n_y} - \kappa_1}{n_y - 1}$$

and

$$\hat{\sigma}^2 = \frac{1}{n_y - 2} \sum_{y=1}^{n_y - 1} (\kappa_{y+1} - \kappa_y - \hat{\delta})^2.$$

Hence the standard error of $\hat{\delta}$, obtained by taking the square root of $\mathrm{Var}(\hat{\delta})$, is:

$$\mathrm{SE}(\hat{\delta}) = \frac{\sigma}{\sqrt{n_y - 1}}, \tag{43}$$

where of course we estimate σ by $\hat{\sigma}$.

$\hat{\kappa}_{n_y+m}$, the forecast values for κ_{n_y+m}, $m = 1, 2, 3, \ldots$ are now found by projecting the expected trend, a change of $\hat{\delta}$ each year:

$$\begin{aligned} \hat{\kappa}_{n_y+1} &= \kappa_{n_y} + \hat{\delta} \\ \hat{\kappa}_{n_y+2} &= \hat{\kappa}_{n_y+1} + \hat{\delta} = \kappa_{n_y} + 2\hat{\delta} \\ \hat{\kappa}_{n_y+3} &= \hat{\kappa}_{n_y+2} + \hat{\delta} = \kappa_{n_y} + 3\hat{\delta} \end{aligned}$$

and so on.

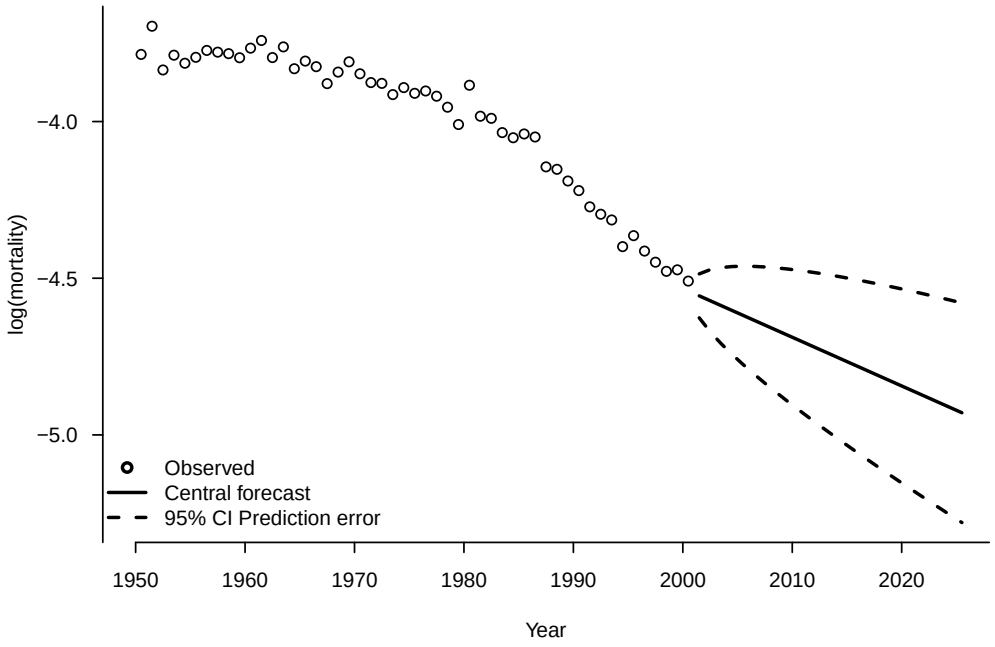


Figure 16: Plot of the Lee-Carter model fitted to males, England & Wales, ages 50–90, from 1950 to 2000 (HMD data) and projected to 2025 for age 60. Also shown are 95% intervals allowing for prediction error (equation (44)).

In general, $\hat{\kappa}_{n_y+m} = \kappa_{n_y} + m\hat{\delta}$, and therefore $\hat{\kappa}_{n_y+m} - \kappa_{n_y} \equiv m\hat{\delta}$ contributes to the *parameter* error:

$$\mathrm{SE}(m\hat{\delta}) \approx \frac{m\hat{\sigma}}{\sqrt{n_y - 1}}.$$

An example of forecasting is carried out by the R script `ForecastLC.R`, which requires the `astsa` library to be installed and loaded (see comments above about installing the `gnm` library). Figure 15 shows the results at age 60, for the model fitted previously.

However, note that this is the standard error arising from the fact that δ and σ were fitted to past observations. It is not the appropriate standard error to use in finding probability intervals for the predicted $\hat{\kappa_y}$, $y > n_y$. These are subject to random error because of the error terms ϵ_y, so the appropriate *prediction* error for projected values is:

$$\mathrm{SE}\left(\sum_{i=1}^{m} \epsilon_{y+i}\right) \approx \sqrt{m}\hat{\sigma} \tag{44}$$

and the 95% *prediction intervals* are given by $\hat{\kappa}_{n_y+m} \pm 1.96\sqrt{m}\hat{\sigma}$. Figure 16 shows the results at age 60, for the model fitted previously.

7.4.5 Discussion of the Lee-Carter Model

The Lee-Carter model is well-suited to forecasting, because the effect of calendar time is confined to the single series κ_y, and this can be modelled using standard time series techniques. In particular this allows us to state some statistical properties of the forecast mortality rates, which overcomes one drawback of deterministic projections.
Beyond that, the method has some disadvantages. It was not originally intended for use in forecasting mortality rates, its purpose was in forecasting expectation of life.

- As mentioned before, the same general 'shape' of mortality rates is imposed at all ages x, because the series κ_y is the sole source of variation over calendar time. This may not be realistic.
- The model assumes that α_x and β_x do not change over time. This may not be realistic. Any particular events in the past that have affected mortality at particular ages (e.g. flu epidemics) may influence the estimated $\hat{\beta}_x$ and therefore be carried forward into the projected mortality. If variability in $\hat{\beta}_x$ is too extreme, it is possible for projected mortality rates to *decrease* with age in a single calendar year.
- As mentioned before, the model has a very large number of parameters, and is inherently unsmooth. This can be overcome by smoothing one or both of α_x or β_x as part of the fitting process (which also helps to avoid the 'cross-over' problem mentioned in the last bullet-point). We would not usually smooth κ_y as part of the fitting process as this is the source of the randomness in the projected mortality.
- The model does not include a cohort term. It is possible that birth cohort may have a stronger influence on changing mortality than calendar year. There is some evidence for this in the UK, though not in all other countries. The influence of birth cohort cannot be represented directly in the Lee-Carter model.

7.5 The Cairns-Blake-Dowd Model

7.5.1 Model Definition

The Cairns-Blake-Dowd (CBD) model is defined as follows:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \kappa_y^{(2)}(x - \overline{x}), \qquad x = x_1, x_2, \ldots, x_{n_x}, \quad y = y_1, y_2, \ldots, y_{n_y} \tag{45}$$

where $\overline{x}$ is the mean age. Rewriting this as:

$$\log(\mu_{x+1/2,y+1/2}) = (\kappa_y^{(1)} - \kappa_y^{(2)}\overline{x}) + \kappa_y^{(2)}x$$

shows that a Gompertz model is fitted in each calendar year y. Therefore, if we assume that the number of deaths in each 'cell' has a Poisson distribution, the CBD model is a GLM (see Section 5.1.2).

The two time series $\kappa_y^{(1)}$ and $\kappa_y^{(2)}$ determine how the two-dimensional Gompertz 'surface' develops over time. Therefore, the idea of projecting future mortality is to model $\kappa_y^{(1)}$ and $\kappa_y^{(2)}$ as a pair of correlated time series, fit them to past data, and then project them into the future.

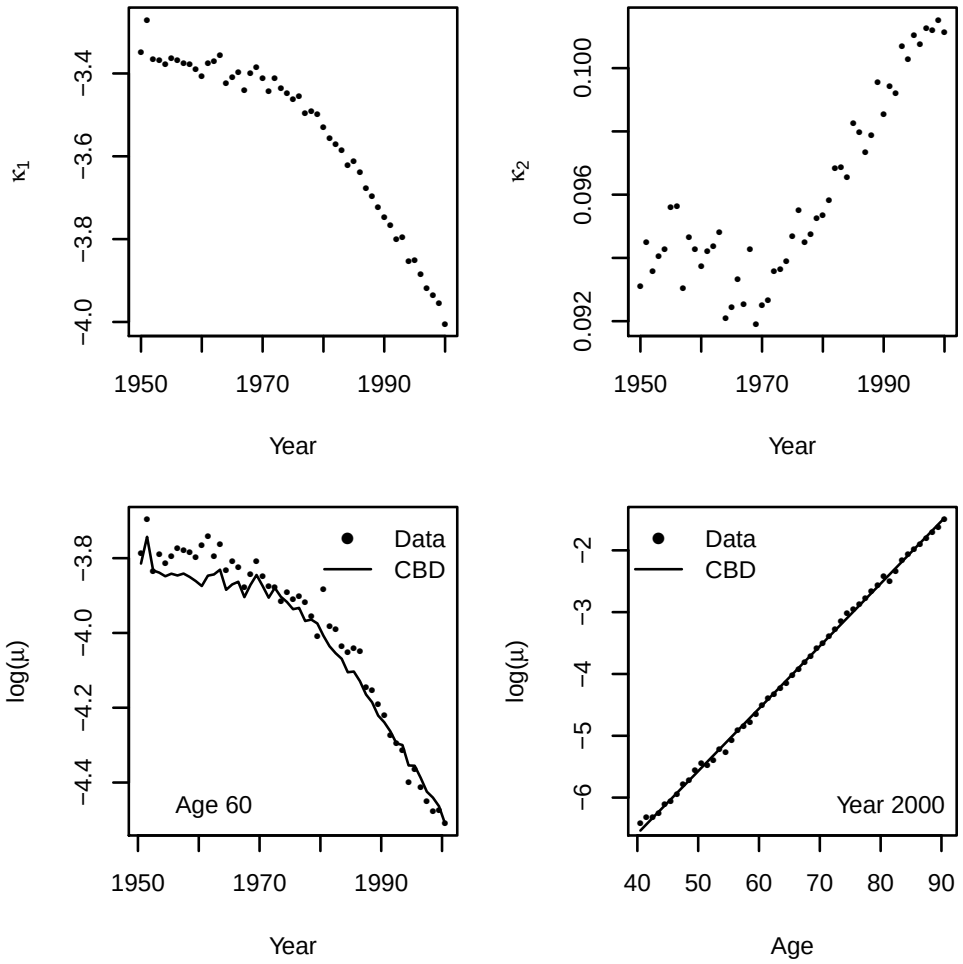


Figure 17: Plots of $\hat{\kappa}^{(1)}$ and $\hat{\kappa}^{(2)}$ for a Cairns-Blake-Dowd model fitted to males, England & Wales, Ages 40–90, from 1950 to 2000 (HMD data). Also shown bottom left is a plot of the fitted force of mortality at age 60, from 1950 to 2000, and bottom right the fitted Gompertz line in 2000.

7.5.2 Fitting the Model

We use the R script `CBD.r` to fit the model. This is closely based on the R program of the same name from Macdonald, Richards and Currie (2018), freely available from the publisher's website: `www.cambridge.org/9781107045415`. The main part of the program is as follows:

```
X1 = kronecker(diag(n.y), rep(1, n.x))
Age.mean = AGE - mean(AGE)
X2 = kronecker(diag(n.y), Age.mean)
X = cbind(X1, X2)
CBD.Male = glm(Dth.V ~ -1 + X, offset=log(Exp.V), family=poisson)
```

where `n.x` $= n_x$, `n.y` $= n_y$ and `AGE` $= 40{:}90$. We omit details of what the `kronecker()` function does; it is just getting the data and parameters lined up correctly for the call to `glm()`. The results are shown in Figure 17.

7.5.3 Projecting Future Mortality

In Section 7.4.4 we treated κ_y as a (univariate) time series and projected its increments as a random walk with constant drift δ.

Here, we have a bivariate time series $\boldsymbol{\kappa}_y = (\kappa_y^{(1)}, \kappa_y^{(2)})$. We do exactly the same and project:

$$\boldsymbol{\kappa}_{y+1} = \boldsymbol{\kappa}_y + \boldsymbol{\delta} + \boldsymbol{\epsilon}_y$$

where $\boldsymbol{\delta} = (\delta^{(1)}, \delta^{(2)})$ is a constant and $\boldsymbol{\epsilon}_y$ is a bivariate Normal random variable. The details of doing this are beyond this course.

7.5.4 Discussion of the Cairns-Blake-Dowd Model

The main advantages of the CBD model are: (a) it fits in the GLM framework; and (b) its simplicity of forecasting. Fitting and forecasting a two-dimensional random walk is not substantially more difficult than fitting and forecasting a one-dimensional random walk (with drift).

Many extensions of the basic CBD model have been proposed (e.g. adding a cohort term, adding a quadratic term) which do substantially improve the fit, but make forecasting much more difficult. This is a trade-off commonly faced in projecting mortality: adding features to a model may improve the fit to past data, but make forecasting difficult or impossible.

7.6 The P-spline Model

7.6.1 Regression with Splines

In Section 5.1.4 we looked at fitting mortality at age 65 over calendar time, using a linear and quadratic model:

$$\begin{aligned} \log(\mathrm{E}[D_x]) &= \log(E_x^c) + a + bx \\ \log(\mathrm{E}[D_x]) &= \log(E_x^c) + a + bx + cx^2. \end{aligned} \tag{46}$$

In principle, we could continue and fit a polynomial of any order we like. However, fitting high-order polynomials generally gives poor results. We may ask, is there anything special

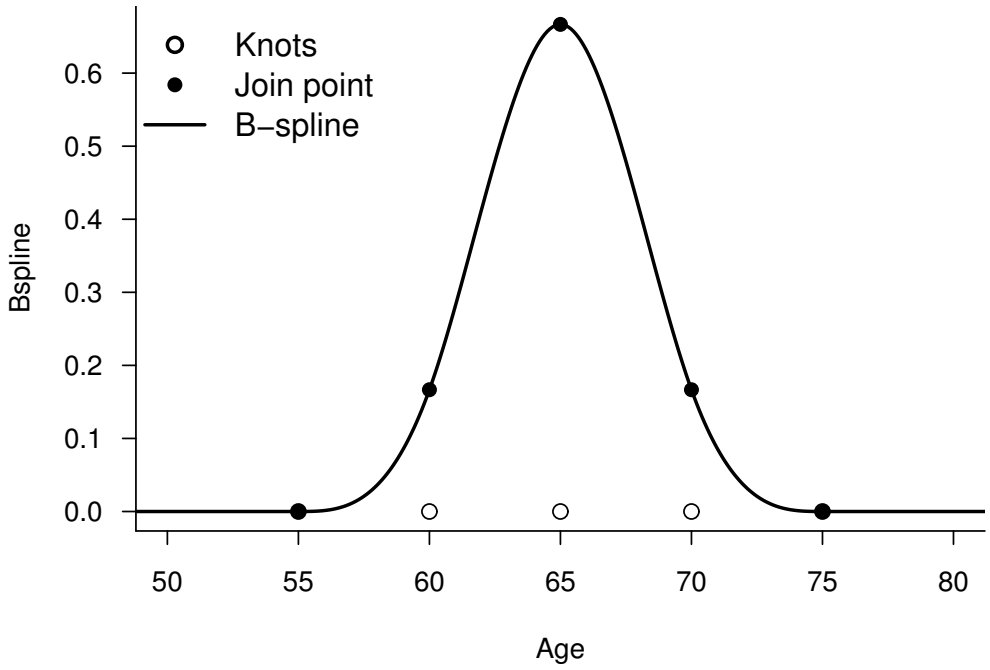


Figure 18: A cubic B-spline with knots at ages 55, 60, 65, 70 and 75. Source: Macdonald, Richards and Currie (2018).

about powers of x for fitting models, or are there better options? That is, can we find a family of functions of x, $f_1(x), f_2(x), \ldots$ such that fitting a model:

$$\log(\mathrm{E}[D_x]) = \log(E^c_x) + \theta_1 f_1(x) + \theta_2 f_2(x) + \ldots + \theta_n f_n(x) \tag{47}$$

to n members of this family gives good results? The answer is yes, there are many such families. One used a lot in actuarial work is *B-splines*.

The problem with using polynomials is that they are *non-local*. Suppose fitting a term in x^4 gives a good fit at ages 30–40. Unfortunately, that x^4 term is also going to be present at ages 80–90, probably with dire results. B-splines are *local* functions: they are non-zero on a defined range, and zero everywhere else. We use cubic B-splines. These are piecewise-cubic functions joined together at a small number of *knots*. At the knots, the function is continuous and has continuous first and second derivatives. Figure 18 shows an example.

We define a *spline basis* to be a set of splines that covers the range of interest. Figure 19 shows a set of evenly-spaced B-splines covering the calendar years 1960–2010. The splines on the left will not influence the fit at the right-hand side, and the splines on the right will not influence the fit at the left-hand side. Compare this with using powers of x as a basis.

We now take a set of basis splines to be our functions $f_1(x), f_2(x), \ldots, f_n(x)$ in equation (47), and use standard R routines to find the regression coefficients $\theta_1, \theta_2, \ldots, \theta_n$. We denote a basis of spline functions by $B_1, B_2, \ldots, B_n$.

7.6.2 P-splines

To graduate using cubic splines, we have to choose how many to use in the basis. We see from Figure 19 that a single cubic spline is defined by five evenly-spaced knots, so if we choose $n + 4$ evenly-spaced knots we have a basis of n splines. How large should n be, and

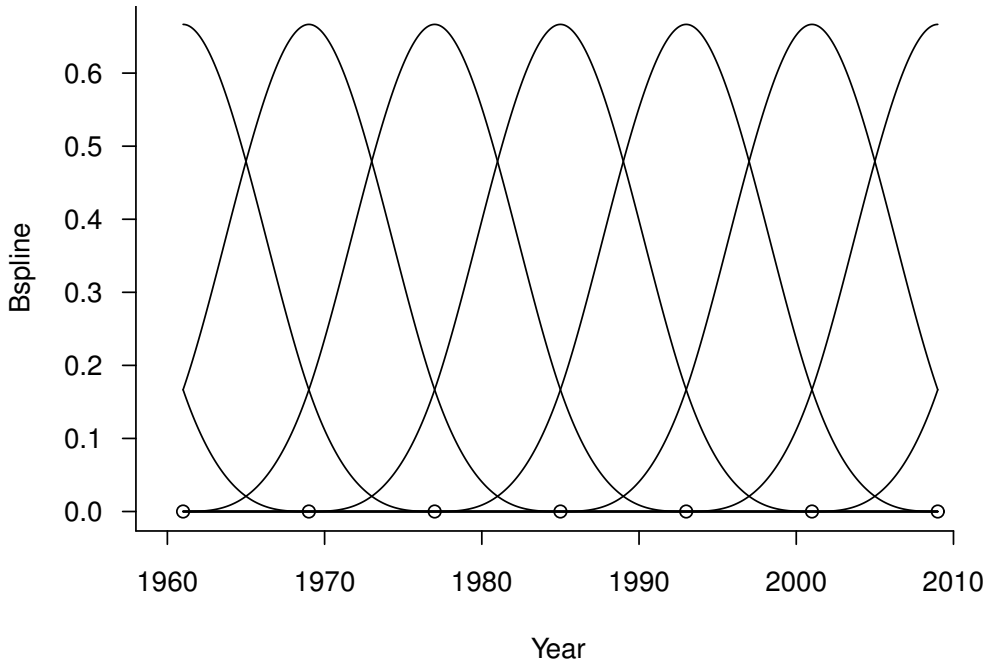


Figure 19: A basis of cubic B-splines covering the period 1960–2010. Source: Macdonald, Richards and Currie (2018).

is there any optimum value of n? If n is small, the fit is smoother, while if n is large the fit follows the data more closely. So we have a trade-off between smoothness and closeness of fit. One way round this is as follows.

- Choose a large number $n + 4$ of evenly-spaced knots, therefore a large number n of basis splines.
- Assume Poisson errors. The usual log-likelihood for μ_x would be:

$$\sum_x \left(d_x \log(\mu_x) - E_x^c \mu_x\right)$$

  and we replace $\log(\mu_x)$ by $\sum_{j=1}^n \theta_j B_j$, giving the log-likelihood $\ell(\boldsymbol{\theta})$ for the parameter vector $\boldsymbol{\theta}$.
- With a large number of basis splines, maximising $\ell(\boldsymbol{\theta})$ would give very close adherance to the data, which we may not want. So define a function $P(\boldsymbol{\theta})$ that measures the smoothness of the fit: small for a smooth fit and large for a rough (wiggly) fit. Then instead of maximising $\ell(\boldsymbol{\theta})$, maximise:

$$\ell_P(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) - \frac{1}{2}\lambda P(\boldsymbol{\theta}).$$

- The function $P(\boldsymbol{\theta})$ is called the *penalty* and this approach is called smoothing with penalised splines or *P-splines*. The key quantity is λ, the *smoothing parameter* (the $1/2$ is just for convenience in certain calculations). Choosing a large value of λ favours a smooth fit over a close fit, choosing a small value does the opposite. So the trade-off between smoothness and goodness-of-fit, and any question of optimisation, is determined by the choice of λ.

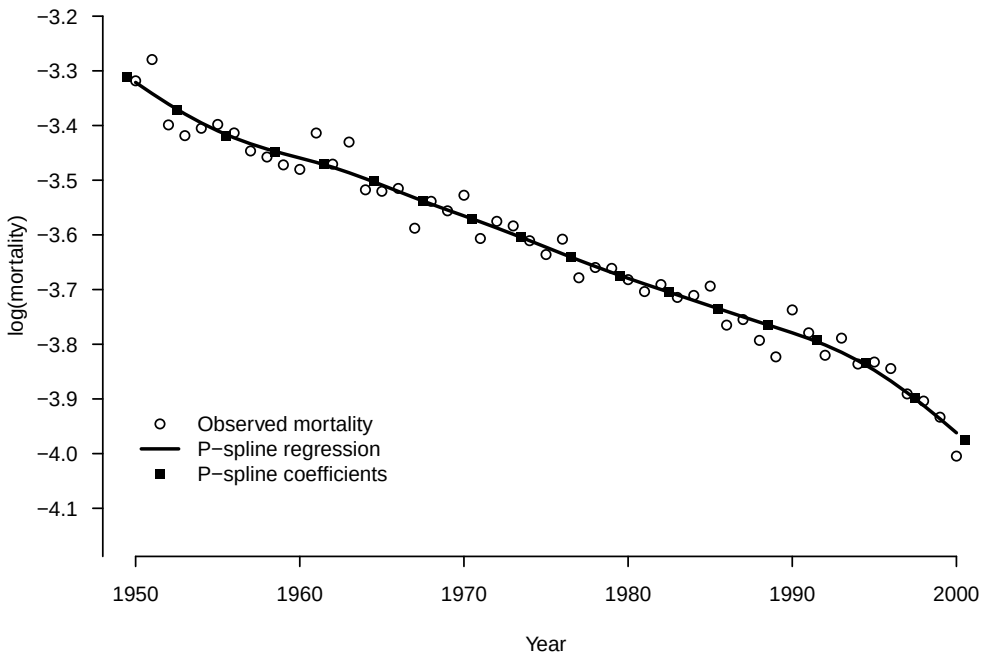


Figure 20: A P-spline graduation of female mortality at age 70 in England & Wales, 1950–2000. Source: HMD data.

- The penalty function most often used is:

$$P(\boldsymbol{\theta}) = (\theta_1 - 2\theta_2 + \theta_3)^2 + \ldots + (\theta_{n-2} - 2\theta_{n-1} + \theta_n)^2. \tag{48}$$

7.6.3 Projecting Future Mortality

Projecting future mortality in the P-spline model exploits features of the fitting process which we just outline briefly. Suppose we have data for 1950–2000, and we want to project this for 30 years, i.e. to 2030.

- Define dummy death and exposure data for 2001–2030. Any positive values will do.
- Extend the spline basis to cover 2001–2030.
- Define a weight vector that gives weight 1 to data in 1950–2000 and weight 0 to data in 2001–2030.
- Calculate the likelihood using only data with weight 1.
- Choose regression coefficients $\hat{\theta}_j$ for the additional basis splines covering 2001–2030 such that their contribution to the penalty term is zero.

The last two points result in exactly the same fitted model in 1950–2000 as in Figure 20. Let $\hat{\theta}_{c-1}$ and $\hat{\theta}_c$ be the last two regression coefficients in the spline basis covering 1950–2000. The first term in the penalty involving 2001–2030 is $(\hat{\theta}_{c-1} - 2\hat{\theta}_c + \hat{\theta}_{c+1})^2$. Requiring this to be zero means $\hat{\theta}_{c+1} = 2\hat{\theta}_c - \hat{\theta}_{c-1}$. Thus $\hat{\theta}_{c-1}, \hat{\theta}_c$ and $\hat{\theta}_{c+1}$ lie on a straight line. Continuing in this way, the projection is a linear continuation of the fitted model.

The last part of `PsplineRegression.r` implements this process and produces Figure 21.

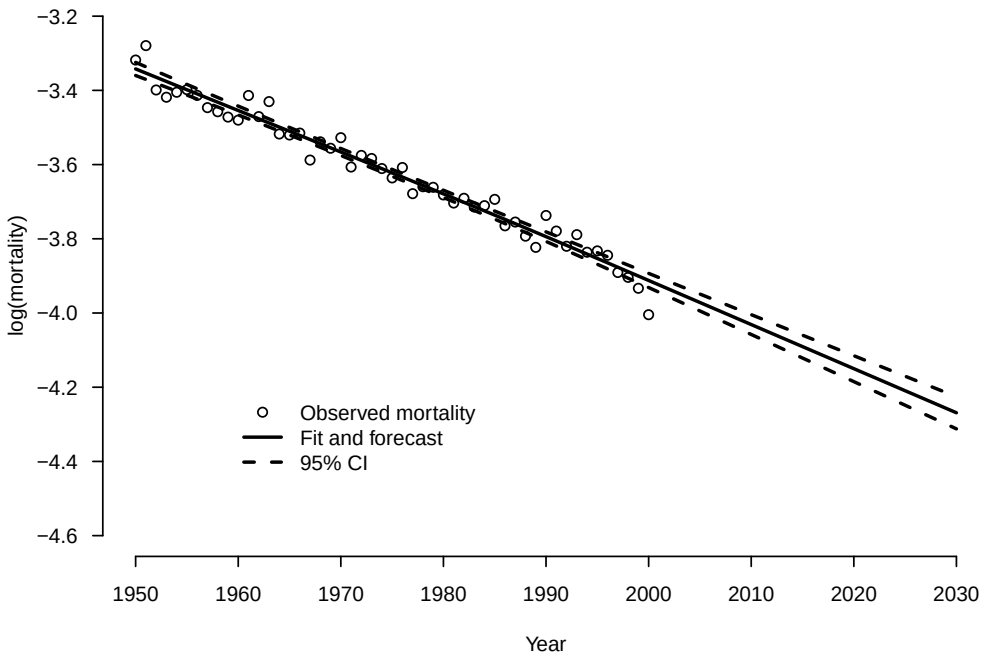


Figure 21: A P-spline graduation of female mortality at age 70 in England & Wales, 1950–2000, with projection to 2030. Source: HMD data.

Projecting future mortality in a consistent way at all ages is done by defining two-dimensional splines, with knots on a grid rather than along a line. This is beyond the scope of this course.

7.6.4 Discussion of the P-spline Model

The cubic P-spline approach is a natural extension of cubic spline smoothing, which has the good features of cubic polynomials as approximating functions, but localised so that 'long-range' effects in the fitted model are minimized. Spline models are regression models and the fitted coefficients are interpretable as regression coefficients. Introducing the penalty in P-splines reduces the choice of knots and basis splines to the choice of the smoothing parameter λ.

Unlike the other methods considered here, fitting to past data and forecasting future mortality are done simultaneously, because the projection is taken care of by the penalty function.

However, there are drawbacks in using the model for projections.

- There is no interpretable basis for allowing the penalty to determine the forecast, it is a mathematical convenience. The resulting forecasts may have objectionable features if certain properties are required *a priori*.

- Forecasts can be very sensitive to the last few years of fitted data. Therefore, as forecasts are updated from one year to the next they can change more abruptly than is desirable for practical work (pricing and reserving).

7.7 The Age-Period-Cohort Model

7.7.1 Cohort Effects

The main reason for considering an Age-Period-Cohort model is that quite strong *cohort effects* have been detected in some countries (and in CMI data), though not in every country. A cohort effect means that persons born around a particular year seem to have mortality improvements significantly different from those of neighbouring cohorts. The UK, USA, France and Japan are countries with cohort effects. In the UK, the cohort born between about 1925 and 1934 has experienced unusually rapid improvements in mortality; it is sometimes called the 'golden cohort'. Explanations that have been suggested include diet in early life (the restricted diet during World War II rationing was actually very healthy), the inception of the NHS in 1948, economic prosperity and declining cigarette smoking.

Cohorts that may experience particularly rapid mortality improvements are financially significant for pension funds and writers of annuities. Therefore models that include a cohort effect have been proposed for actuarial work in recent years (they have been used elsewhere for longer).

7.7.2 Model Definitions

A good reference for this material is Cairns, A.J.G., Blake, D., Dowd, K., Coughlan, G.D., Epstein, D., Ong, A., and Balevich, I. (2009). *A quantitative comparison of stochastic mortality models using data from England and Wales and the United States.* North American Actuarial Journal **13(1)**, 1-35.

The simplest age-period-cohort (APC) model is:

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x + \kappa_y + \gamma_c$$

where β_x is a parameter for each age x, κ_y is a parameter for each calendar year y, and γ_c is a parameter for each birth cohort c. Note that if c is the calendar year of birth then $c = y - x$, and we can write any cohort effect as $\gamma_c = \gamma_{y-x}$.

More generally, if we let $\beta_x^{(i)}$ be the ith age effect, $\kappa_y^{(i)}$ be the ith calendar year effect, and $\gamma_{y-x}^{(i)}$ be the ith cohort effect, for $i = 1, 2, \ldots, n$, we can write an APC model as:

$$\log(\mu_{x+1/2,y+1/2}) = \sum_{i=1}^{n} \beta_x^{(i)} \kappa_y^{(i)} \gamma_{y-x}^{(i)}.$$

For example the Lee-Carter model (with no cohort effect) can be written as:

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x^{(1)} + \beta_x^{(2)} \kappa_y^{(2)}$$

with $n = 2$ and $\kappa_y^{(1)} = \gamma_{y-x}^{(1)} = \gamma_{y-x}^{(2)} = 1$. An extension of the Lee-Carter model with a cohort effect is the Renshaw-Haberman model:

$$\log(\mu_{x+1/2,y+1/2}) = \beta_x^{(1)} + \beta_x^{(2)} \kappa_y^{(2)} + \beta_x^{(3)} \gamma_{y-x}^{(3)}.$$

The CBD model can be written as:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \beta_x^{(2)} \kappa_y^{(2)}$$

where $\beta_x^{(2)} = (x - \overline{x})$, and this can be extended in various ways by adding cohort terms, for example:

$$\log(\mu_{x+1/2,y+1/2}) = \kappa_y^{(1)} + \beta_x^{(2)}\kappa_y^{(2)} + \gamma_{y-x}^{(3)}$$

(which is known as Model M6).

7.7.3 Identifiability Constraints

We noted above that we can write any cohort effect as $\gamma_c^{(i)} = \gamma_{y-x}^{(i)}$. This means that, like the Lee-Carter model, an APC model is generally unidentifiable. There will be many sets of parameters that maximise the likelihood. To get around this we need to impose constraints, which to a certain extent are arbitrary. Constraints that work for the Renshaw-Haberman model are:

$$\sum_y \kappa_y^{(2)} = 0 \qquad \sum_x \beta_x^{(2)} = 1 \qquad \sum_{x,y} \gamma_{y-x}^{(3)} = 0 \qquad \sum_x \beta_x^{(3)} = 1.$$

Constraints that work for Model M6 are:

$$\sum_{x,y} \gamma_{y-x}^{(3)} = 0 \qquad \sum_{x,y} (y-x)\gamma_{y-x}^{(3)} = 0.$$

7.7.4 Fitting Age-Period-Cohort Models

As long as suitable constraints are imposed to ensure identifiability, APC models can be fitted using maximum likelihood in the same way as the other models discussed here. There is an increased level of complexity, and some R packages do exist that fit special cases of APC models, but we will not pursue that here.

7.7.5 Projecting Future Mortality

With the Lee-Carter model time series approaches were used to project κ_y and therefore projections of future mortality. With the CBD model, time series approaches were used to project $\kappa_y^{(1)}$ and $\kappa_y^{(2)}$ simultaneously. Similarly, in an APC model time series approaches can be used to project all the $\kappa_y^{(i)}$ and $\gamma_{y-x}^{(i)}$ simultaneously. This is an advanced topic that we will not pursue here.

7.7.6 Discussion of Age-Period-Cohort Models

The main advantage of an APC model is that if cohort effects are thought to exist and may be of financial importance, then a model that includes them and can project them is desirable.

APC models are more difficult to fit and to interpret than the other models discussed here.

- They are generally unidentifiable, and constraints on the parameters are needed (or some other solutions have been suggested).
- They are very demanding of the data. Generally speaking, we are interested in projecting the mortality of the incomplete cohorts (say, persons approaching retirement) but we need data covering complete cohorts (up to 100 years) to fit the model.

- Some commentators have suggested that cohort effects are not real, rather that they are an artifact of the modelling process.

7.8 Sources of Forecast Error

Forecasting mortality for long periods (40–50 years) is subject to many uncertainties. Macdonald, Richards and Currie (2018) list the following seven sources of error.

- **Model risk.** We may have chosen the 'wrong' model (this is not to suggest that there is a 'right' model). Our example of fitting the CBD model to males in 1950–2000 did not look very convincing before 1970; a different model might give a better fit.

- **Parameter error.** This is the usual sampling variance of parameter estimates in any parametric model.

- **Stochastic or prediction risk.** The stochastic error in the forecast rates of mortality, if we use time series methods which have in-built uncertainty of their own (see equation (44)).

- **Basis risk.** The actuary may wish to project the future mortality of a pension scheme, or a line of life insurance business. The available data will almost always be too little to fit any projection model, so the projections will be based on one of the available large data sets (usually national statistics). To the extent that this differs systematically from the experience the actuary is interested in, the projections will be wrong. It may be possible to model the difference and adjust the projections, which is very similar to graduation by reference to a standard table (see Section 5.3).

- **Volatility.** Short-term forecasts may be wrong because they do not predict random fluctuations, such as severe winters, hot summers or flu epidemics.

- **Idiosynchratic risk.** Heterogeneity in the portfolio of interest may not be reflected in the projections. For example, longevity is strongly associated with wealth, and proxy information correlated to wealth may be known to the actuary (e.g. benefit amount or income). Separate projections allowing for this would be desirable, but these are probably impossible to obtain.

- **Mis-estimation risk.** The underlying data may not support the estimation of mortality rates or projections with any degree of confidence. For example, even the national statistics of a small country may not support mortality projections.

Tutorial 7: Mortality Projection

1. In Section 7.4.3 the call to the R function `gnm()` resulted in fitted parameters `Alpha.gnm`, `Beta.gnm` and `Kappa.gnm`. Show that the transformations below:

```
Kappa.m <- mean(Kappa.gnm)
Beta.m <- mean(Beta.gnm)
Alpha.hat <- Alpha.gnm + Kappa.m * Beta.gnm
Beta.hat <- Beta.gnm / (nrow(Dth) * Beta.m)
Kappa.hat <- nrow(Dth) * Beta.m * (Kappa.gnm - Kappa.m)
sum(Kappa.hat); sum(Beta.hat)
```

   result in estimates `Alpha.hat`, `Beta.hat` and `Kappa.hat` that satisfy the identifiability constraints in Section 7.4.2.

2. Give an expression for the standard error of projected values of $\log(\mu_{x+1/2,y+1/2})$ under the Lee-Carter model, where κ_y is projected as a random walk with drift (see Section 7.4.4) allowing for both stochastic error (i.e. that due to fitting $\hat{\delta}$ to past data) and prediction error (i.e. that due to projecting κ_y).

3. Consider a basis of B-splines $B_1, B_2, \ldots, B_n$ with evenly-spaced knots, as in Figure 19. Suppose the regression coefficients are all equal, $\theta_1 = \theta_2 = \ldots = \theta_n = \theta$. By considering the first derivative of the fitted model $f(x) = \theta \sum_i B_i(x)$ show that $f(x)$ is a constant.

4. (a) Fit a Lee-Carter model to the HMD data for England & Wales, males ages 50–90, years 1940–1980. Plot the values of $\hat{\boldsymbol{\alpha}}_x$, $\hat{\boldsymbol{\beta}}_x$ and $\hat{\boldsymbol{\kappa}}_y$.

   (b) By projecting $\hat{\boldsymbol{\kappa}}_y$ as a random walk with drift (see Section 7.4.4) find projected rates $\log(\mu_{x+1/2,y+1/2})$ at ages 50–90, years 1981–2005. Choose an age in the range 50–90 and plot 95% prediction intervals of the projected rates at that age.

   (c) Compare the projected rates at your chosen age with the actual crude rates from the England & Wales data in 1981–2005. Comment on the results.

   (d) Repeat (b) and (c) for some other ages. Do you think the Lee-Carter method would have successfully predicted future mortality in England & Wales had it been used by an actuary in 1980?

5. (a) Fit separate 1-dimensional P-spline models to the HMD data for England & Wales, females ages 50, 70 and 90, years 1940–1980. Plot the results.

   (b) Project each of the fitted models in (a) to find rates $\log(\mu_{x+1/2,y+1/2})$ at ages 50, 70 and 90, years 1981–2005. Plot the results with 95% confidence intervals.

   (c) Compare the projected rates at ages 50, 70 and 90 with the actual crude rates from the England & Wales data in 1981–2005. Do you think the P-spline model would have successfully predicted future mortality in England & Wales had it been used by an actuary in 1980?

8 Exposed to Risk

8.1 Introduction

Exposed to risk arises naturally in models of mortality; it is the 'denominator' in the calculation of $\hat{\mu}_x$ or $\hat{q}_x$. In particular,

- **Central Exposed to Risk**, E_x^c, is the observed waiting time in a multiple-state or Poisson model, and
- **Initial Exposed to Risk**, E_x, is the exposed to risk which applies in the binomial model of mortality. Often, E_x must be calculated in some approximate fashion, as the actuarial estimate $E_x \approx E_x^c + \frac{1}{2}d_x$.

We then estimate

$$\begin{aligned} \hat{\mu}_x &= \frac{d_x}{E_x^c}, \\ \text{or} \quad \hat{q}_x &= \frac{d_x}{E_x}. \end{aligned}$$

Of course, when calculating $\hat{q}_x$ or "initial" type mortality rates we can also find E_x^c within a multiple state framework, estimate $\hat{\mu}_x$, and then estimate $\hat{q}_x$ from $q_x = 1 - e^{-\mu_x}$. This method is generally preferred since we do not require to make additional assumptions to calculate the initial exposure, and because estimates based on the Markov model are more efficient in a statistical sense.

NB: The transformation from $\mu \to q$ could be **before or after** a graduation has been made.

Furthermore, the Markov model extends in a straightforward way to more complicated studies (e.g. illness and death models). It is notable that all of the recent large graduations in the UK have first used E_x^c to calculate $\hat{\mu}_x$, and then found $\hat{q}_x$ from this value of $\hat{\mu}_x$. See, for example, various CMIB reports.

In this course, we will concentrate on the computation of **central exposed to risk** from typical actuarial data. If, for some particular reason, we must use the binomial model and so an initial exposed to risk measure is required we will usually calculate this from the approximate relationship

$$E_x \approx E_x^c + \tfrac{1}{2}d_x.$$

NB: For most ages with adequate numbers of exposed to risk the two methods for calculating q_x described above would give similar numerical results.

Historically, the study of the calculation of exposed to risk was very important since a census style approach to its calculation usually had to be adopted. In the computer age we know policy details on a daily basis, i.e., on any day we know for each policyholder i) date of birth, ii) date of entry, iii) date of exit, iv) reason for exit. It is thus a trivial task to calculate the exact exposed to risk at any age, and the corresponding number of deaths. Nevertheless, there are various issues that need to addressed (i) the definition of age, x, (ii) the possibility of a clash in the definitions used to define age, x, for those that are exposed to risk and those that die (see §8.3).

8.2 Homogeneity

We emphasise a fundamental assumption: all our lives are identical, at least as far as mortality is concerned. We approximate this ideal state by subdividing our lives into more or less homogeneous groups, while recognising that some (perhaps much) heterogeneity will remain.

Common factors in life insurance are

(a) Age - this is almost always used.

(b) Sex - again this is almost always used.

(c) Type of policy - connected to the reason for insuring, e.g. annuities.

(d) Smoking status.

(e) Duration in force - selection effect, since mortality increases with duration.

(f) Occupation of policyholder.

Comments

(a) Not all these factors will be used since marketing and/or administrative considerations lead to shorter proposal forms.

(b) A further point is that if we over-divide our data we start having the problems associated with too little data.

8.3 The principle of correspondence

In a mortality investigation we want to estimate μ_x or q_x. We do this by bringing together

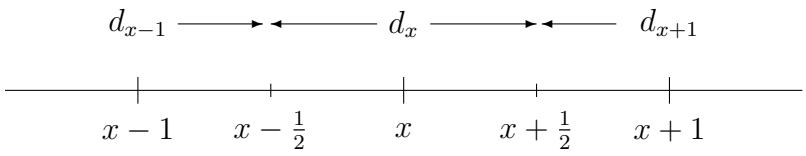
- data on exposures, and
- data on deaths.

Problems can arise if these data come from two different sources. For example, we might get the deaths information from the **claim file**, and the exposure information from the **premium collection file**. Ideally these two sources should use the **same definition of age**, but there is no guarantee that this will be so.

Exact age and age labels. We distinguish sharply between the **exact age** of a policyholder (usually computed to the nearest day) and the **age label** used to describe a policyholder. All the problems with exposed to risk come from the use of age labels, particularly when the label used to measure the exposed to risk at a particular age and the label used to count the number of deaths at that age are different. An example illustrates the problem.

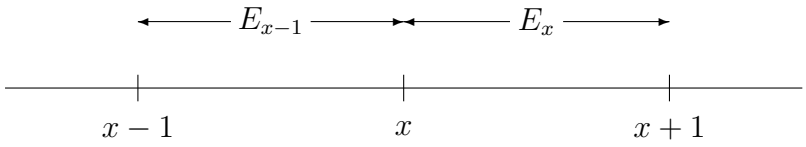
Example: The life office may compute deaths and exposures by

- Death is recorded as being at 'age nearest birthday at date of death'.



Thus, any death in $[x-\frac{1}{2}, x+\frac{1}{2})$ contributes to d_x.

- Exposure is recorded as 'age last birthday'.



Thus, exposure in $[x, x+1)$ contributes E_x. In particular, a life that dies age $x+0.75$ exact contributes to d_{x+1} but at the time of death is exposed to risk age x.

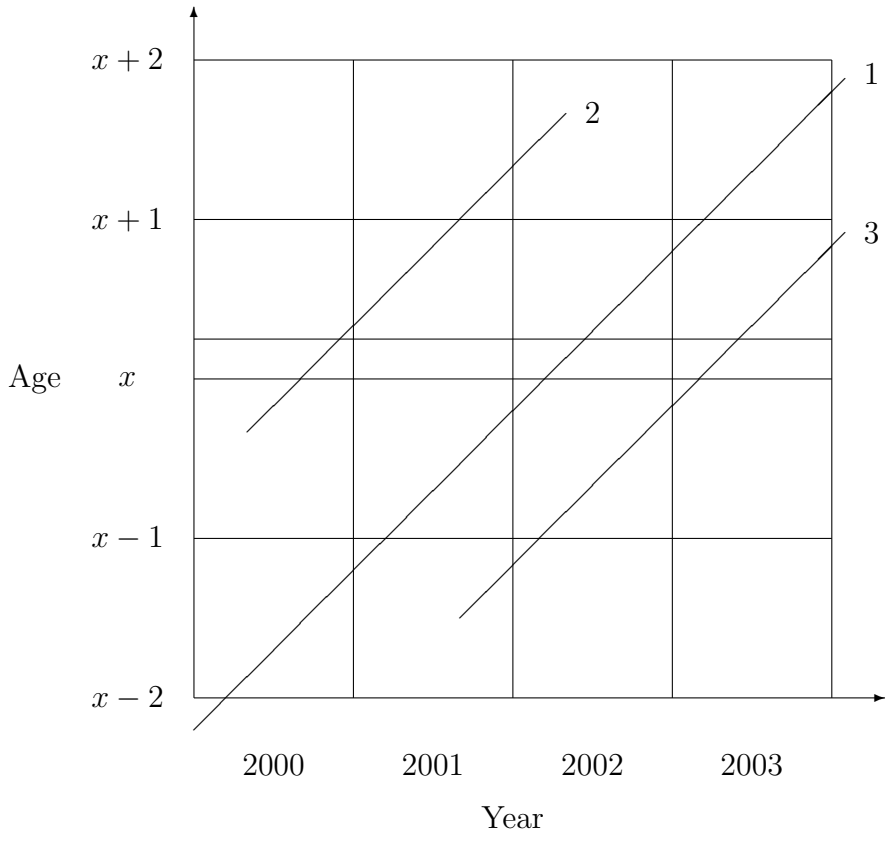
Principle of correspondence: death data and exposure data must use the same age definition, that is, if a life is counted as being exposed to risk of death at age label x then if he/she were to die he/she would be counted in the deaths **at the same age label.**

When the death and exposure data do not correspond (as in the above example) one or other is adjusted so that the above principle is not violated. Normally the census/exposure data are adjusted since they are more extensive and regular than deaths data.

Golden Rule: We will always adjust the exposure data to fit the deaths data. The reason for this is that d_x is subject to more random variation than E_x^c so it seems sensible to adjust the smoother variable; we have more chance of getting the "correct" adjustment.

8.4 Exact calculation of E_x^c

In principle, the calculation of E_x^c is trivial given the appropriate data on the individual lives observed in the study. In the example below, the study period is the four year period 1 January 2000 until 31 December 2003.



The diagram (known as a **Lexis diagram**) plots exact age (y-axis) against calendar time (x-axis); each life observed is represented by a 45° line. (See www.demog.berkeley.edu/ bmd/lexis.html for a similar explanation).

- **Life 1** first comes under observation on 1 January 2000 at age $x - 2.2$ and remains under observation until 31 December 2003 at which time he/she is aged $x + 1.8$, i.e., 4 years exposure split between different age intervals.
- **Life 2** first comes under observation on $2000 + 0.3$ at age $x - 0.4$ leaving (by death or some other decrement) in $2002 + 0.4$ at age $x + 1.7$, i.e., 2.1 years exposure split between different age intervals.
- **Life 3** enters the study on $2001 + 0.6$ at age $x - 1.5$ and remains until the end of the investigation on 31 December 2003 at which time he/she is aged $x + 0.9$, i.e., 2.4 years exposure split between different age intervals.

A **horizontal cross section**, say at age $x + \frac{1}{4}$, will cut a number of "life lines". This will be the number of lives attaining age $x + \frac{1}{4}$ during the entire study period.

Similarly a **vertical cross section** gives information on the number of lives between certain ages on a particular date.

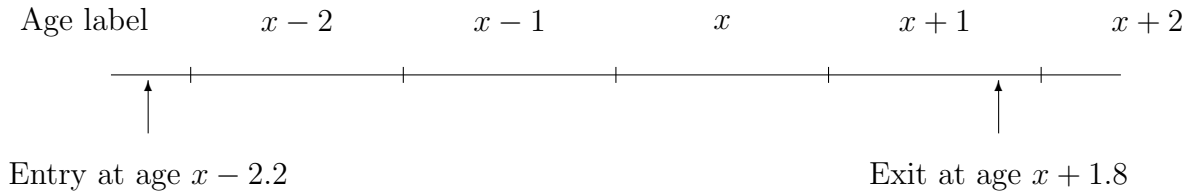
To calculate E_x^c we require for each life observed during some period $[0, T]$

- date of birth
- date of entry
- date of leaving

We then simply accumulate exposure for each age according to the definition used in our investigation. This is the **exact (or direct) exposure method**. If we add to our data the reason for the cessation of observation then we have d_x as well, and we are finished.

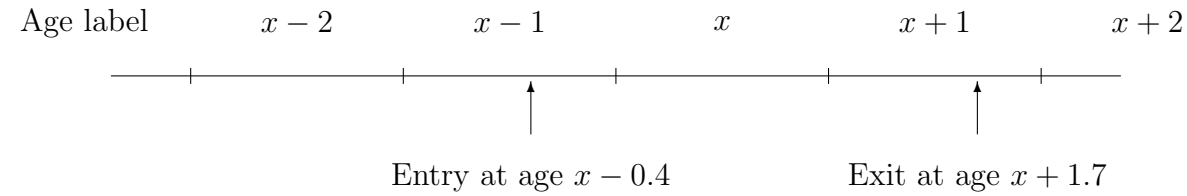
Example: Age label 'age last birthday'

Suppose we use the age label, 'age last birthday' in the above example. In the following diagrams, the age label x applies to the interval $[x, x + 1)$, etc. Then we have for Life 1:



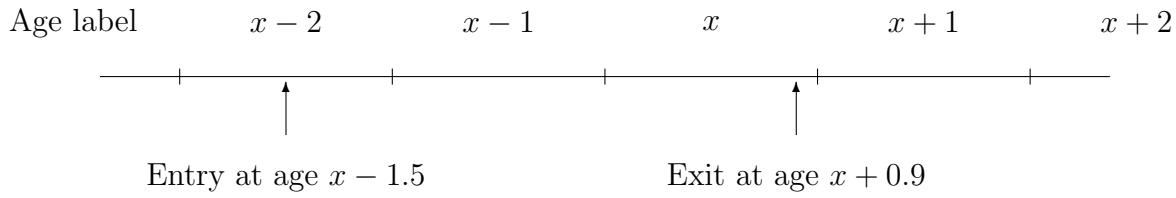
and we can simply read off the contributions to the exposures: 0.2 to E_{x-3}^c, 1 to each of E_{x-2}^c, E_{x-1}^c and E_x^c and 0.8 to E_{x+1}^c (making a total of 4 years).

For Life 2 we have:



and again we can simply read off the contributions to the exposures: 0.4 to E_{x-1}^c, 1 to E_x^c and 0.7 to E_{x+1}^c (making a total of 2.1 years).

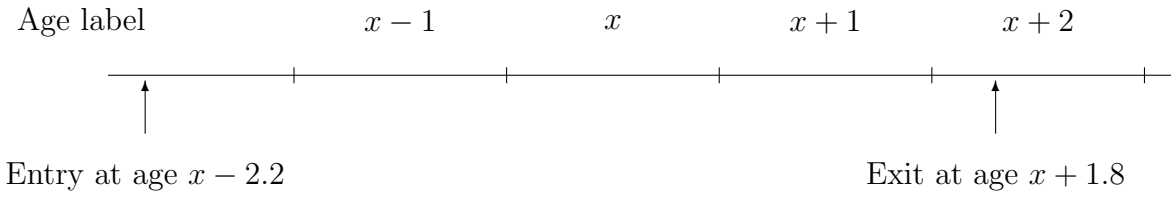
Finally, for Life 3 we have:



and again we can simply read off the contributions to the exposures: 0.5 to E_{x-2}^c, 1 to E_{x-1}^c and 0.9 to E_x^c (making a total of 2.4 years).

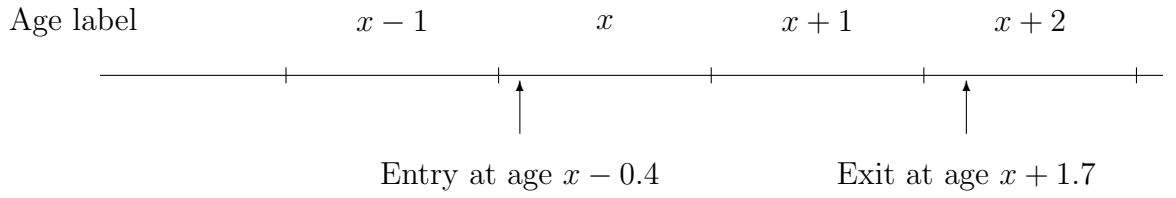
Example: Age label 'age nearest birthday'

Suppose we change the definition of the age label to 'age nearest birthday' then the age label x applies to the interval $[x-0.5, x+0.5)$, etc., as indicated in the following diagrams. The contribution to the exposures is computed for Life 1 as follows:



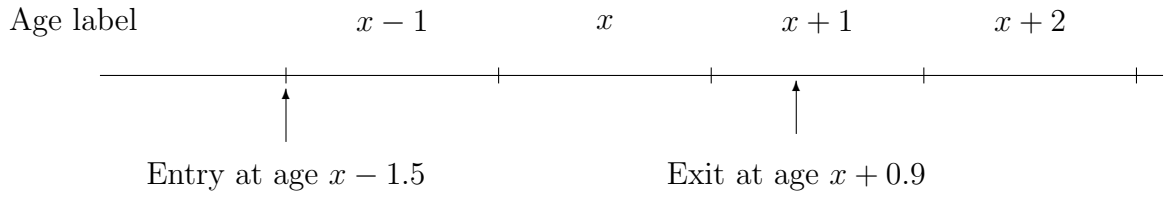
and again we can simply read off the contributions to the exposures: 0.7 to E_{x-2}^c, 1 to each of E_{x-1}^c, E_x^c and E_{x+1}^c and 0.3 to E_{x+2}^c (making a total of 4 years).

For Life 2 we have



and the contributions to the exposures are 0.9 to E_x^c, 1 to E_{x+1}^c and 0.2 to E_{x+2}^c (making a total of 2.1 years).

Finally, for Life 3 we have



and the contributions to the exposures are 1 to E_{x-1}^c, 1 to E_x^c and 0.4 to E_{x+1}^c (making a total of 2.4 years).

8.5 Rate intervals

8.5.1 Life year rate intervals

Insurance companies use various definitions for age labels. So far, we have looked at the following definitions: (the familiar) age last birthday, age nearest birthday and age next birthday. With such definitions, the rule for a change in the age label refers directly to the policyholder's birthday. For example, with age nearest birthday a policyholder's age changes at exact ages $x - 0.5$, $x + 0.5$, etc. With age last birthday and age next birthday the age label changes on the birthday. A definition that leads to a change of age label defined in terms of the birthday of the policyholder is known as a **life rate interval**. The following table is a useful summary:

Age Definition	**Age Interval**	$\hat{\mu}$ **Estimates**	$\hat{q}$ **Estimates**
x last birthday	$[x, x+1)$	$\mu_{x+\frac{1}{2}}$	q_x
x nearest birthday	$\left[x-\frac{1}{2}, x+\frac{1}{2}\right)$	μ_x	$q_{x-\frac{1}{2}}$
x next birthday	$(x-1, x]$	$\mu_{x-\frac{1}{2}}$	q_{x-1}

Other definitions of age labels occur: **calendar year rate intervals** and **policy year rate intervals** are two possibilities. We discuss these briefly in turn.

8.5.2 The calendar year rate interval

Suppose the age label is determined with reference to a **fixed point in the calendar year**. In this case, the age label will change on a **fixed date** in the year **for all individuals**: a **calendar year rate interval** results. Here are some examples:

- Age last birthday on the preceding 1 January.
- Age nearest birthday on the preceding 30 June.
- Age attained on the birthday in the current calendar year.

One common way for a calendar year rate interval to arise is if the only information held is (i) the year of birth, and (ii) the year of death. The rate interval is then the calendar year running from 1st January to 31st December. One way to see this is that a life is included in d_x regardless of when in the year the death occurs. One would like to think that lack of information on this scale no longer happens!

One difficulty with the Calendar Year rate interval is that the number of lives exposed to the risk of death is **not a continuous function**. This happens because all individuals change age labels **at the same time** as they pass a particular date in the calendar year. As a result, census formulae are more complicated than for rate intervals where lives move in and out of different age labels continuously over the year (for example, the life year rate interval where lives enter and leave the rate interval continuously - birthdays are roughly uniform over the year).

8.5.3 The policy year rate interval

With this rate interval, the recorded age label of an individual is determined with reference to the anniversary of the date the person took out the policy (or joined the pension scheme, etc.) Here are some common examples:

1. age on birthday in the previous policy year,
2. age next birthday at entry plus number of completed policy years,
3. age last birthday at the nearest policy anniversary.

Example: Consider a life with: date of birth 14/03/46, date of death 19/09/92, and date of policy start 12/05/79. What is the age at death if age is defined as age last birthday on 1 April in the policy year of death?

Note first that the age label will change on the policy anniversary so we are dealing with a **policy year rate interval**. The argument is

- The policy year of death starts on 12/05/92
- 1 April in policy year of death is 01/04/93
- Age last birthday on 01/04/93 is 47 ■

In the majority of cases policy anniversaries are spread uniformly over the year and this can be assumed to be the case **unless policy information contradicts this assumption**; for example, individuals can only join a scheme on 1 January or 30 June in any year.

The remainder of this chapter is concerned with what we do when the data on exposed to risk is not complete. Again, this should not arise with modern computing facilities but two cases occasionally arise.

(a) What happens when the exact dates of entry to and exit from observation have not been recorded? In section 8.6 we use **census information** to estimate the exposed to risk.

(b) What happens when the definition of age used for the census data does **not correspond exactly** to the age interval x to $x+1$ (for integer x) with the definition of age used for the deaths? In section 8.7 we show how to estimate the exposure we require.

8.6 Calculation of E_x^c from census data

A **census** is simply a head count at a particular time instant. Data submitted to the CMIB is usually of census form. For example, it is convenient for companies to submit the total number of policies in force on a particular date, say January 1st.

We assume for the moment that

- the **age label** x means 'age x last birthday'.

Suppose we carry out observations from time $t = T_1$ to $t = T_2$ and define:

- $P_{x,t}$ to be the number of lives under observation between age x and $x+1$ exactly at time t, i.e. those with age label x last birthday.

This corresponds to the vertical section discussed in the context of the Lexis diagram.

If we knew $P_{x,t}$ for all $t \in [T_1, T_2]$ then we would calculate

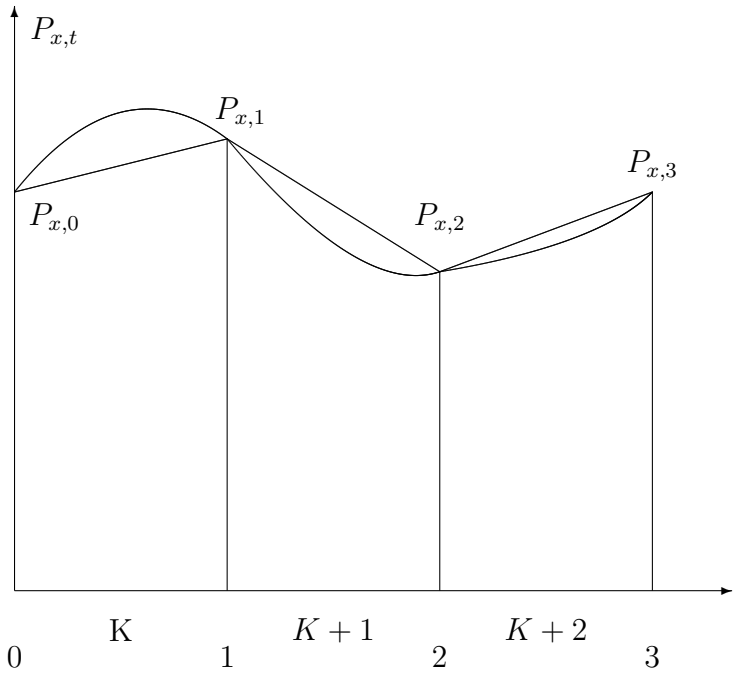
$$E_x^c = \int_{T_1}^{T_2} P_{x,t}\, dt. \tag{49}$$

Suppose, however, that we only know $P_{x,t}$ for certain t, say the 1st January or valuation dates, and that our **data** have the following form:

- **Death data**. We have d_x, the total number of deaths aged x last birthday, during each of the $N+1$ calendar years $K, K+1, \ldots, K+N$.

- **Exposure data**. We observe $P_{x,t}$, the number of lives under observation, aged x last birthday at time t, where $t =$ 1st January, in calendar years $K, K+1, \ldots, K+N+1$. The $P_{x,t}$ are known as the **census data**.

We must, therefore, approximate the integral (49) from the more restricted information available. Many approximate integration formulae could be used, e.g., Repeated Simpson's Rule, but, in practical work, the usual assumption is that $P_{x,t}$ is **linear between census dates**. In exam questions a linearity assumption is adequate as long as the approximations and assumptions made are fully explained.



NB: With the definition age x last birthday the function $P_{x,t}$ is **continuous** in t (assuming we have a large number of policyholders). Lives will enter and leave the category "age x" in a continuous fashion. The above graph of $P_{x,t}$ results. However, we only know $P_{x,t}$ at $t = 0, 1, 2$ and 3. The linearity assumption results in the approximation shown, and the

trapezium rule gives

$$\begin{aligned} E_x^c &= \int_{T_1}^{T_2} P_{x,t}\,dt \\ &= \int_0^3 P_{x,t}\,dt \\ &\approx \tfrac{1}{2}\left(P_{x,0}+P_{x,1}\right)+\tfrac{1}{2}\left(P_{x,1}+P_{x,2}\right)+\tfrac{1}{2}\left(P_{x,2}+P_{x,3}\right) \\ &= \tfrac{1}{2}P_{x,0}+P_{x,1}+P_{x,2}+\tfrac{1}{2}P_{x,3} \end{aligned}$$

or in general (with this age definition)

$$E_x^c = \int_K^{K+N+1} P_{x,t}\,dt \approx \tfrac{1}{2}\sum_{t=K}^{K+N}(P_{x,t}+P_{x,t+1}).$$

Approximations to the exposed to risk of this form are called **census formulae**, and the method is called the **census method**.

Calculation of μ_x and q_x

The calculated value of E_x^c can be used for the direct estimation of $\mu_{x'}$ where x' represents the age at which the estimate applies. **The age x' will vary depending on the particular label used for age definitions.** The calculation of the ages at which estimates apply is a common feature of exam and tutorials questions.

Initial rates of mortality (q type) are ideally calculated from values of μ_x, or directly using the approximate formula

$$E_x = E_x^c + \tfrac{1}{2}d_x \Rightarrow q_{x'} = \frac{d_x}{E_x}.$$

Example: Suppose (a) the total number of deaths aged 20 next birthday during the period 2004 to 2007 inclusive was 25, and (b) we have the following census data:

Date	Number of lives 20 next birthday
1/1/04	50,600
1/1/05	51,210
1/1/06	48,410
1/1/07	47,730
1/1/08	48,020

We apply the trapezium rule and estimate

$$E_x^c = \tfrac{1}{2}\times 50600 + 51210 + 48410 + 47730 + \tfrac{1}{2}\times 48020 = 196660$$

and so

$$\hat{\mu}_{20} = \frac{25}{196660} = 0.00001271.$$

NB: The subscript '20' in $\hat{\mu}_{20}$ is an **age label**. In this example $\hat{\mu}_{20}$ estimates the force of mortality in the age range $19 \to 20$ and so (approximately) $\hat{\mu}_{20}$ estimates the force of mortality at exact age $19\frac{1}{2}$, the midpoint of the age range.

If we require an estimate of q type rates, we calculate $\hat{q}_{20} = 1 - e^{-\hat{\mu}_{20}}$. The age interval is still $19 \to 20$ and so $\hat{q}_{20}$ estimates q_{19}, the initial mortality rate at age 19.

8.7 Adjustment of census data

Ideally, our values of d_x and E_x^c will use the same definitions of age. If the definitions differ then we should, if possible, adjust the census data, since the death data carries "most of the information", particularly when mortality rates are small. The CMIB uses "**age nearest birthday**" in its work. However, some offices submit census data in the "**age last birthday**" form. We must adjust the census data. We introduce the following:

Notation: Let $P_{x,t}^C$ denote the number of lives aged x at time t according the census definition of age. Similarly, let $P_{x,t}^D$ denote the number of lives aged x at time t according the definition of age for deaths.

Problem: The principle of correspondence problem is as follows: we know $P_{x,t}^C$ at certain census times $t_1, t_2, \ldots$ and we need to estimate $P_{x,t}^D$ from $P_{x,t}^C$.

Example: Suppose we have

- d_x = number of deaths aged x **last birthday** at date of death, and
- $P_{x,t}^C$ = number of lives aged x **nearest birthday** at time t.

Different age definitions are being used and so we must adjust the census data. Consider the unknown $P_{x,t}^D$ lives. These lives are age x last, i.e. they are between age x and $x+1$ exact. So, assuming a uniform distribution of birthdays throughout the year, $\frac{1}{2}$ of these lives will be aged x nearest and $\frac{1}{2}$ will be aged $x+1$ nearest. Thus,

$$P_{x,t}^D \approx \tfrac{1}{2}P_{x,t}^C + \tfrac{1}{2}P_{x+1,t}^C,$$

and we have expressed the **unknown** $P_{x,t}^D$ in terms of the **known** $P_{x,t}^C$. In general, we have

$$\begin{aligned} P_{x-1,t}^D &\approx \tfrac{1}{2}P_{x-1,t}^C + \tfrac{1}{2}P_{x,t}^C \\ P_{x,t}^D &\approx \tfrac{1}{2}P_{x,t}^C + \tfrac{1}{2}P_{x+1,t}^C \\ P_{x+1,t}^D &\approx \tfrac{1}{2}P_{x+1,t}^C + \tfrac{1}{2}P_{x+2,t}^C \end{aligned}$$

and similarly for younger and older ages. We now estimate E_x^c as

$$E_x^c = \int_0^T P_{x,t}^D \, dt \approx \tfrac{1}{2}\int_0^T \left(P_{x,t}^C + P_{x+1,t}^C\right) \, dt$$

and evaluate this integral using the approximate integration formula developed before.

NB: The $P_{x,t}^D$ estimates are now approximately consistent with the death data. The principle of correspondence is no longer violated.

Answering exposed to risk questions

1. Examine the **death data** and determine the **rate interval**. Ask the question "when does a life change from age label x to $x+1$?"

2. Examine the **census data** and determine whether or not they are consistent with the death data. Ask the question: "would a life in $P_{x,t}^C$ who died on the census date always be included in the d_x data?"

3. If necessary, adjust the census data to correspond with the death data **stating any assumptions you are making**.

4. Use a census formula to estimate E_x^c stating any assumptions you are making, usually assuming that $P_{x,t}$ is **linear between census dates**. It is important not to apply the census formulae given above mechanically to all questions. In particular the formulae require modification if the census dates are **not at equally spaced annual intervals**.

5. Calculate $\hat{\mu}_x = \dfrac{d_x}{E_x^c}$ or possibly $\hat{q}_x$ as required.

6. Determine the ages to which the estimates apply

   - $\hat{\mu}_x \rightarrow$ mean age at the middle of the rate interval.

   - $\hat{q}_x \rightarrow$ mean age at the start of the rate interval.

Follow the above steps and the majority of exam marks will be obtained!

Example: You have been provided with the following data from death registrations in 2000 and 2001: there were 205 deaths at age 60 in 2000 and 215 deaths at age 60 in 2001. Age of death is defined as the **age last birthday**.

You have also been provided with the following census data for lives aged 60 and 61 **nearest birthday** on each census date.

Census date	Lives aged 60	Lives aged 61
1 January 2000	10180	9980
1 January 2001	10770	10230
1 January 2002	11130	10980

(a) Estimate the force of mortality, μ_{60}, and hence estimate q_{60}. State the ages to which your estimates apply.

(b) Estimate the initial exposed to risk E_{60}, and hence estimate q_{60}. Compare the two estimates of q_{60}.

Solution: (a) We would like to compute the central exposed to risk

$$E_x^c = \int_0^2 P_{x,t}^D \, dt$$

where $P_{x,t}^D$ represents the exposed to risk of death aged x at time t where age is defined using the age definition for deaths. We have two problems:

- the age definitions for death and census are not consistent, and
- we only have census information at times $t = 0$, 1 and 2.

We need to approximate $P_{x,t}^D$ (the exposed to risk using the definition of age for deaths) in terms of $P_{x,t}^C$ (the exposed to risk using the census definition of age). If births are **uniformly distributed** throughout the year, then at any time t on average half the lives aged x last birthday (definition D) will be aged x nearest and half will be aged $x+1$ nearest (definition C). Thus,

$$P_{x,t}^D \approx \tfrac{1}{2}\left(P_{x,t}^C + P_{x+1,t}^C\right).$$

Hence

$$E_x^c = \int_0^2 P_{x,t}^D \, dt \approx \tfrac{1}{2} \int_0^2 (P_{x,t}^C + P_{x+1,t}^C) \, dt.$$

Next, we assume that $P_{x,t}^C$ is **linear between census times** for all x. This allows us to approximate the integral using the trapezium rule. We find

$$\begin{aligned} E_x^c &= \tfrac{1}{2}\left[(\tfrac{1}{2}10180 + 10770 + \tfrac{1}{2}11130) + (\tfrac{1}{2}9980 + 10230 + \tfrac{1}{2}10980)\right] \\ &= \tfrac{1}{4}(10180 + 2 \times 10770 + 11130 + 9980 + 2 \times 10230 + 10980) \\ &= 21067.5. \end{aligned}$$

There were 420 deaths in the two years so $\hat{\mu}_{60} = 420/21067.5 = 0.019936$. Whence, $\hat{q}_{60} = 1 - e^{-0.019936} = 0.019734$. The age definition for deaths is age x last birthday, so age x applies to the year of life exact age x to exact age $x+1$. Hence, $\hat{\mu}_{60}$ applies at exact age 60.5, and $\hat{q}_{60}$ applies to exact age 60.

(b) We can also use the actuarial estimate to estimate E_{60} as $E_{60}^c + \frac{1}{2}d_{60} = 21277.5$. The estimate of q_{60} is $420/21277.5 = 0.019739$ (very close to that obtained in (a)).

8.8 Comparison of life/calendar/policy rate intervals

The rate interval is determined by the way in which deaths (or another decrement of interest) are recorded. Each type of rate interval has advantages and disadvantages. Here is a summary:

1. With life year rate intervals the main advantage is that individuals have the same age at the start of the rate interval. We do not have to make an assumption about the distribution of actual ages over the rate interval, but we do need exact age details.

2. With calendar year rate intervals less data is required to complete an investigation. The ages of individuals can be calculated at the beginning of the year and will not need to be recalculated for deaths during the year. However, distributional assumptions are required to calculate the average age at the start/middle of the rate interval (introducing an approximation).

3. With policy year rate intervals the main advantage is that they give the most accurate results for select investigations (i.e. the study of the variation in mortality with duration of policy). Policy anniversary gives the exact information required to calculate duration of policy exactly. Average ages at the start/middle of the rate interval still have to be calculated but this will be less of a problem than if we had calculated average duration.

   In addition, policy durations need to be recorded for administrative reasons (e.g. premiums rising each year, etc.).

Tutorial 8: Exposed to Risk

1. A mortality study lasts from January 1, 2004 until December 31, 2006, i.e., for three years. The following is the experience of three lives in the study:

   - Life A comes under observation on $2004 + 0.4$ at exact age $x - 1.2$ and remains under observation until the end of the study.
   - Life B comes under observation at the start of the study at exact age $x + 0.2$ and remains under observation until he dies at age $x + 2.1$.
   - Life C comes under observation on $2004 + 0.8$ at exact age $x - 0.7$ and remains under observation until he dies at time $2006 + 0.7$.

   (a) Calculate the contributions to the exposed to risk of these lives under the age label 'age last birthday'.

   (b) Repeat (a) but with the age label 'age nearest birthday'.

   (c) Repeat (a) but with the age label 'age next birthday on preceding January 1'.

2. A mortality study is conducted over the two year period 2001 to 2002. Suppose that the number of deaths aged 60 (where the age label is 'age nearest birthday') is 25 in 2001 and 29 in 2002. Suppose further that census information on the number of lives aged 60 nearest is available as follows: 2018 lives on January 1, 2001, 2037 lives on January 1, 2002, and 2052 lives on January 1, 2003.

   (a) Estimate E^c_{60}.

   (b) Estimate $\hat{\mu}_x$ and hence estimate $\hat{q}_x$.

   (c) Use the actuarial estimate to estimate $\hat{q}_x$ and compare this answer with that obtained in (b).

   (d) At what exact ages do $\hat{\mu}_x$ and $\hat{q}_x$ apply?

   (e) Suppose that additional census information is available as follows: 2032 lives on July 1, 2001 and 2042 lives on July 1, 2002. What now is your estimate of E^c_{60}, and what are the new estimates of $\hat{\mu}_x$ and $\hat{q}_x$?

3. A census type investigation into the mortality of a certain class of policyholders is conducted, where age of death is defined by: age x last birthday. Census data are available on 1 January where age is defined by: age x nearest birthday. There were 55 deaths aged 50 in 1997 and 49 deaths aged 50 in 1998. The census data are as follows:

Census date	Number of lives		
	Age 49	Age 50	Age 51
1 January 1997	16678	18231	19345
1 January 1998	16976	16845	17823
1 January 1999	16995	17534	17982

   Use these data to estimate the force of mortality μ_{50}. At what exact age does your estimate apply? *Ans:* $E^c_x = 35607$, $\hat{\mu}_{50} = 0.00292$.